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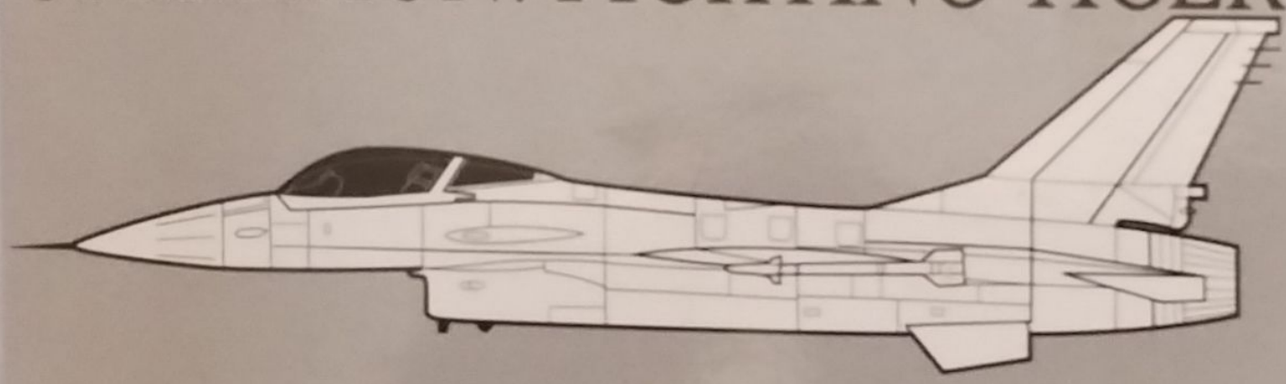
FALCON 3.0™

OPERATION: FIGHTING TIGER™



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OPERATION: FIGHTING TIGER

FLIGHT MANUAL

Spectrum HoloByte

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SECTION

I

INTRODUCTION

"The General who is skilled in defense hides in the most secret recess of the earth; he who is skilled in attack flashes forth from the topmost heights of heaven." — Sun Tzu

It has been suggested that with the disintegration of the Soviet Union and the unprecedented international cooperation exhibited in the Gulf War, the world is embarking on a new order. But if its birth in the flames of Operation: Desert Storm is any indication, the new world order will be no more peaceful than the old. There are still many intractable disputes between nations to provide the tinder for future armed conflagrations. And if the new order presumes international involvement in all military disputes, the armed forces of the United States will be kept quite busy as representatives of the last remaining military superpower. *Operation: Fighting Tiger* includes three theaters with campaigns based on longstanding disputes. In two of these theaters, Korea and the Kashmir province claimed by both India and Pakistan, the disputes have already erupted into war at least once. Armed conflict in the third, the Kurile Islands near Japan, has only become possible in the last few years.

In addition to new theaters, *Operation: Fighting Tiger* includes new aircraft, new weapons, naval task forces and some new missions to challenge you. This manual contains complete descriptions of each theater — the history, the forces involved, the events leading up to the war and the objectives you must accomplish to help your side win. In addition, there are sections on mission planning, electronic warfare as it relates to the F-16, the FSX (a modified version of the F-16 which your squadron will fly in the Kurile campaign) and a reference section detailing the new aircraft, weapons, ships and SAM systems in the game.

Since *Operation: Fighting Tiger* is an add-on module for *Falcon 3.0,* almost all of the cockpit controls, the Campaign screens and other major portions of the game work exactly the same. A few enhancements have been made which will affect the whole game (which are described in the **Falcon 3.01** section), but most relate directly to one or more of the new theaters.

To get started, we recommend you first read **Section II: Falcon 3.01**. Before playing any of the new campaigns, you should at least skim **Section III: Theaters of Conflict** and **Section VI: Electronic Warfare**. You should also read **Section VII: The FSX** and **Section V: Tactical Operations** before playing the Kurile Islands campaign. We think you will find that the other

chapters contain a great deal of useful information which will increase your enjoyment of the game and improve the success of your campaigns, but you don't need to digest all this information at one time. The following is a summary of the contents of each section:

Section II: Falcon 3.01

The **Falcon 3.01** section outlines the differences you will find between *Falcon 3.0* and *Falcon 3.01*. In addition, the section documents many miscellaneous new features which will affect play in all the theaters. Some of these features have been added to *Falcon 3.0* since its original release and are documented here for your convenience; others will only be found in *Falcon 3.01*. The features described include VASI lights, changes to the ILS, a previously undocumented mode of the HFR radar, and the modem command dialog box for setting up communications.

Section III: Theaters of Conflict

The **Theaters of Conflict** section describes the three new theaters of *Operation: Fighting Tiger*. For each theater, the section provides a short history of the conflict which brings your squadron to the area and a brief on the current military situation. Your overall objectives for successfully completing the campaign are also outlined.

Section IV: Mission Planning

The **Mission Planning** section discusses many of the facets of planning a successful combat mission. It includes several charts and diagrams which will help you win your campaigns.

Section V: Tactical Operations

The **Tactical Operations** section describes each of the new mission types in *Operation: Fighting Tiger*. It also includes a list of all the missions in the game and the tasks you must accomplish to successfully complete them. The new missions include SEAD (suppression of enemy air defenses), HAVCAP (high-asset value combat air patrol), train interdiction, airborne assault interdiction and anti-ship. This section also explains the rules of engagement found in the Kurile Islands campaign.

Section VI: Electronic Warfare

In *Falcon 3.01*, the electromagnetic environment of the game is crowded with new radar threats. The **Electronic Warfare** section explains the new TWI codes for ground- and sea-based radars and how these can help your pilots live longer. It also gives a few tips on evading SAMs and hunting down their launchers.

Section VII: The FSX

The **FSX** section describes both the FSX aircraft and its SEA radar. While closely based on the F-16, the FSX has some noteworthy differences. This section also explains how to aim and fire the AGM-84A Harpoon anti-ship missile.

Section VIII: Reference

The **Reference** section describes all the new aircraft, ships, weapons and radar systems which have been added to the game as part of *Operation: Fighting Tiger*. It also contains a glossary of acronyms and military terminology.

SECTION

II

FALCON 3.01

INTRODUCTION

To embark on *Operation: Fighting Tiger*, you must first install the game into your *Falcon 3.0* directory. Once installed, you will find the *Operation: Fighting Tiger* theaters right alongside the original theaters in the War Room. When you are ready to assign a squadron to one of the new campaigns, simply click on the Theater button to select the desired theater in the same manner as before. The new theater names are Kurile Islands, Korea and Pakistan.

Operation: Fighting Tiger contains enough changes to the game as a whole to almost be considered a completely new version of *Falcon 3.0*. Therefore, the official version number of *Operation: Fighting Tiger* is "*3.01.*" Installing *Operation: Fighting Tiger* will automatically make the change. This chapter covers the differences between *Falcon 3.0* and *Falcon 3.01*. Any last-minute changes to the game can be found in the file README.TXT located on *Operation: Fighting Tiger* Disk 1. Type the following command to read this text file: `TYPE README.TXT | MORE`. The Readme file can also be read with any word processing program, text editor or text viewer.

Installation

Please refer to the Installation Card included with the game for instructions on installing *Operation: Fighting Tiger* into your *Falcon 3.0* directory. Once *Operation: Fighting Tiger* is installed, run the game exactly as before by typing `FALCON3`.

System requirements

Minimum requirements:

- *Falcon 3.0* (installed)
- 12MHz 80286 IBM (or compatible) computer or IBM PS/2 Model 30–286 or higher
- 1MB RAM
- DOS 5.0, DR DOS 5.0 or DR DOS 6.0
- 3MB of available hard disk space (14MB total are needed for *Falcon 3.0* and *Operation: Fighting Tiger*)
- VGA graphics card and VGA color monitor
- One 1.2MB 5¼" disk drive or one 1.44MB 3½" disk drive
- At least 602K (616,448 bytes) of available conventional memory

These additional items will allow you to access all the features of *Falcon 3.01*:

- Ad Lib, Sound Blaster, or Roland LAPC-1 or MT-32 sound card
- Expanded memory (EMS) manager such as QEMM or EMM386
- At least 4MB RAM (If you have at least 1MB of EMS, you will be able to access all features which require extra memory; you just might not be able to use them all at the same time.)
- Null-modem serial cable or 2400 baud Hayes compatible modem for two-player mode; or Novell Netware-compatible local area network (LAN) for multi-player mode.
- 80x87 math coprocessor required for the High Fidelity flight model option

Note that the only system requirements of *Falcon 3.01* above and beyond the original requirements of *Falcon 3.0* are 2K additional (602K total) conventional memory required to run the program and 3MB additional hard disk space. If you have difficulty running *Falcon 3.01* because of the increased memory requirement, please refer to the Troubleshooting section of the **Release Notes and Communications Handbook** which came with *Falcon 3.0*. It contains tips for freeing up additional conventional memory. A good expanded memory manager is very helpful for this process.

Parts list

Your *Operation: Fighting Tiger* package should contain the following items:

- Either two 1.2MB 5¼" or two 1.44MB 3½" disks
- Installation card
- Flight manual
- Navigation map
- Registration card

FALCON 3.01 VS. FALCON 3.0

When you first run *Falcon 3.01*, you should notice a few changes right away, like our new company logo and a new title screen. If you have any doubts as to whether you are playing *Falcon 3.01* or *Falcon 3.0*, look for the version number on the lower right-hand corner of the title screen. It should read V3.01. The following list summarizes those differences between *Falcon 3.0* and *Falcon 3.01* which affect the entire program, not just the new *Operation: Fighting Tiger* theaters. Please refer to **Section III: Theaters of Conflict** to learn about the Kurile Islands, Korea or Kashmir campaigns.

New digitized sounds for old and new radio messages — There are several new radio messages and new digitized speech samples to go with them. In addition, many of the original radio messages which previously used the "radio static" sound cue have been given digitized speech samples. Note that a few radio messages still use the generic "radio static" sound cue. A list of the new messages and what they mean is given in **Section IV: Mission Planning**.

New wingman commands — Three new wingman commands have been added to help you better control your flight. These commands allow you to spread out your formation, to tighten up your formation or to send your flight home without you. These commands are described both later in this chapter and in **Section IV: Mission Planning**.

New planes in Red Flag — All the new planes added for *Operation: Fighting Tiger* will also be found in Red Flag. Note that you can only fly the F-16 in Red Flag. If the FSX is added to a Red Flag mission, it will be flown by a non-squadron pilot. Turn to **Section VIII: Reference** for information on all the new aircraft in the game.

Color changes — The sky and terrain colors for the new theaters have been adjusted. In addition, the takeoff time for the last mission of the day has been pushed back to 11 P.M. in all theaters. You will find that the world is much darker at this hour and you will have to rely more heavily on your cockpit instruments to fly and fight.

New TWI codes — The TWI has new codes to designate the many new surface radar systems in the game. Changes in the codes will affect the original theaters, but the air defense weapons present in these theaters have not changed. Refer to **Section VI: Electronic Warfare** for a definition of the new TWI codes. In addition, the TWI will now detect radar systems that are in search mode, not just radars which have a lock on your aircraft.

Landing aids — A new, more realistic mode has been added to the ILS. In addition, VASI lights have been added to the airfields in the new theaters. See the section on "Instrument and visual landing aids" later in this chapter for more information.

More weather — The weather can now be overcast. Check the "New features" section later in this chapter for details.

Improved debriefing — The Debriefing Screen provides more information, including whether you or any of your squadron pilots crashed into the ground and, if you fail a mission, an explanation why. See **Section V: Tactical Operations** for an explanation of the new debriefing information.

Communications — To play *Falcon 3.01* over a modem, network or direct connection, all players must be using the same version of the game. There are too many differences between *Falcon 3.01* and *Falcon 3.0d* (or earlier) for reliable communications to be established, even in the old theaters.

EMS used for digitized sounds — *Falcon 3.01* uses EMS slightly differently than *Falcon 3.0*. Although there are many more digitized radio messages, this feature does not require any more EMS than it did before (1MB). You may notice, however, that the longest ACMI tape that you can record is now a bit shorter. Also, if you have only slightly more than the minimum required expanded memory for digitized speech, you might find that some EMS options on the System Setup Screen which were available to you in *Falcon 3.0* can no longer be turned on when you run *Falcon 3.01*.

AVTR v1.1 — ACMI in *Falcon 3.01* uses version 1.1 of the AVTR. You will not be able to replay tapes recorded with a different version of the AVTR.

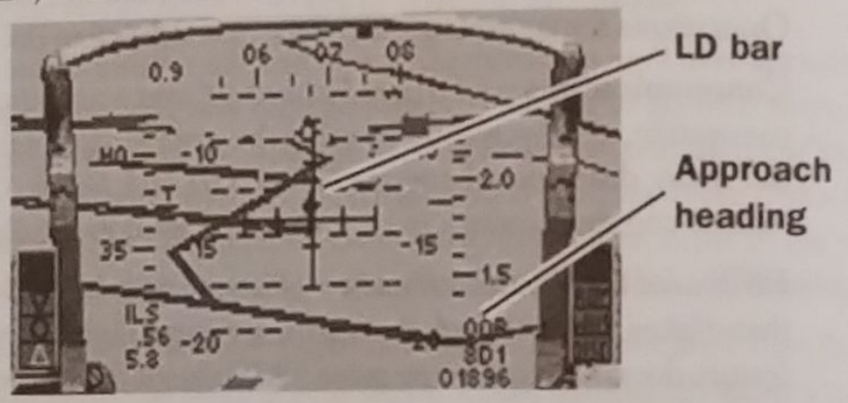
NEW FEATURES

The following section explains many of the new features of *Falcon 3.01*. A few of the features described in this section were added to the game in the later versions of *Falcon 3.0* (either A, C or D); they are documented here for your convenience. Features which have special relevance to *Operation: Fighting Tiger* are not included here, but are described in an appropriate section elsewhere in this manual.

Instrument and visual landing aids

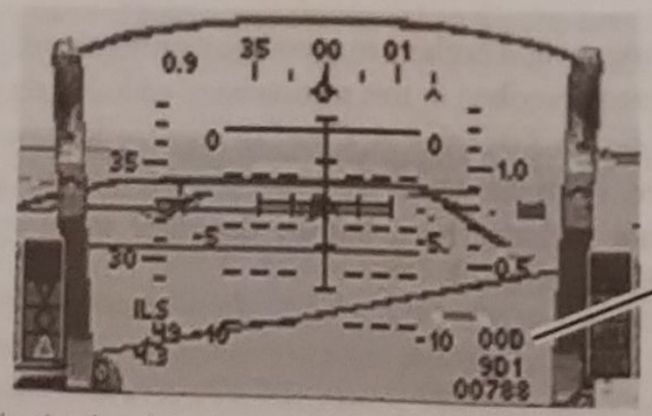
ILS modes

The ILS now has two modes, Directional and Beacon. Pressing \ turns on the ILS; Shift \ will toggle between the two modes. In both ILS modes, the correct heading for the final approach to the runway located at your LAND waypoint is displayed in the lower right-hand corner of the HUD (on top of the stack of numbers). In Directional mode, this heading is followed by the letter "D"; in Beacon mode, the heading is followed by the letter "B".



In the ILS Beacon mode, you can have the LD bar lined up without being on the proper approach heading to your runway.

In the Beacon mode, the ILS locks onto a radio beacon at the end of this runway. The beacon is omnidirectional so centering the localizer deviation (LD) bar will steer your aircraft directly for this point from any direction. The glide slope indication will only be valid if you are approaching the runway on the indicated heading. The Beacon mode is helpful for finding the runway and making the first pass of a racetrack landing, but it is not the best mode to use on the final approach.



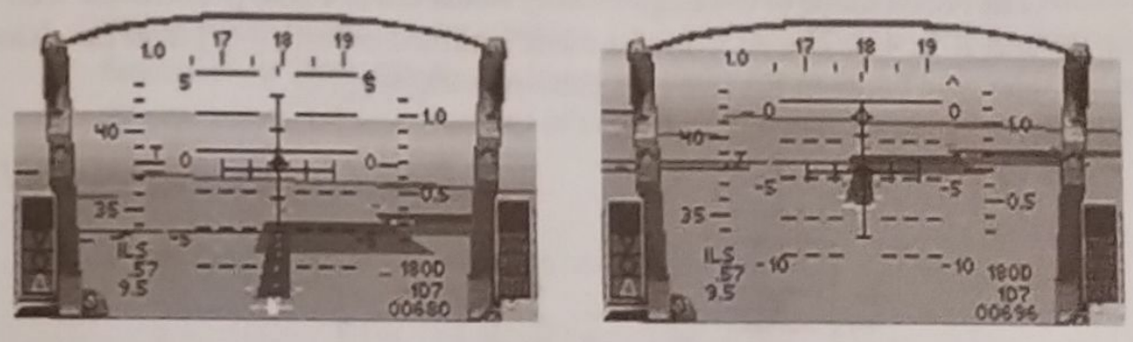
In the ILS Directional mode, lining up the LD bar will put your aircraft on the proper approach heading.

The Directional mode is used to make the final approach to the runway. Lining up the GSD (glide slope deviation) and LD bars will put your aircraft

on a 3° glide slope right down the middle of the runway. If the needles are not lined up, steer your aircraft toward them. Do it early. Because this ILS beam is directional, you have to be within about 20° of the proper heading to receive the correct signals. Outside this envelope, the GSD and LD indications will be invalid.

ILS view shift

When the ILS HUD mode is turned on, the view out of the cockpit is shifted downward so that you can more easily see the runway when your aircraft has the high angle of attack proper for landing. Another smaller downward shift occurs when the landing gear are lowered. Both view shifts can be toggled off and on with Shift Scroll Lock.

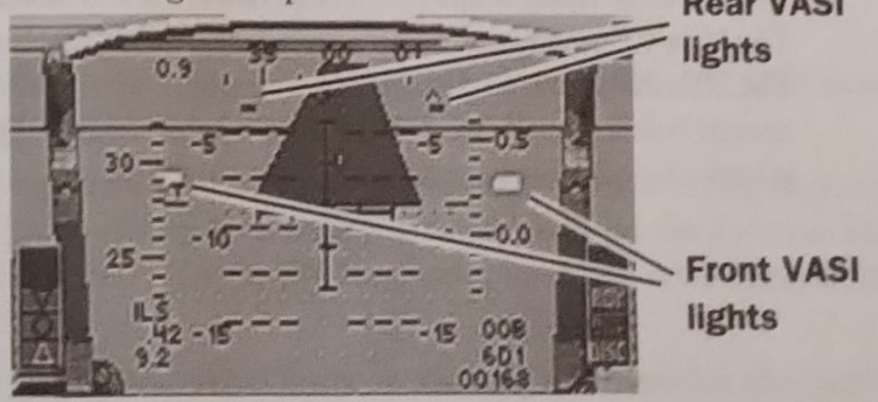


Normal view **ILS view shift**

These two views illustrate the ILS view shift.

VASI lights

Visual Approach Slope Indicator lights are visual landing aids which let you determine at a glance whether you are on, above or below the correct glide slope. VASI lights consist of two pairs of bicolored lights spaced about 50 feet apart on the approach end of the runway. The lights are constructed so that the color you see as you land is either red or white depending on whether you are above or below the glide slope.



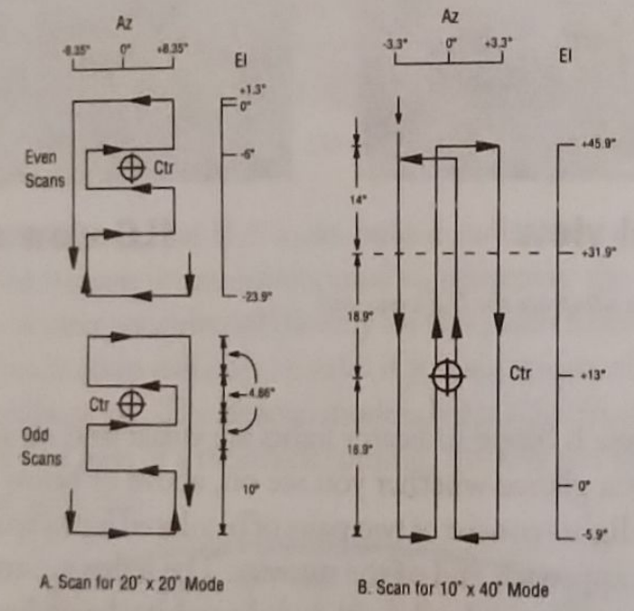
VASI lights: "Red over white, all right. White over red, you're dead!"

The color combination of the front and rear VASI lights will indicate your position relative to the glide slope. If both pairs of lights appear red, then you are below the glide slope. If both pairs appear white, then you are above the glide slope. You are on the correct glide slope when the front pair of lights appears white and the rear pair appears red. If you are above the glide slope, push forward on the stick. If you are below the glide slope, level out until you intercept it again. VASI lights will be found only in the new *Operation: Fighting Tiger* theaters.

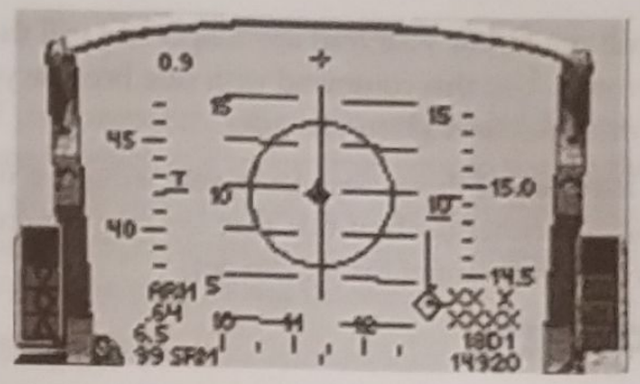
Radar

ACM mode scan patterns

The ACM mode of the High Fidelity Radar has two scan patterns: 20° x 20° and 10° x 40°. The ACM radar mode is turned on with F6. You can then alternate between the two scan patterns with Shift F6.



The 20° x 20° mode scans 20° of both elevation and azimuth in one pattern. The 10° x 40° mode scans 10° of azimuth and 40° of elevation per scan pattern. The 20° x 20° scan is angled slightly downward, so it is better for searching out aircraft below your F-16. The 10° x 40° scan is angled slightly upward, so it is better for searching out targets above your F-16.



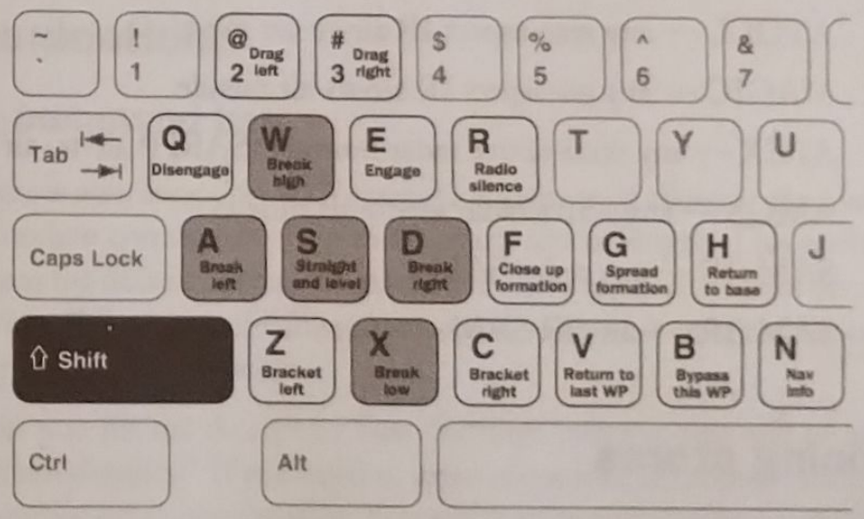
When you are using the 10° x 40° ACM scan, a vertical line will appear centered on the HUD.

Radar settings

The radar range, scan elevation and scan azimuth settings can all be cycled in the opposite direction by pressing Shift in addition to the appropriate function key. For example, to change the radar range directly from 20 nm to 40 nm, press Shift F8 instead of pressing F8 three times.

Wingman commands

Three new wingman commands have been added.



The new commands are: Shift G for spread formation, Shift F for close up formation and Shift H for return to base.

The spread formation command will double the distance between the planes in the flight; the close up formation command will halve the distance. The flight will not spread out so far that successive planes lose visual contact with each other.

The return to base command instructs your flight to abandon the current flight plan and return to base by the most direct means possible. Your

wingmen will not follow your lead any longer nor will they respond to any more commands. Use this command with care because you cannot change your mind once you have given the order.

When you issue one of these commands, your wingmen will respond with a radio message to either confirm that they are carrying out your order or to inform you that they cannot carry them out.

Missile tracking

Incoming missiles can be tracked in Padlock, Track and Wingman Views. Whenever you first switch to Padlock or Track View, the closest missile targeted at your aircraft, if there is one, will be tracked. You can switch the view to padlock or track nearby aircraft by pressing [T]. Pressing [T], however, will not switch the view to padlock or track a missile. To reestablish missile tracking, you must again press the view key ([8] or [9]) for the view you are using. You can also track any missiles targeted at your wingmen in the same way using the Wingman View.

The following names will appear in the Padlock Status Window when you have padlocked a missile (regardless of the specific designation of the missile):

- ATOLL — any rear-aspect IR air-to-air missile
- MAGIC — any all-aspect IR air-to-air missile
- APEX — any semi-active radar homing (SARH) air-to-air missile
- AMOS — any active radar air-to-air missile
- SAM (R) — any radar SAM
- SAM (IR) — any IR SAM

Jettisoning stores

The jettison controls have been changed to allow you to selectively jettison stores:

[Ctrl][K] jettisons all stores except for wingtip missiles and the ECM pod (if you loaded one prior to takeoff).

[Ctrl][F] jettisons all external fuel tanks (both centerline and wing).

[Ctrl][C] jettisons the ECM pod only. This is the only way to jettison an ECM pod.

Remember: "F for fuel, C for countermeasures."

Weather

In some of your missions, you might notice a thick overcast. This is a new weather pattern which will appear from time to time in the game. The cover will vary in altitude and thickness, depending on the atmospheric conditions. If the current weather forecast on the Report Screen is listed as "Cloudy," there is a chance that the weather will be overcast.

Difficulty levels

The Intelligence option is no longer available on the Difficulty Levels Screen. Instead, a Campaign option has been added so that you can adjust the degree of difficulty of your campaigns. The three Campaign options are *ESY* (easy), *STD* (standard) and *HRD* (hard). The more difficult the setting, the less often you will receive replacements and resupplies, the more mobile air defenses will appear in enemy columns and around enemy targets, and the more aircraft you will encounter in enemy formations. The Campaign option will also affect the Realism Value found in the top left-hand corner of the Difficulty Levels Screen. The campaign level in *Falcon 3.0* would be between the *STD* and *HRD* settings on this scale.

Communications

Modem command box

There is now an option in Communications mode allowing the manual entry of modem commands of up to 60 characters in length. This enables you to bypass the default Hayes AT modem setup string and use your own if you are having difficulty establishing a connection or wish to use a phone number longer than 18 characters.

After you hit the Accept or Save Settings buttons, you will be asked "Use modem defaults?" If you need to input some special modem commands or a long phone number, press [N]. You will then be able to enter a series of modem commands one line at a time. Starting with the default modem commands, a series of command line boxes will appear in the the dialog box. Although the default modem commands are automatically entered into the first few boxes, you may alter them if you wish. To replace these commands with your own, press [Backspace] to erase the default command and then type the desired alphanumeric string in its place (up to 60 characters). You can also use the default commands in addition to your own. Press [Enter] to send the command appearing in the box to the modem. When you are ready to exit the command entry session, press [Enter] at a blank command line box.

For example, to use the command mode to enter a long phone number, press [N] to bypass the automatic hookup, but press [Enter] to send the default commands until you reach an empty command line box. Type in the phone number, including any credit card codes, and then press [Enter] twice to send the phone number to the modem and exit the command entry session. The modem will then try to connect.

Communication problems

If the connection is accidentally broken in head-to-head mode, there is a possibility that squadron data will get corrupted. For safety's sake, create a new squadron when entering head-to-head mode. This will provide the most reliable results in head-to-head mode, as well as protect existing squadrons that are involved in a single player campaign.

Miscellaneous

The low fuel HUD readout has been moved. It now replaces the Master Arm Indicator instead of the AGL readout.

A Copy Leader button has been added to the Load Armament Screen in the campaign mode. This button replaces the Report button when you are arming any aircraft other than the flight leader. Activating the button will load the current aircraft with the same weapons as the leader.

In the campaign, the BOMB waypoint action code is only available when you are assigned a bombing mission.

OTHER NOTES

Joysticks

Users of joysticks with a throttle have the option of turning off the joystick's throttle and using the [+] and [-] keys on the keyboard instead. During joystick calibration, you will be given the option of using the throttle wheel or not. Simply press [N] for "No." If you change your mind in the future, you have but to recalibrate your joystick.

If you use the throttle wheel for throttle control, you must first advance the throttle several clicks and then press the [+] key on the keyboard to start the engine. You will then be able to control the engine RPM with the joystick throttle wheel.

For setups containing a joystick, ThrustMaster WCS and MAXX pedals, choose "Joystick" instead of "ThrustMaster" at the System Setup Screen.

Dwight Ennis has created a RAM-resident program called "The Pilot's Edge," which activates the coolie hat on the ThrustMaster Flight Control System. With this program the coolie hat and the other buttons on your Flight Control System can be customized to perform virtually any single-key function of the keyboard. You could, for example, set up the coolie hat to change views and change the speed brake button to activate the IFF. "The Pilot's Edge" can be purchased from ThrustMaster (503/639-3200, fax: 503/635-5528) or Dwight Ennis (408/263-8037, Prodigy NRFR51A). Contact them for further details and price.

Sound

Systems with only 1MB of total RAM and no sound card should set Sound and Music to PC SPEAKER, Radio Messages to NONE and Background Sounds to SIDEWINDER on the System Setup Screen. This will allow you to hear guns, lock-on tones, enemy lock and launch warnings, stall warnings and a caution tone. Digitized voices are not supported on machines without expanded memory (EMS).

Mouse

If you are using QEMM and a Logitech mouse, you may experience some problems with moving the pointer. The pointer may move sluggishly and draw multiple images of itself on the screen. If that is the case, this can be corrected by entering your CMOS setup and turning off the BIOS Shadow RAM. Consult the manual for your computer to see how to shut off BIOS Shadow RAM.

Stacker

Falcon 3.01 works with Stacker, but Stacker must be loaded into high memory. If you are using EMM386 as your DOS 5.0 high memory manager, replace the `DEVICE` command that loads Stacker with `DEVICEHIGH` in your `CONFIG.SYS` file. On the line that loads Stacker, simply add the four letters `HIGH` after the word `DEVICE`. If you are using another memory manager, read the manual to learn how to load device drivers high.

SECTION

III

THEATERS OF CONFLICT

INTRODUCTION

The campaigns in *Operation: Fighting Tiger* are based upon the precept that planning and execution are more important than brute force. The Gulf War provides an excellent example of the truth of this principle. Point by point, the army of Iraq was roughly equal to the forces of the Coalition (where it may have suffered in quality, it at least had quantity). In addition, conventional military wisdom dictates that at least a three-to-one advantage is necessary for a successful attack against prepared defenses like those built by Saddam Hussein's army. On paper, the Coalition did not have the power to dislodge the Iraqi army from Kuwait. Yet, when the shooting started, every element of the Iraqi military was reduced to an inept rabble. The main reasons for this were a well-designed plan of attack and brilliant execution.

Success in the campaigns of *Operation: Fighting Tiger* will also demand good planning and good execution. Defense cuts and politics constrain the amount of force the U.S. can commit to any theater, so you will never have the advantage of overwhelming numbers. Instead, your commanders must earn the advantage by multiplying the effectiveness of the troops they do command. It is imperative to the success of the battle plan that you accomplish the missions that you are assigned — you might get a medal for shooting down six enemy fighters, but this will mean little to the troops at the front line who run out of ammunition because their supply convoy was destroyed by the two bombers you failed to intercept. It is also imperative that you wisely utilize the resources that you command. This will require your judgment in deciding the right amount of force to use: too much and you are wasting vital resources, too little and the mission is doomed. You must plan and lead your squadron's missions to achieve both a high mission success rate and low combat losses.

As you fight to success in your missions, you will see the results. The allied offensive will pick up momentum and the enemy positions will fall. If you routinely fail to accomplish your missions, however, the campaign will quickly become a disaster. Mediocre efforts will meet with mediocre results: stalemate. In many campaigns, you will have to overcome an initial enemy advantage before your side can switch to the offensive.

The following chapter describes the theaters of operation that make up *Operation: Fighting Tiger*. Each section gives a short history of the conflict which brings your squadron to the region, the fictional events leading up to the hostilities and a briefing outlining the balance of forces.

KURILE ISLANDS

"Today is victory over yourself of yesterday; tomorrow is victory over lesser men." — Miyamoto Musashi

History

Since World War II, relations between Russia and Japan have been troubled by Cold War politics and by the Soviet occupation of a few rocky islands which Japan claims as its own rightful property. The disputed islands are part of a chain stretching from the northern Japanese island of Hokkaido to the Kamchatka peninsula of the Russian Republic. Over the past two centuries, the islands in this chain have been bought, traded and stolen several times by both countries. The current dispute concerns only the four islands closest to Japan: Etorofu, Kunashiri, Shikotan and Habomai.

Relations between Russia and Japan first began in the early 19th century when Russia sought to expand its foreign trade. At first willfully isolationist, the feudal society of Japan quickly learned imperial ambitions and joined the international competition for power going on in the region between Russia, China and a handful of European nations. Just as quickly, Japan became very good at this competition, dispatching its rivals in a series of wars to gain control of large tracts of Russia and China as well as all of Korea. Japan continued pursuing its imperial ambitions until its defeat in World War II.

In 1875, when Russia had the upper hand, Japan was forced to trade the resource-rich Sakhalin Island to Russia for the barren Kurile chain. In 1905, however, Japan defeated Russia in a war, took back Sakhalin Island, and also took Manchuria and Korea. At the end of World War II, Russia again had the upper hand and extracted agreements from the other Allied powers promising Sakhalin and the Kurile Islands to Russia in exchange for entering the Pacific war. Russia reoccupied the islands in the last few days of World War II when Japan could no longer put up any resistance. At issue, however, is whether the four disputed islands are technically part of the Kuriles. None of the treaties signed before or after World War II satisfactorily defines which islands are part of the Kuriles; in fact, the postwar treaties were intentionally vague on the subject because the Allies had second thoughts about giving the islands to the Soviets, but did not wish to renege on their promises either. Japan claims that the four islands are not part of the Kuriles and so should be returned. The Russian position is that the Potsdam declaration formally ending the war left Japan in control of "only such minor islands as we [the Allies] determine." The Russians claim that the islands were legally ceded to Russia and that Japan should accept the consequences of losing the war.

Legal issues aside, the Kurile Islands had great strategic importance to the Soviet navy and so it was perhaps inevitable that the issue was never resolved in Japan's favor, although the Soviets periodically vacillated on the subject because of their need for trade and technology from Japan. The breakup of the Soviet Union created an opening for resolution of the matter, but Japanese politicians may have made it difficult on their newly democratic Russian counterparts by directly linking Japanese economic aid with the return of the islands. Yeltsin and his fellow politicians must still deal with a conservative faction and the military. They will also eventually face reelection. As the Russians learn democracy, they will undoubtedly find that nationalistic stands on such diversionary issues as the Kurile Islands make up for much lack of substance in campaign platforms. Still, Yeltsin has gone as far as demobilizing the Russian troops stationed on the islands to pave the way for talks to be held in the fall of 1992.

Kurile campaign

The Kurile campaign takes place after the turn of the century. Negotiations between Russia and the Japan went well, with the Russians agreeing to return the Kurile Islands to Japan in exchange for an infusion of economic aid. Economic strife continued in Russia, however, and Yeltsin was eventually ousted by a military coup. The new and paranoid military dictators believed that giving up the Kurile Islands was a serious mistake which would create a

strategic weakness in the barrier of islands protecting the Sea of Okhotsk and dangerously restrict Russian naval access to the Pacific Ocean. The transfer of the islands was still in progress at the time of the coup, so there are troops from both nations on the island of Kunashiri at the start of the campaign (only Russian troops occupy the other islands). Prior to your squadron's deployment to Hokkaido, the Russians reversed their troop withdrawal and began sending reinforcements to the islands. Japan, in turn, has begun strengthening its own positions on Kunashiri, with support from the United States. There has already been some fighting on Kunashiri between the Japanese SDF (Self Defense Force) and the Russian army, as well as clashes between Japanese and Russian military aircraft, but no direct hostilities have yet occurred between U.S. and Russian forces.

At the beginning of the Kurile Islands campaign, the United States is not officially involved in the conflict. Therefore, the normal peacetime rules of engagement (ROE) between U.S. and Russian forces are still in effect (although the political situation could change before your squadron enters the war). This means that your actions against Russian forces will be constrained until open hostilities break out. As far as the politicians in Washington are concerned, the dispute over the Kurile Islands is a private matter between Japan and Russia. These Washington politicos claim that the conflict doesn't invoke the mutual protection treaty between the U.S. and Japan because the Kurile Islands are not clearly Japanese territory. The commander of the U.S. Pacific Command sees the matter a little differently, however. With Russia again ruled by a militaristic regime, the possession of the southern Kurile Islands is crucial to U.S. strategic interests because it would allow the West to keep track of Russian ships and submarines moving in and out of the Pacific Ocean. Thus, you might notice some ambivalence in your orders — you could be assigned missions in the area of hostilities, but won't be allowed to operate freely. It is mostly to the Russian's advantage to keep the U.S. out of the war, so you could find your squadron deliberately inserted between hostile Russian and Japanese forces. The Russians won't be able to get to the Japanese without going through you, but if they shoot first, it might force the U.S. to officially enter the conflict. After all, if the rules of engagement are to be enforced, you might as well use them to your enemy's disadvantage.

The existence of strict rules of engagement does not release you from the responsibility of successfully carrying out your mission, however. If you are ordered to provide air cover for Japanese troops landing on the islands, you must still prevent Russian strike aircraft from bombing them. Military concerns often conflict with political considerations, and, in this case especially, you serve two masters: the Air Force brass and the Washington politicos. Because your military commanders do not like operating under strict

rules of engagement any more than you probably do, they will allow you some leeway. They might even overlook a few bullet holes in the tail feathers of a MiG. If you go too far, however, your commanders will not be able to save you from being court-martialed. Of course, it just might take a few bullet holes to dissuade a Russian pilot intent on bombing Japanese troops. (Less violent means of persuasion include flying to the control elbow of the enemy plane and achieving a radar lock to spook the enemy pilot. Only a pilot with ice water in his veins would continue an attack with an F-16 ready to put a missile up his tailpipe.)

The campaign will not always start out at the same level of hostilities, however. There may have already been altercations between U.S. and Russian forces before your squadron is ready for action. The rules that apply to your mission will be given at the end of the mission briefing. Breaking these rules could result in the court-martial and subsequent discharge under other than honorable circumstances of the responsible pilot. While the Russian forces are allegedly operating under rules of engagement also, their interpretations of the rules might be somewhat different than your own, so be careful. The rules of engagement are described in **Section V: Tactical Operations**.

In Japan, your squadron is fortunate enough to be assigned the FSX version of the F-16 because of a convoluted political deal between the United States and Japan. After the U.S. Congress balked at approving the FSX program because of the necessary transfers of defense and aerospace technology — areas where America is still a world leader — a deal was struck so that General Dynamics would produce some FSXs at its plants in Texas. Part of the deal was that the U.S. Air Force would purchase the American-built FSXs to fill the gap created by delays in the Advanced Tactical Fighter (ATF) program. These FSXs are to be stationed at U.S. bases in Japan until replaced by the ATF and then resold to the Japanese air force. The deal helped a flagging U.S. defense industry suffering from the outbreak of peace, preserved a few American jobs and allowed the politicians of both countries to avoid domestic criticism for purchasing foreign-built aircraft. The deal also produced a fine fighter aircraft. An additional benefit of the plan is that it gives the U.S. Air Force in Japan maximum integration with the Japanese air force because all the weapons and hardpoints are fully compatible. This will allow American FSXs to operate from any base in Japan without logistical problems. The long-range anti-ship capability of the FSX will be useful too.

Theater briefing

Geography

The Kurile Islands form a rocky chain running from Hokkaido to the Kamchatka peninsula and mark the border between the Pacific Ocean and the Sea of Okhotsk. The four southernmost islands of the chain, Etorofu, Kunashiri, Shikotan and Habomai, are the objects of the territorial dispute. The islands themselves are not much more than barren chunks of rock, but they fence one of two vital straits giving the Russian fleet access to Pacific waters. The Tsushima strait to the south, which runs between South Korea and Japan, could be easily blocked in the event of hostilities. If Russia were to relinquish the four disputed islands, free movement of its fleet through the northern strait would also be jeopardized, particularly in the winter when the ice pack moves south.

Logistics

Neither combatant's economy is at much risk from the war. Hokkaido has some industry, but Japan's major production centers are safely to the south. Military supplies for both sides are produced and stockpiled outside the theater. Supplies for troops on the Kurile Islands are subject to interdiction, however, and Russia in particular might find it difficult to maintain supply lines.

Order of battle

Russia has deployed a strong garrison force occupying well-prepared fortifications on each of the disputed islands. In addition, Etorofu and Kunashiri have airfields capable of supporting all types of Russian military aircraft. Currently only a squadron of MiG-23s is stationed at Etorofu, but additional aircraft are being deployed as part of the Russian reinforcement. The only other major enemy airbase in the theater is on Sakhalin Island, but Russian planes, including the new MiG-31, can easily reach the area from bases outside the theater. In addition, the Admiral Kuznetsov aircraft carrier, which carries two squadrons of MiG-29s and another of Su-27s, is operating in the northern Pacific and could enter the theater at any time. Intelligence data suggests that the Russians might also unleash an experimental fighter aircraft designated the UMF. Little is known about this weapon, but it is apparently a robot aircraft which blurs the line between guided missile and fighter. All the known information about this vehicle is given in **Section VIII: Reference**.

Satellite reconnaissance has revealed the muster of a Russian naval task force off the Kamchatka peninsula in the northern Pacific. Led by the aircraft carrier Admiral Kuznetsov, the task force also contains at least three Kirov-class battle cruisers, three Ivan Rogov-class amphibious warfare ships, several destroyers, and a few troop transports and oilers.

In a strategic gamble, much of the Japanese SDF on Hokkaido has been diverted to the task of invading the Kurile Islands, leaving Hokkaido itself vulnerable. There are a couple of points along the north coast of Hokkaido suitable for landings by Russian Lebed hovercraft, and it is possible that Russia could take advantage of these, rather than counterattacking the Kurile Islands directly. Allied strategy relies on air power to stop the Russian fleet, whatever course it takes.

Your squadron has been assigned to fly the new FSX fighter, a highly modified version of the F-16 jointly developed by General Dynamics and Mitsubishi Heavy Industries. The FSX is stronger, more maneuverable and more powerful than a stock F-16, but uses the same cockpit configuration and weapons control software. The FSX is also compatible with the Harpoon anti-ship missile, which should help you deal with the Russian fleet. A full description of the FSX and a briefing on using the Harpoon missile can be found in **Section VII: The FSX**.

Army	Russia	Japan	U.S.
Manpower	26,000	24,000	5,600
Tanks	680	400	35
APCs	1,200	100	20
Artillery	300	300	15
Air Defense Guns	220	45	24
SAMs	320	160	65
Air Forces	**Russia**	**Japan**	**U.S.**
Fighter	MiG-31, MiG-29, Su-27, MiG-25 MiG-23, MiG-21, MiG-19, UMF	FSX, F-15, F-4	FSX, F-16, F-15
Bomber/Attack	MiG-27, Su-24, Su-25, Tu-22	F-1	A-10
Attack Helicopter	Mi-24		
Transport Aircraft	400	60	30

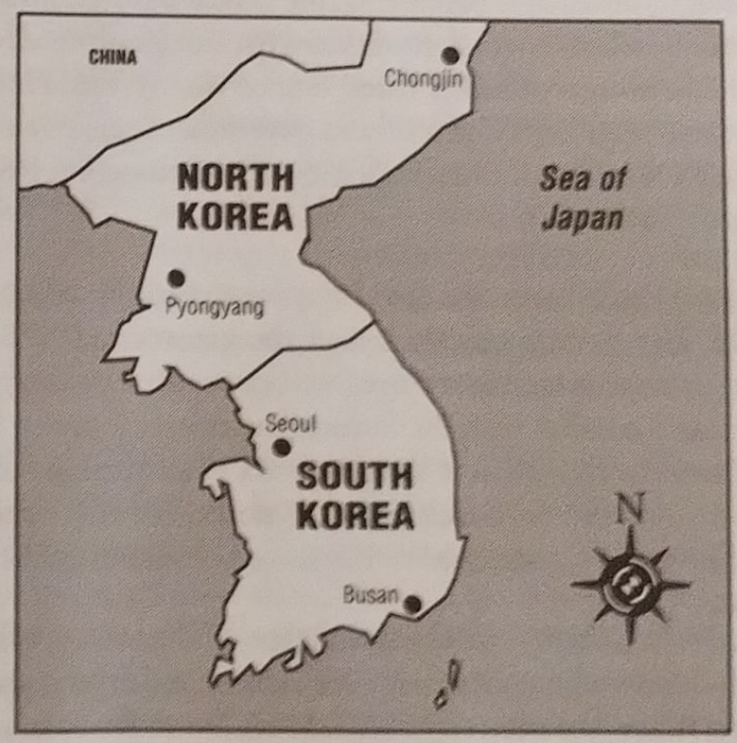
Squadron objectives

As part of the U.S. Air Force contingent in Japan, your squadron will be assigned the following objectives for the campaign:

- Protect Japanese SDF units operating on Kunashiri Island
- Support Japanese SDF assaults on the other Kurile Islands
- Stop the Russian fleet
- Repulse any Russian amphibious attacks
- Protect civilian and military assets on the island of Hokkaido

KOREA

"There is no security on this earth. Only opportunity." — Douglas MacArthur



History

Undivided, Korea is one of the oldest existing nations, having maintained the same borders, language and racial makeup since the seventh century. In the 19th century, however, a chain of events began which lead to the division of the country. Korea became a pawn in an imperial rivalry between China, Japan and Russia, or, as a Korean proverb puts it, "a shrimp crushed in the battle of whales." Japan eventually won, dispatching its rivals in a series of wars between 1894 and 1905. In 1910, Japan formally annexed Korea, and for the next 40 years Korea became a labor colony of Japan, producing rice and industrial products in support of the Japanese war machine.

World War II and defeat ended Japan's occupation of Korea. In the week prior to Japan's surrender, the Soviet army rushed down the Korean peninsula to the 38th parallel (as well as moving to seize other Japanese territories like the Kurile Islands). In accordance with agreements reached among the Allied powers at the Yalta Conference, the Soviet Union accepted the surrender of Japanese forces to the north of the 38th parallel, and the United States accepted the surrender of Japanese forces to the south. The division of Korea was intended as a temporary arrangement, but two impediments prevented the reunification of the country. During the long Japanese occupation almost all Korean institutions, public or private, had ceased to exist, so there was no government to be returned to power. In addition, the wartime alliance between the Soviet Union and the United States proved to be a marriage that could not last. Instead of cooperating, the U.S. and the U.S.S.R. fostered rival governments, which not only entrenched two antagonistic ideologies but also entrenched two sets of ruling elites, one of whom would lose power if the country were reunified. The southern government called itself the Republic of Korea (ROK). The northern called itself the Democratic People's Republic of Korea (DPRK).

The support the Soviets and the U.S. gave to their fledgling states differed markedly. The Soviets quickly armed the government of Kim Il Sung in preparation for an armed invasion of the South, while the U.S., wary of being dragged into a conflict with the Soviets, intentionally under-armed the new South Korean army so that it could only act as an internal police force. This also left it vulnerable to attack from the north, but U.S. strategists did not consider Korea very important to the security interests of the U.S.

In 1949, the U.S. Army withdrew from the ROK, leaving behind only a few hundred advisors who assisted with the training and organization of the ROK army. Less than a year later, on June 25, 1950, North Korea launched a massive attack with seven infantry divisions supported by an armored division of Soviet-built T-34 tanks. ROK forces were undermanned and under equipped. In three days, Seoul, the ROK capitol, was overrun; in two weeks, the DPRK army controlled half of South Korea. Under the auspices of the United Nations, American troops were quickly airlifted to reinforce the ROK army, but even so, by August 26th, the DPRK was within 40 miles of Busan and within 40 miles of pushing the U.S. Eighth Army into the sea. The beleaguered U.S. and ROK forces managed to hold the perimeter around Busan long enough for more reinforcements to arrive and the war bogged down into a stalemate.

To break the deadlock, General Douglas MacArthur made a risky but brilliantly successful amphibious assault on Inchon, 200 miles behind North Korean lines. A hastily thrown together U.S. X Corps cut off the DPRK invasion force from its supplies, while the U.S. Eighth Army mounted a counteroffensive from Busan. The offensive shattered the DPRK army in two weeks. U.S. forces pushed past the 38th parallel and advanced deep into North Korea with the aim of unifying the country.

Once again U.S. strategic ambivalence about Korea surfaced. U.S. strategists considered the Korean conflict to be "the wrong war, in the wrong place, with the wrong enemy." The last thing they wanted was to waste resources needed to fight the Soviet Union in Europe by fighting the Chinese in Korea. General MacArthur, who disagreed with the State Department's focus on Europe, practically dared the Chinese communists to enter the war by marching his troops toward the Chinese border with open flanks and scant reconnaissance. The Chinese took advantage of MacArthur's blunder and, using sheer numbers to make up for inferior weapons and training, drove the U.N. troops back below the 38th parallel. The Chinese army captured Seoul, with only a lack of supplies preventing it from advancing farther. Fresh reinforcements allowed the U.N. forces to retake Seoul and establish a defensive front at the 38th parallel. The situation was again at a stalemate, and the U.N. faced a long, grinding war of attrition against a huge Chinese army.

In the spring of 1951, the Chinese resolved to inflict such heavy casualties that the U.N. would abandon Korea. Their tactic, however, was to assault well-equipped, well-prepared U.N. positions en masse, which resulted only in huge numbers of casualties on the Chinese side. A brief U.N. counteroffensive established the current border between North and South Korea. The cease-fire was signed on July 27, 1953.

The cease-fire agreement created the Demilitarized Zone (DMZ), a ribbon of bare ground lined with barbed wire and concrete bunkers running from one coast to the other. The two sides have maintained an uneasy truce, but shots across the DMZ are not at all uncommon, and the South periodically discovers tunnels being dug underneath the DMZ by the North. The U.S. has maintained a strong military presence in the South as deterrence to any further incursions by the North.

Because of the military standoff, the two governments carried their rivalry into the realms of economics and international politics. At first, the North won this rivalry handily. The central planning of its communist system enabled the country to quickly rebound from the destruction caused by the war, achieving an annual growth in GNP of almost 20%. Its economic success and its staunch

anti-American stance made North Korea a leader and role model for other countries in the Third World, while the South was an economic basket case despised for its dependence on the U.S. North Korea has also been more stable politically. Kim Il Sung, the first leader of the DPRK, still presides over the North with a Stalin-style cult of personality. In contrast, the South has flip-flopped between authoritarian military rule and democracy.

In the mid-1960s, the South began to improve its economic performance under the military government of Park Chung Hee, and with economic success came renewed demands for democracy — and renewed government crackdowns on dissension. Because of the repression, President Park was assassinated by his own CIA and the country returned to civilian rule — briefly.

Because of U.S.-Soviet detente in the early 1970s, both Koreas worried that the superpowers would decide the "Korean issue" without them. In 1971, the first North-South dialogue began. The two sides agreed on unification in principle and even produced the "Three Great Principles of Unification," but the talks got nowhere since neither government wished to relinquish power. The dialogue, which officially ended in 1975, was effectively over within a year after it had begun.

The 1980s were good to the South as it built a strong export-based economy. So good, in fact, that out of desperation the North sent agents to assassinate the South Korean president. The attempt failed, but left relations between the two countries at their worst since the cease-fire. During the early 1980s, the Reagan administration provided generous military assistance and allowed the sale of F-16s and other high technology military hardware to the South. The 1988 Olympics held in Seoul were symbolic of the reversal of international prestige between the South and the North as well. The economy of the North simply failed to keep up. Its policy of self-reliance gave the DPRK a degree of protection from the cycles of recession and inflation plaguing the world economy, but limited its economic growth. The demise of the Soviet Union also hurt the North, removing an important benefactor, trading partner and supplier of high technology weapons.

In 1987, South Korea held popular elections and elected a (more or less) civilian government. Reunification was an important issue in the campaigns of all the candidates, but the great ideological gulf between the two halves of Korea belies any election year posturing on the issue. In the North, Kim Il Sung is slowly transferring power to his son, Kim Jong Il, but he has at least one ambition left unfulfilled and only one means of fulfilling it.

The second Korean war

The Korean campaign in *Operation: Fighting Tiger* is a reprise of the first Korean war. Faced with the South's rising economic fortunes and the worldwide demise of communism, North Korea realizes that it will never be able to dominate a political reunification of Korea. Its window of opportunity for winning a military reunification is also closing as the South modernizes and strengthens its defenses. The North decides that it must act immediately or lose the ability to act at all, and so devises a plan to take the South by surprise.

As a subterfuge to put the South off its guard, North Korea agrees to participate in nationwide elections for a referendum on reunification. To demonstrate its good faith, the North offers to demobilize its forces along the DMZ, but really uses the resultant troop movements to mask preparations for its invasion. On election day, Northern agents instigate riots and violence to add to the normal confusion. Before the votes can be counted, the DPRK launches a midnight attack, led by special forces commandos who destroy key facilities and assassinate military and civilian leaders. The main attack is covered by heavy barrages of long-range artillery and improved Scud missiles (much more accurate than those used by Iraq).

The campaign begins the morning after the North Korean forces launch their assault. Few military installations have escaped damage in the initial special forces attacks, intelligence is skimpy and the North Korean army is already at the outskirts of Seoul. The infamous tunnels that the North continually dug under the DMZ played a small but critical role in the attack, allowing DPRK commandos to infiltrate well into South Korea before being detected. The confusion created by these rear-area assaults led to the evaporation of the ROK defensive line along the DMZ. Even those units that maintained their cohesion during the attack were forced to withdraw under the risk of being enveloped by the advancing DPRK army.

So far the assault has gone exactly according to the DPRK script. Their attack is divided into three major prongs, one down each coast and the third splitting the country in two. The largest force is concentrated on the west coast where the capital, Seoul, and most of the major military installations are located. In addition, the DPRK copied MacArthur's amphibious landing at Inchon to support the attack on the southern capital. As the situation stands, the DPRK western force is threatening to take Seoul, the central force has advanced to Chuncheon and the eastern force has taken Kangnung Airbase. If Seoul falls, several key ROK and U.S. airbases will be threatened and the DPRK will have the advantage of high capacity railroads to rapidly supply and reinforce its army.

With the ground forces in disarray, air power is the only effective weapon left to the ROK and the U.S. A hasty defense has been organized in Seoul, but the defenders lack the artillery to blunt the DPRK advance. If U.S. and South Korean F-16 squadrons cannot stop the North Koreans, the ROK army will have to fall back all the way to Cheongju and abandon its three biggest airbases. Even if the defenders are able to hold Seoul, the middle prong of the DPRK attack can outflank the ROK if they capture Weonju. And Weonju itself is vulnerable to a flank attack from the western prong of the DPRK army. (In other words, you will have your hands full.)

If Seoul falls, the next major battles will probably center on Daejeon and then Daegu. In the first Korean war, Daegu anchored the Busan perimeter, the last line of defense for U.N. forces. It is likely that major elements of each prong of the DPRK attack plan to meet at Daegu in an attempt to break through any defensive perimeter the ROK may establish before it can solidify. The central and eastern axes of the DPRK attack will probably meet little resistance on the ground until they reach Daegu, but the mountainous terrain they must pass through is perfect for ambush from the air. The DPRK will be racing against the reorganization of the ROK army. Your job will be to provide the time for that reorganization.

Your squadron is stationed at Osan Airbase, 30 miles south of Seoul, and you were fortunate enough not to lose any pilots or planes in the commando attacks on the base during the night. The base is secure, for now, and your orders are waiting.

Theater briefing

Geography

Korea is a peninsula of rough mountains jutting out of the Asian continent. Along the west coast are fertile plains filled with rice paddies. Korea has a monsoon climate with cold, dry winters, warm, wet summers and pleasant springs and falls. The mountainous terrain is good for defense, but there are three main invasion corridors into the south: one down the middle and two in the west. The country has many rivers which present major obstacles against invaders. Except in the frozen winters, the western plains are waterlogged and mostly unsuitable to vehicle traffic, so armor is often forced to operate in narrow columns and becomes vulnerable to air strikes and ambushes.

Logistics

Both countries have well developed, virtually self-sufficient defense industries which could sustain a protracted engagement even without much outside assistance. The North is somewhat better endowed with natural resources, but neither country is wealthy. The economy of the North emphasizes self-reliance and is fairly robust, although much less productive than that of the South. The economy of the South is based on exports, and although its international competitiveness has been hurt by rising wages, this has been more than offset by the increased domestic demand created by those higher wages. In 1990, the GNP of the North was about 20 billion dollars, while that of the South was about 158 billion dollars. If the productive capacity of South Korea is damaged during the war, Japan would provide a ready source of fuel, spare parts and ammunition to ROK and U.N. forces. China will probably provide some assistance to North Korea.

North Korea's military success could prove its logistical undoing. Every day, the North must funnel over 100,000 tons of supplies to its forces in the south and each advance stretches its supply lines farther. Although the South has a smaller army, defensive forces generally use more ammunition than offensive forces, so the South's supply requirements are similar. The South can rely on pre-placed stockpiles, however.

Order of battle

The military of North Korea has always had a single goal: the invasion of South Korea. It maintains several thousand tanks and APCs, hundreds of bomber and strike aircraft, dozens of FROG and Scud missiles (which they manufacture themselves), a hundred amphibious assault craft and several divisions of highly trained special forces. In contrast, the South Korean military has three-quarters the manpower and less than half the equipment of the North. It has no dedicated bombers or strike aircraft, although its three squadrons of F-16s can fill that role along with the aging F-5. There is also a strong U.S. military presence in the country. The main problem facing the South in the campaign is not manpower, but organization, since the surprise attack has left the ROK army scattered and in disarray. The most important U.S. airbases are Osan and Kunsan. The U.S. Navy adds significantly to the allied air power, providing carrier-based F-14s and F/A-18s. Long-range bombers such as the B-52 and the F-111 can also reach the theater of operations.

The DPRK army closely follows the Soviet model with its emphasis on mobility and high rates of advance. Air defenses are integrated down to the platoon level; you can expect to see SA-6 and SA-7 SAMs and many ZSU-23-4 Shilkas. Installations in the North are more heavily protected with permanent SAM installations armed by long-range SA-4s. In addition, the North has a sophisticated air-defense radar net which can provide ground control to interceptors and hand off targets to SAM systems.

Army	North	South	U.S.
Manpower	1,000,000	650,000	31,500
Tanks	4,100	1,550	150
APCs	4,340	2,080	100
Artillery	5,800	4,000	80
Air Defense Guns	8,000	600	80
SAMs	280	310	150
Air Forces	**North**	**South**	**U.S.**
Fighter	MiG-29, MiG-23, MiG-21, J-7	F-16, F-4, F-5	F-16, RF-4C
Bomber/Attack	Su-25, Q-5		A-10
Attack Helicopter	Mi-24		AH-64
Transport Aircraft	An-12, Sa-342	UH-60	C-130, UH-60

Squadron objectives

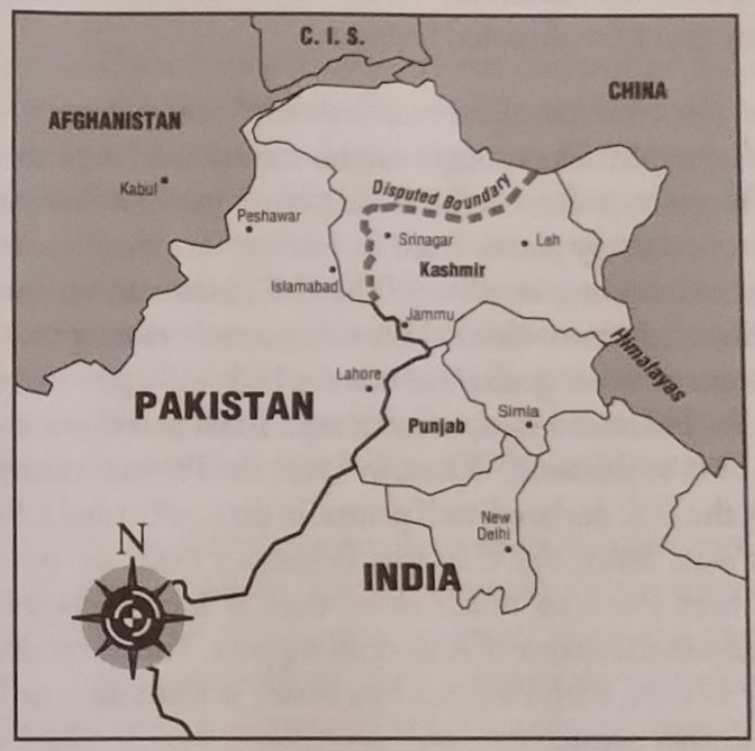
The situation on the ground is bleak, but from your perspective in the skies, things might not seem too bad. The North Koreans only have a few Class A fighters, and their ground troops are moving so fast they have outrun their larger SAM launchers. In the mountainous regions of Korea, there should be plenty of opportunity to ambush enemy columns and vanish before they can bring their air defenses to bear.

The Air Force has been assigned the following objectives:

- Buy time for ROK and U.S. ground forces to regroup with hard-hitting combat air support and interdiction missions
- Protect the rear battle area and civilian factories from raids by DPRK bombers
- Achieve complete air superiority through attacks on DPRK airbases, radar installations and by shooting down the DPRK's best planes
- Strike the DPRK command structure at installations deep within North Korea

KASHMIR

"What difference does it make to the dead, the orphans and the homeless whether the mad destruction is wrought under the name of tyranny or the holy name of liberty or democracy?" — Gandhi



History

After World War II, Britain freed most of the colonies it had acquired during the 19th century. When Britain partitioned the Indian subcontinent, it realized that the Moslems and Hindus in the area had too many differences to live together peaceably. Therefore, Britain divided the territory into two nations, India and Pakistan, and allowed each province in the region to join either nation as it wished. While this may seem a reasonable proposition, the provinces were not in themselves homogeneous. The result was a mass migration of Hindus from Moslem areas and Moslems from Hindu areas, frequently under violent compulsion from the majority group. Almost 15 million people lost their homes; over 1 million people were killed. All Hindu provinces joined India; all Moslem provinces joined Pakistan, save one. This lone Moslem state was Kashmir.

The inclusion of Kashmir into India became an immediate point of contention between the two newly fashioned nations. Pakistan claimed rightful

possession of Kashmir as a Moslem state, charging that the Hindu Maharajah who controlled Kashmir had colluded with Britain and India to prevent the natural accession of Kashmir to Pakistan. India's position on the matter was that Pakistan threatened the independent state of Kashmir, which then voluntarily joined India to avoid being forcibly annexed by Pakistan. Pakistan and India fought a war over the province for two years. In the end, the United Nations mediated a cease-fire, giving Pakistan the northern third of Kashmir and India the rest. Neither side was satisfied with the result, and both countries still claim the entire disputed territory.

After the war, Pakistan allied itself with the United States to counterbalance the military might of its stronger neighbor and joined the U.S.-led South East Asian Treaty Organization (SEATO). In return for military aid, Pakistan let the U.S. operate spy planes from its territory (a valuable service in the days before reconnaissance satellites). The U.S., however, was never completely resolute in its relations with Pakistan, apparently valuing these relations only as the means to other goals. Nor was the U.S. willing to make an enemy out of India for Pakistan's sake, even though India had close relations with the Soviet Union at the time. When, in 1965, the Pakistani army again invaded Kashmir, the U.S. declared itself neutral in the conflict and refused to sell arms to either side. Since the U.S. was Pakistan's main weapons supplier, this decision hurt Pakistan much more than it hurt India and disillusioned Pakistan about the nature of American support. The war ended in a stalemate after only 17 days, with Pakistan's successes in Kashmir countered by Indian victories farther south.

Another consequence of Britain's method of partition also resulted in conflict between Pakistan and India. The nation of Pakistan was created with two geographically separated regions, one to the west of India, the other to the east. The capital of the country was established in the western region. East Pakistan grew to resent being governed from afar, feeling that its own economic development was always subordinated to that of the west, and revolted in 1971. The revolution probably would have failed if India had not intervened. India defeated the Pakistani army in both the east and the west to give East Pakistan (now called Bangladesh) its independence. Once again the U.S. failed to live up to Pakistan's hopes, making only the symbolic gesture of sending an aircraft carrier group to the Indian Ocean. Pakistan dropped out of SEATO, developed military relations with China and purchased many Chinese-built tanks and aircraft.

After the Iranians overthrew the pro-Western government of the Shah and the Russians invaded Afghanistan, Pakistan assumed new importance in the

strategic plans of the U.S. Relations were quickly patched up with a 3.2 billion dollar military aid package, and Pakistan is becoming what the Shah's Iran had been: America's policeman in the Gulf. Its well-trained, professional armed forces regularly send pilots, technicians, advisors and instructors to many Gulf nations. Pakistan also helped the U.S. funnel weapons to the Afghan rebels fighting the Soviet puppet government and, during the Gulf War, sent a brigade of about ten thousand troops to assist in ousting Iraq's army from Kuwait.

The breakup of the Soviet Union has not diminished Pakistan's strategic importance to the U.S. In fact, Pakistan's importance might even increase because many of the former Soviet republics have strong Islamic factions which could align themselves to another U.S. enemy, Iran. Because of Pakistan's integral involvement in the defense of many Arab and Moslem nations, American influence in Pakistan is magnified throughout the Gulf region. At the same time, India's importance is somewhat diminished since the U.S. no longer needs to worry about pushing the Indian government closer to the Soviet Union. Thus, if the U.S. is once again called upon to aid Pakistan against India, it might be more inclined to do so than it has been in the past.

India's governance of Kashmir has been beset by problems. Unable to take Kashmir by force, Pakistan fights the war by proxy: training, arming and providing safe haven to separatist militants waging a guerrilla campaign against the Indian government. As a result of the unrest, local democratic control in Kashmir was suspended in 1990, and governance direct from New Delhi has been influenced strongly by the needs of the security forces. These security forces have had some success, capturing or killing many top militants, but only at the cost of repressing the Moslem population. Still, the militants are becoming increasingly desperate, resorting to terrorist attacks and the kidnapping of tourists to keep pressure on the Indian government, as well as intimidation and murder to stifle dissent among their war-weary fellow Moslems. The Indian government, however, is also in a desperate state, having only one tool, violence, to deal with the situation — a tool it has publicly admitted will not solve the problem.

In addition to waging a proxy war in Kashmir, Pakistan and India are fighting directly, if quietly, across the Siachen glacier in the Himalayas. The battle-ground is at such an elevation that the fighting requires specially trained and equipped soldiers. Every year a dozen soldiers on each side are killed. While the area has no economic or strategic importance, neither government can afford the political liability of conceding one square foot of ground to its enemy.

A jaunt down the road...

The militaries of both countries are proud of their traditions and tend to overestimate their own capabilities. Pakistani generals have even been overheard describing a war with India as a "jaunt down the road to New Delhi." All that is needed to start a war in this region is half an excuse.

The Kashmir campaign in *Operation: Fighting Tiger* begins after separatist guerillas in Kashmir have ambushed an Indian security battalion. Although the militants retreated across the Pakistani border, the Indian commander was unwilling to let the attack go unpunished and continued the pursuit, leading to a clash between Indian and Pakistani forces. The Indian army seized the opportunity of open hostilities to attack rebel bases across the border and launched air strikes and armored thrusts into Pakistan-held areas of Kashmir. Naturally, Pakistan responded by mobilizing its own forces, and the skirmish escalated into a general war.

While the U.S. does not want to get involved, it cannot afford to renege on its military commitments to Pakistan a third time, especially since India is technically the aggressor in this case. On the other hand, it would be difficult for the President to drum up support for a large war effort from either Congress or the American public since India is not an enemy of the U.S. Therefore, the U.S. sends only a few squadrons of fighter and attack aircraft, a carrier task force and a supply of weapons and ammunition for the Pakistani army. Among this contingent is your squadron of F-16s. It is Washington's hope that this force will be sufficient to assist the Pakistani army to victory or at least to stave off defeat. Augmenting the number of U.S. forces involved would carry a heavy liability in domestic politics, but defeat carries an equal liability in international politics and could seriously weaken U.S. influence in the Gulf. Your squadron is the most potent weapon available to the U.S. forces in Pakistan, so the success of the war effort is in your hands.

Theater briefing

Geography

Although the province of Kashmir is quite mountainous, most of the battle area to the south is a flat plain interrupted only by the Indus river and its tributaries. This plain is ideal for large tank battles, but provides few opportunities for terrain masking to aircraft. In the north the situation is reversed. There are many mountains and valleys for aircraft to hide in, but the terrain is rough going for ground troops.

Logistics

India's defense industry is virtually self-sufficient thanks to a policy of licensing weapon technology for indigenous manufacture. Pakistan is dependent on the U.S. and China for weapons, spare parts and ammunition. Both countries have diverse economies which will probably be only marginally impacted by the fighting. Pakistan's military production and supply centers are more vulnerable than India's, but both countries' supply lines to the battlefield are equally vulnerable.

Order of battle

The militaries of both India and Pakistan trace their heritage to the army of the British Empire, and this has probably exacerbated the rivalry between the two nations. In World War II, soldiers from this region were much feared by the Germans, and for good reason. It is said that the Gurkhas would infiltrate Nazi barracks and silently slit the throats of the soldiers sleeping in every other bunk before vanishing into the night. Even today, the favorite weapon of Indian and Pakistani soldiers is probably a long, curved knife.

Pakistan has a very credible army and air force, supplied by the U.S., China and Europe, and can mobilize its total force very quickly. India is potentially much stronger than Pakistan, but would take longer to mobilize its army to full strength. India also has a much longer border to defend, as well as other hostile neighbors like China and Burma, which would prevent it from committing its entire army against Pakistan. India's weapons come mainly from Russia and Europe. A short war favors Pakistan because its forces are situated closer to the battlefield. In a long war, India's greater total force should eventually wear down the Pakistanis.

Army	India	Pakistan	
Manpower	1,265,000	565,000	
Tanks	3,100	1,980	
APCs	1,200	800	
Artillery	4,000	1,600	
Air Defense Guns	2,750	400	
SAMs	740	750	
Air Force	**India**	**Pakistan**	**U.S.**
Fighter	MiG-29, Mirage 2000, MiG-23, MiG-21	F-16, J-7	F-16
Bomber/Attack	Jaguar, MiG-27	Mirage III, Mirage 5, Q-5	A-10
Attack Helicopter	Mi-24		AH-64
Transport Aircraft	An-12, Sa-342	C-130	C-130, UH-60

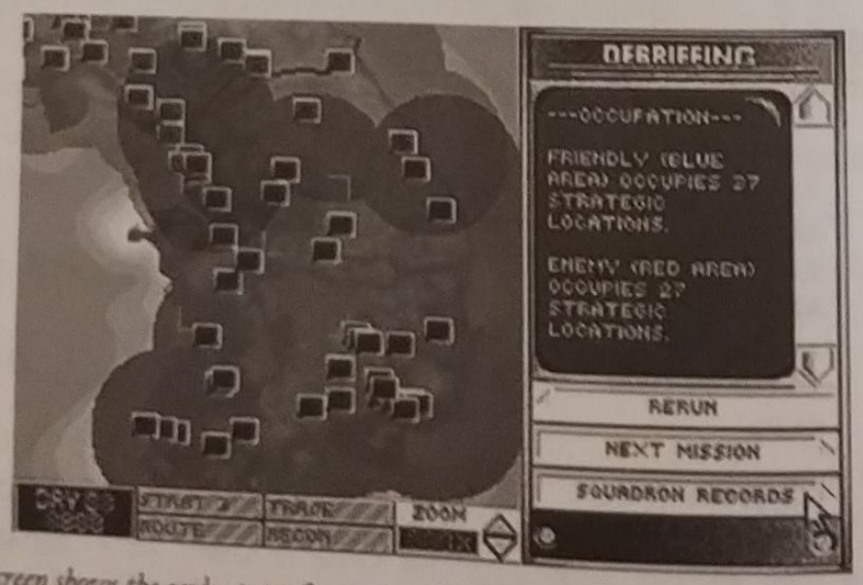
Squadron objectives

The U.S. Air Force has been assigned the task of supporting the Pakistan army. While U.S. forces are officially deployed in a defensive capacity, it is recognized that some offensive missions will be necessary. The following objectives must be achieved:

- Achieve air superiority
- Interdict Indian forces before they can enter the battle
- Support Pakistani armor in both defense and offense
- Destroy Indian radar and SAM installations
- Protect Pakistani rear areas from Indian air raids

HOW A CAMPAIGN WORKS

At the beginning of every campaign, the opposing sides each hold a number of strategic positions in the theater. The most important of these sites, which can be the seat of the government, key military bases, economic centers or even just a strategic crossroads, make up each side's "centers of gravity." Without its centers of gravity, a nation loses both its ability and its will to continue fighting. Therefore, to achieve victory, your side must capture the opponent's centers of gravity while protecting its own. Each side has three such centers, but their exact locations vary from campaign to campaign so that no two wars in the same theater will be exactly alike. The Theater briefing at the beginning of the campaign will outline your general objectives based upon the centers of gravity for that campaign.



This screen shows the early state of a campaign in Korea. Friendly forces control 37 of the strategic locations on the map (which are represented by the map icons), while the enemy controls 27. Three of these strategic locations per side are the centers of gravity which form the keys to victory. (The map icons are not displayed on this screen in the game. They have been superimposed here to illustrate the relationship between the colored areas of influence and the strategic locations.)

The strategy used by the troops on the ground to take these positions is out of your control as the commander of an F-16 squadron, but because your squadron is such an important part of the allied war effort, how well you do your job will strongly influence how well the army can do its job. If you fail all your missions, your side will quickly be forced into a defensive position and then into retreat. If you succeed in your missions, your opponent will be the one who conducts "retrograde maneuvers."

Every morning of the war a situation report is compiled. If you succeeded in most of your missions the day before, then you can expect to see more allied gains than losses. However, if your side has been on the defensive, it might take a day or two to start taking advantage of your successes.

A campaign will continue until one side has taken all the centers of gravity from the other or until 15 days of fighting have elapsed. If neither side has achieved a substantial advantage at the end of 15 days, the politicians step in and the campaign ends in stalemate.

SECTION

IV

MISSION PLANNING

INTRODUCTION

This chapter discusses the elements of planning a mission, including weapon choice, waypoint setup, fuel consumption, formations, putting together multiple flight strike forces and target attack techniques.

The conflicts behind the *Operation: Fighting Tiger* campaigns have been brewing for a long time, giving your opponents ample time to prepare extensive anti-aircraft defenses. Sound planning is essential to counter this increased threat to the health of your pilots. More than ever before, you will be challenged to utilize your resources efficiently. Good mission planning can multiply the effectiveness of these resources and should result in high success rates and low combat losses. Poor or even just mediocre planning will probably yield equally poor results. Mission success, if it comes at all, will come only at a frightful cost in men and material. Your squadron might not even survive the campaign.

The concepts of good planning are simple: study your enemy, know your own capabilities, then use your strengths against your opponent's weaknesses. The key to a successful offense is to concentrate overwhelming force on the point you choose to attack. The key to a successful defense is aggressiveness.

The strengths of your squadron are many. The F-16 is one of the finest aircraft in the world and can carry a large payload of sophisticated weapons. You lead some of the most talented, best trained pilots any air force has ever produced. With these assets you can always seize the advantage of attack. You will be able to choose the the moment, the method and the direction of attack. Even defensive missions can be prosecuted aggressively, using the superior radar and avionics of your F-16 to ambush enemy aircraft.

The challenge you must face before ever getting into the cockpit is to put these concepts into action. You must analyze your mission and your target to determine which weapons to carry, what method of attack to use, what support missions are needed and how many planes to assign to each flight. You must plot your waypoints to accomplish your goals, setting altitude, speed and action code. Fuel consumption must be considered, as well as the individual skills of your pilots. Finally, you must decide which of your flights will most need your leadership in the air. The campaign will set up basic mission plans for you automatically. The information in this chapter will help you optimize your missions with more advanced tactics.

ANALYZE THE TARGET

The first step to mission planning is the analysis of the target. You must determine what weapons will do the job, what method of attack best suits the terrain and air defenses around the target, how much fuel it will take to fly to the target and back, what defenses lie between you and accomplishing your mission, and whether these justify adding support flights to the mission for escort and air defense suppression.

For example, if the target is a flight of enemy aircraft, you will need air-to-air missiles; if the target is a large building, you will need GBUs or general purpose bombs; and if the target is a tank column, you will need Maverick missiles. The ordnance chosen often limits the method of attack, so you must choose the safest method given the limitations of your resources. If the target is a command center deep in enemy territory, you might need extra fuel, an ECM pod and perhaps even escort flights. If the target is an infantry column attacking allied forces along the front lines, you can dispense with the extra fuel and carry more ordnance. The quality of the information on the target's location is also important. If the intelligence is good, as it usually is about fixed or slow moving targets like buildings and ships, the mission can be planned to a high degree of detail. If the intelligence is poor or the target is moving fast, as is usually the case with enemy aircraft and troop columns, the mission must be planned in less detail, but in such a manner as to allow the flight leader to determine the appropriate tactics on the spot.

The air defenses protecting a target can include enemy aircraft, point defense SAMs and AAA, long-range SAMs near the target and any air defenses along the routes to and from the target. Enemy aircraft are ubiquitous: no matter what the mission, you should always plan on encountering enemy aircraft. The disposition of SAMs and AAA around fixed targets can be determined with the Recon and Trace functions on the mission planning screens. The air defenses of the Russian fleet have been closely studied, and this information is available in **Section VIII: Reference**. Intelligence is generally not available about the air defenses travelling with enemy troop columns, but ZSU-23s, SA-6s and SA-7s are standard mobile anti-aircraft weapons. Enemy air defenses can preclude certain tactics, the use of certain weapons and can make the assignment of support flights for air defense suppression and escort necessary.

CHOOSE THE RIGHT WEAPONS

The right weapon for the job is the weapon that is effective against the target, can be used with the desired method of attack and, of course, is available. The amount of weight and drag the weapons will add to the aircraft must also be considered because a heavily laden aircraft is a vulnerable aircraft. On air-to-ground missions take only the amount of ordnance you will need and be sure to save some space for A-A missiles, especially if flying deep into enemy territory. You might have to fight your way both there and back. In the Gulf War, pilots found that they could often successfully dogfight their way to the target without dumping their air-to-ground ordnance. In a conflict with stiffer air defenses, this might not be possible. The following table lists targets appearing in *Operation: Fighting Tiger* and the weapons which can be used against them.

Target	Primary Weapon	Secondary Weapon
Aircraft carrier	Harpoon (2)	Maverick (3), GBU (2)
Battle cruiser	Harpoon (2)	Maverick (3), GBU (2)
Amphibious warfare ship	Harpoon (2)	Maverick (3), GBU (2)
Destroyer	Harpoon (1)	Maverick (2), GBU (1)
Berezina	Harpoon (1)	Maverick (2), GBU (1)
Air cushion landing craft	Maverick (1)	LAU (5)
LST	Maverick (1)	LAU (5)
Large building	GBU (2), Mk84 (2)	Mk83 (4)
Factory, medium-sized building	GBU (1), Mk84(2)	Mk83(4)
Small building	Mk83 (1)	Mk82 (2)
Bridge	GBU (3)	Mk84 (3)
Runway	Durandal (4)	None
Hangar	GBU (1)	Mk83 (2)
Bunker	GBU (1)	None
POL tank	Mk83 (1)	Mk82 (1)
SAM launcher, radar vehicle	HARM (1), Shrike (1)	Maverick (1)
Tank	Maverick (1)	Mk82 (1), Mk82HD (1)
APC	CBU (1)	Mk82 (1), Mk82HD (1)
Truck, Scud launcher, personnel	CBU (1)	Mk82 (1), Mk82HD (1), LAU (2), Strafe (10)

The Primary column lists the weapons best suited to destroy the target. The Secondary column lists weapons that are less effective, but can be used if the primary weapon is not available. The number of weapons it normally takes to destroy a target is shown in parentheses. Note that the method of attack can rule out the use of certain weapons. For example, a 2000 lb general purpose bomb would not be a wise weapon to use on a low-level attack.

CHOOSE AN APPROPRIATE METHOD OF ATTACK

Air-to-ground missions

Ground attacks can be either standoff or close-in, high-level or low-level. They can be pre-planned in great detail with waypoints and action codes, or the method of attack can chosen at the target site and conducted by the flight leader using the wingman commands. The best method of attack is usually a compromise between avoiding the worst enemy air defenses and conforming to the capabilities of available weaponry.

Standoff attacks

As its name implies, a standoff attack is launched outside the range of enemy air defense weapons. By definition, a standoff attack is the safest of any attack method, but only a few weapons have full standoff capability. There are roughly three range umbrellas of anti-aircraft defenses to deal with. The smallest umbrella is that of AAA and SA-7 SAMs. Even general purpose iron bombs can be dropped from outside this umbrella, and a standoff method of attack can be used with any weapon except LAUs, Durandals, high-drag bombs and guns. The second umbrella, that of medium-range SAMs like the SA-6, the SA-N-9 and the SA-N-4, extends about 15–20 nm. Only the Harpoon anti-ship missile and the HARM can be launched from outside this range. The largest umbrella, that of the SA-N-6 and SA-4 SAMs, approaches the range of the Harpoon at about 50 nm. A standoff attack can still be conducted inside these larger umbrellas, but you must use time rather than distance to avoid enemy SAMs. The key is to fly underneath radar to avoid detection until reaching maximum weapon range. Then, pop up just high enough to acquire the target, fire your weapons as fast as possible and dive back into cover before the enemy SAM operators have time to lock onto your aircraft and fire their own weapons.

The following table lists the effective ranges for available standoff weapons.

Weapon	Range
Harpoon	60 nm
HARM	45 nm
Shrike	12 nm
GBU	12 nm
Maverick	5 nm

Remember that "standoff" is a term relative to the range of enemy defenses. What might be considered a standoff weapon in one situation might not be so considered in another.

Close-in attacks

High-level bombing

High-level attacks with GBUs, Mavericks, Shrikes or slicks (Mk82s, 83s and 84s) use altitude to stay out of the umbrella of short-range SAMs and AAA. If there are radar-guided SAMs at or near the target, a high-level attack is suicidal unless an additional flight is added to the mission tasked specifically with attacking enemy air defenses. Once the Iraqi radar-guided SAM threat had been neutralized in the Gulf War, F-16 pilots dropping bombs normally ingressed at 25,000 ft and released their ordnance by 15,000 ft to avoid the remaining infrared-guided SAMs and AAA barrages. The penalties for dropping bombs from this high altitude were loss of accuracy, difficulty in acquiring and identifying targets, and poor battle damage assessment (BDA) after the attack.

Low-level bombing

Low-level bombing is generally the most hazardous approach, exposing the flight to all types of anti-aircraft weapons. Its advantages are in accuracy and that, although the flight is exposed to more anti-aircraft fire, it will be exposed to less lethal types than it would on a high-level attack. The key factor is the prevalence of SA-6s and SA-4s around the target area. If there are many such weapons, a low-level attack could be the only kind possible. If there are SA-6s at the target itself, HARM or Shrike anti-radiation missiles, perhaps

carried by a dedicated SEAD flight, will be needed to destroy them before the main attack begins.

The ordnance and target can also dictate a low-level approach. Durandals require such accuracy that they should be dropped from low altitude. It is also difficult to hit moving ground troops from high altitude with free-fall bombs. An additional hazard with low-level bombing is shrapnel damage from your own bombs. High-drag bombs (Mk82HD) are used to minimize this risk. The following table gives the minimum safe release altitude for the available free-fall ammunition dropped with a straight and level fly-through.

Weapon	Minimum safe release altitude
Durandal	50 ft
Mk82HD	100 ft
CBU	475 ft
Mk82	950 ft
Mk83	1,225 ft
Mk84	1,500 ft

These figures are derived from an F-16 flying straight and level at 550 kts before and after weapon release. The altitude can be safely decreased by flying at a higher airspeed or by performing at least a 4G pull-up immediately after release.

Air-to-air missions

The conditions of air-to-air combat change too rapidly to plan the actual intercepts in detail. Instead, the flight leader must determine the appropriate tactics on the spot. There are, however, a few things you can do in the mission setup that will help your pilots chalk those MiG silhouettes onto the side of their aircraft. On a fighter sweep, for example, you could add a trailer flight to the mission so that any bogeys that try to sneak up on the lead flight will expose themselves to the second. On defensive missions you could add a second line of defense. Alternately, you could be more aggressive and send a flight out ahead of the defensive line to actively search for enemy aircraft to destroy. On scramble missions, you know the rough heading of the enemy aircraft so you could send a second flight to circle around and attack the enemy force from the flank or rear. When you set up your waypoints, however, be sure to avoid concentrations of enemy anti-aircraft defenses.

SET UP FLIGHTS

Number and size

There is a trade-off between the number of flights you can create and the size of those flights. In combat, it is often better to use a number of small flights rather than a single large one. The advantages are that in assigning each flight a particular portion of the mission, you can set up its waypoints and weaponry so that each flight complements the others. The disadvantage is that you can only lead one flight. The pilots in the other flights will follow the flight plan you create to the letter, so effective planning becomes even more crucial to success.

For some air-to-air missions, such as a fighter sweep, you might want to lead a four-ship formation, but that is probably the largest flight you should assign. On air-to-ground missions, it is definitely better to use two-ship formations. With extra flights available, you can assign a flight or two to take out anti-aircraft batteries and another to deal with enemy fighters. Having several flights attacking a target from different directions will make the job of enemy SAM operators more difficult since they will not be able to track a single formation and pick off the aircraft one by one.

Formations

The formations you can choose for your flights can be classified into two types: flat and stacked. Generally the flat formations, wedge and vic, are better for air-to-ground missions because all the aircraft in the flight will fly at roughly the same altitude and can acquire their targets and release their weapons at the

same time. This is especially important if you are flying low to avoid radar. The flat formations are probably also better for fighter escorts of low flying strike aircraft so that the escort does not give away the position of the strike force to enemy radar.

For other air-to-air missions, the stacked formations are probably best because it allows each aircraft in the flight to search a different area of the sky for bogeys.

When flying in a vertically stacked formation, it is important to fly at a sufficient altitude so that your wingmen are not forced into the ground. While they will not blindly fly into the ground, they will be forced out of formation and their attention will be focused on not becoming a lawn dart. They will be less likely to detect incoming bogeys and less able to react to threats.

Formation	Altitude separation	Wingtip-to-wingtip separation	Nose-to-tail separation
V Stack	480 ft	0 ft	480 ft
C Stack	960 ft	0 ft	0 ft
H Stack	480 ft	960 ft	0 ft
Ladder	720 ft	0 ft	720 ft
Wedge	0 ft	720 ft	0 ft
Vic	0 ft	720 ft	675 ft
Echelon	720 ft	480 ft	480 ft

In the standard formation, the separation between each aircraft is about 960 ft. The table translates this distance into the altitude, wingtip-to-wingtip and nose-to-tail separation distances which result from the geometry of each formation. The formation commands will change the distance between aircraft.

Formation commands

There are two new wingman commands which you can issue in flight to change the distance between the aircraft in your formation:

Shift G — Spread formation

The spread formation command will double the distance between adjacent aircraft in the flight. The command can be issued repeatedly, but your wingmen will not spread out so far that adjacent aircraft lose visual contact.

Shift F — Close up formation

The close up formation command will halve the distance between adjacent aircraft in the flight. This command can also be issued more than once, but your wingmen will not fly closer than is safe. When changing the distance between aircraft, the flight will maintain the same formation geometry.

These commands can be used to allow a stacked formation to fly at a lower altitude. They can also be used to spread out a formation for better radar and visual coverage. Your wingman will inform you with a radio message if they cannot carry out your command.

SET UP WAYPOINTS

Altitude

The best altitude for a given waypoint depends on many factors. On air-to-ground missions, altitude should be kept as low as possible when there is a significant threat of radar-guided SAMs. Otherwise, it is better to operate above the ceiling of IR SAMs and AAA. The altitude of ground attack waypoints should also be chosen with consideration of the blast effects of the weapons to be employed.

Air-to-air missions should be flown at a fairly high altitude so that your flight does not give up a speed or altitude advantage to its adversaries. Long-range SAMs are still a threat, however, and you should plan your waypoints so that you avoid flying in areas with intact air defenses.

The actual altitude flown by your wingmen during the mission can vary from the altitude you assign on the Set Waypoints Screen because of the formation chosen for the flight, the action the flight is carrying out or the weapons being dropped. For example, your wingmen will not drop Mk84s from 100 ft, no matter what your orders!

Speed

The factors that should be considered in assigning a speed to a waypoint are fuel consumption, maneuverability and the dictum: "Speed is life." The best maneuvering speed for an F-16 is around 450 kts, and in situations where combat is likely it would be better to maintain about 50 kts above this speed. (Note that the FSX has a corner speed around 420 kts.) A higher speed also makes you a more difficult target for enemy air defenses. If you are cruising at low altitude, you might also want to maintain a higher speed in case you are bounced by enemy fighters, even though doing so will consume considerably more fuel. At a low speed you may not have time to build up enough energy to evade your attackers, who will have both a speed and an altitude advantage.

There are no strict limits on the airspeed at which ordnance can be launched, but most bombing tables the Air Force publishes for the F-16 list speeds between 450 and 650 kts. Higher speeds make it more difficult to put free-fall ordnance on target and give you less time to acquire targets with guided weapons. Slower speeds make your aircraft more vulnerable to AAA.

Warning! The speed of landing waypoints must not be set higher than 300 kts or your wingmen might crash upon touchdown. (The correct landing speed is automatically set in the campaign but not in Red Flag.)

Actions

Most of the action codes which can be assigned to a waypoint are discussed in the *Falcon 3.0* manual, but two new codes have been added for *Operation: Fighting Tiger*.

SEAD: Suppression of Enemy Air Defenses. This action is similar to CAS, but instead of attacking any type of ground unit, the flight will focus its attacks on anti-aircraft weapons and radar systems. This action code can be used to attack mobile, fixed and maritime anti-aircraft systems. The flight can employ any air-to-ground weapons for the job. Refer to **Section VI: Electronic Warfare** for information on specialized avionics systems, weapons and tactics employed for SEAD missions.

ANTISHIP: Naval Interdiction. When performing the ANTISHIP action, the flight will attack the particular class of ship chosen on the Set Waypoints Screen. The ship classes are: KUZNETSOV, KIROV, ROGOV, UDALOY and BEREZINA. When you choose the ANTISHIP action for a waypoint, the Target button will become the Class button. Pressing the Class button will cycle through the available ship classes.

The ANTISHIP action will only be available when an icon for an enemy task force is displayed on the map (which is the only time enemy ships are in the area). The particular task force which will be attacked by the flight performing the ANTISHIP action will be highlighted on the map.

When carrying out the ANTISHIP action, the flight will fire their weapons as soon as they have both acquired a target and are within the maximum weapon range. After weapon delivery, the flight will switch to the next waypoint in the mission plan.

The waypoints are one of two means available to you for controlling your wingmen and the only means for controlling any independent flights you create. Your wingmen will do their best to follow the commands assigned at each waypoint, but they will not blindly attempt things that are impossible or suicidal. When carrying out air-to-ground waypoint actions, your pilots will use their best weapons first and will continue firing until they either run out of ordnance or they reach the waypoint marker. If the flight runs out of weapons before reaching the waypoint, it will break off and go to the next waypoint in the mission plan. (If you are the flight leader, however, your wingmen will continue to follow you.)

Fuel consumption

In general, the internal fuel of your F-16 should be sufficient for most of your missions, but if your F-16 is heavily loaded or your flight path is over 350 nm, you might need to add external fuel. The FSX variant of the F-16 carries more internal fuel, but also has a more powerful engine which burns more fuel, so the range of the FSX is comparable to the F-16. The table below is a quick reference to determine whether or not you should carry external fuel tanks.

F-16 Range	Internal	Internal + Centerline	Internal + Wing	Internal + Centerline + Wing
Drag = 22 GW = 20,113 lbs	450 nm	575 nm	—	—
Drag = 100 GW = 23,500 lbs	420 nm	540 nm	710 nm	—
Drag = 200 GW = 26,000 lbs	370 nm	475 nm	625 nm	730 nm
Drag = 400 GW = 31,500 lbs	290 nm	370 nm	490 nm	575 nm

These figures are for a typical F-16 combat mission flown at 8,000 ft and averaging 450 kts. The table is valid for the Complex and High Fidelity flight models. For the Moderate and Simple flight models, increase the ranges given in the table by 15% and 25% respectively.

To use this table, measure the total distance of your flight path with the map scale rulers and note the average airspeed and altitude. Take the total distance of your flight path and multiply it by the adjustment factors found in the Airspeed and Altitude Correction table below. The result is your corrected flight path.

Airspeed	350 kts	450 kts	550 kts	630 kts	—
Multiply your flight path distance by:	.95	1	1.1	1.2	—
Altitude	**500 ft**	**8,000 ft**	**20,000 ft**	**30,000 ft**	**40,000 ft**
Multiply your flight path distance by:	1.2	1	.6	.4	.3

Continue your mission setup and assign your pilots to the mission. When you get to the Load Armament Screen, check your weapons load and make any desired adjustments. Note the gross weight and drag index of your aircraft given in the lower right-hand corner of the Load Armament Screen. Find the entry in the F-16 Range table that most closely matches the gross weight and drag index for your weapons load. Compare the range given under the amount of fuel you have loaded with your corrected flight path. If the range is less than your corrected flight path, you should load extra fuel tanks on your aircraft.

As an example, consider a mission with the following profile:

Flight path distance: 275 nm
Average airspeed: 550 kts
Average altitude: 500 ft
Aircraft gross weight: 25,000 lbs
Aircraft drag: 190

The corrected flight path for this mission would be:

275 nm x 1.1 (airspeed correction) x 1.2 (altitude correction) = 363 nm

The range entry for internal fuel closest to the aircraft configuration for this mission is 370 nm. Internal fuel should be adequate for this mission, but since the corrected flight path is so near the limit, you might want to carry an external tank anyway in case something unexpected happens during the mission.

Note that this table is only meant as a guide to help you plan your missions. If your mission involves much dogfighting, you can burn up a lot of fuel without covering a lot of ground. You must still keep an eye on your fuel guage while you are in the air. Remember: your need for fuel will always be inversely proportional to the amount left in your fuel tank.

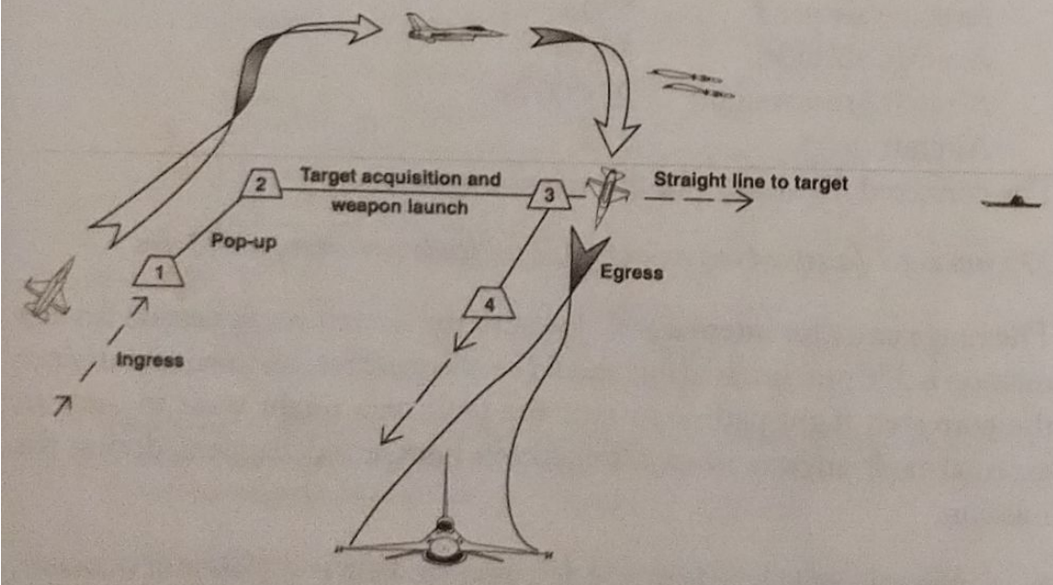
Afterburner usage will significantly increase the rate of fuel usage, so you should try to avoid using your afterburners for an extended period. If you do, the above fuel calculation might be inadequate.

WAYPOINT LAYOUT

Standoff attacks

The idea behind a standoff attack is to fire your weapons at maximum range so that you keep as much distance between yourself and the air defenses around your target as possible. The longer the range of the weapon, the better this works.

There are four waypoints associated with a typical standoff mission. These waypoints allow a flight to approach its target underneath the enemy air search radar, pop up at the maximum weapon range to acquire the target, fire its weapons and break off back to low altitude, all before the enemy has time to react. While human pilots might be able to perform this mission with only a little help from preset waypoints, computer-controlled wingmen need explicit directions for all phases of the mission. The following diagram depicts the waypoint setup for a standoff attack. Note that the diagram only illustrates those waypoints which are specific to a standoff attack; a full flight plan would also include waypoints for takeoff, landing and perhaps for altering the path of the flight to avoid enemy air defense positions on the way to the target.



1. Ingress waypoint
Action: NOTHING
Altitude: Low
Speed: 500 kts

The most important characteristic of the ingress waypoint is its altitude. The altitude should be low enough that your flight is shielded from enemy radar for as long as possible. With long-range weapons, such as the Harpoon, it

should be possible to avoid detection altogether until ingress is completed. See the Ground Radar Range table in **Section VI: Electronic Warfare** to determine an acceptable altitude for ingress.

2. Pop-up waypoint
Action: NOTHING
Altitude: Weapon launch altitude
Speed: 500 kts

At this waypoint, the flight will climb to proper altitude for target acquisition and weapons launch. The waypoint should be placed just outside the maximum weapon range for the chosen launch altitude and far enough from the ingress waypoint to give the flight enough distance to climb from the ingress altitude to the launch altitude (about 1 mile per 5,000 ft of climb). If the target is protected by pulse-doppler radar (as is the case for most ships and SA-4 installations), the ingress and pop-up waypoints should be placed so that the flight makes its climb along a path perpendicular to the radar system. This is because pulse-doppler radar cannot detect aircraft travelling at right angles to the radar beam. With luck, the first (and last) thing the enemy radar operator will see is your flight's turn onto its final bomb run.

For self-propelled weapons, like the Harpoon, the best launch altitude is simply a matter of target acquisition. You must be at least 1,900 ft above sea level for your SEA radar to pick up targets 80 nm away. The best launch altitude is that at which you can detect the target at the maximum range of the weapon, which would be 1425 ft for the 60 nm range of the Harpoon. (Be sure to add a few hundred feet to compensate for any atmospheric effects.) For free-fall weapons, like GBUs and iron bombs, the launch altitude has a great influence on the range of the weapon. Note that the formation chosen for the flight and the number of aircraft in the flight affects the altitude of each individual plane. In the vertically stacked formations at standard spacing, each wingman will fly up to 960 ft below the previous wingman. Thus the last wingman of a seven plane flight flying in stacked formation would be almost 7,000 ft below the flight leader. In this case, the altitude of the pop-up waypoint must be increased by 7,000 ft so that all planes in the flight are able to acquire the target. A better solution is to use a flat formation for standoff missions. If you are the flight leader, you can also tighten up the formation with the new wingman commands: close up Shift F and kick out Shift G. The action code for this waypoint should be NOTHING so that your wingmen aren't distracted while they set up for their primary goal.

3. Attack waypoint
Action: BOMB, ANTISHIP, SEAD, CAS
Altitude: Same as pop-up waypoint
Speed: 450 kts

This waypoint should be placed on a direct line drawn between the pop-up waypoint and the target. It should be placed far enough from the pop-up waypoint that all members of the flight have enough time to fire their weapons, but no closer to the target than is safe. The minimum safe distance will depend on the defenses at the target. AAA and SA-7s have a range of about 1–2 nm; SA-6s, SA-8s, SA-N-4s and SA-N-9s have a range around 15–20 nm; and SA-4s and SA-N-6s have ranges of 44 and 54 nm. You may not always have weapons which will allow you to keep out of range of all the enemy defenses, but you should be able to avoid at least some of them. Keeping the distance between the pop-up and the attack waypoints as short as possible reduces the exposure of your flight to enemy defenses, but gives your flight less time to fire its weapons. (The better trained your pilots, the faster they can acquire targets and fire their weapons.) If you give your wingman mixed weapons loads, they will fire the longest range weapons first and continue firing the other weapons until they are winchester or they reach the attack waypoint and break off.

4. Egress waypoint
Action: NOTHING
Altitude: Low
Speed: Fast

This waypoint should be placed away from the target, but close enough so that the flight will immediately hit the deck. Your wingmen will only change their altitude at a rate sufficient to bring them to the assigned altitude by the time they reach the waypoint. Thus, if you put the egress waypoint too far away from the target, the flight will descend at a leisurely pace, giving enemy SAM shooters plenty of time to knock them out of the sky. If the primary anti-aircraft threat is a SAM guided by pulse radar, the egress waypoint should be placed so that the flight descends along a path perpendicular to the threat.

If you are leading the flight, you can order a re-attack by resetting the waypoint counter to the ingress waypoint with Shift V. Under normal conditions this can be hazardous since the enemy has been alerted to your presence. Any other flights involved in the mission will not repeat the attack with you.

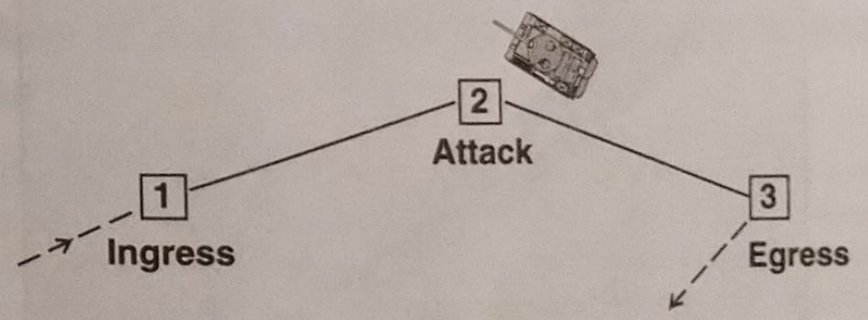
Because of the precision with which a standoff mission must be carried out, it is important that you, as flight leader, fly at the planned altitudes and airspeeds and fly exactly to the waypoints. If you do happen to miss a waypoint by a small margin so that the waypoint counter does not increment, you can order your flight on to the next waypoint by pressing Shift B.

Close-in attacks

If you cannot use the standoff method of attack, you will have to make either a low altitude or a high altitude close-in attack. The advantages of a close-in attack are simplicity and flexibility. There is less that can go wrong. In a standoff attack, a SAM launch at an inopportune moment can disrupt the whole mission. The disadvantage of a close-in attack is that you must get much closer to the target and its anti-aircraft defenses.

Low altitude bombing

A low altitude bombing attack should be planned with a waypoint pattern similar to that shown in the diagram:



The plan for a low altitude bombing attack is very simple. Overflying the target is inevitable.

1. Ingress waypoint
Action: NOTHING
Altitude: The altitude chosen for ingress should depend upon the anti-aircraft threats present along the ingress route. A relatively high altitude could be selected to save fuel, but protection from air defenses should be given priority.
Speed: 450-500 kts

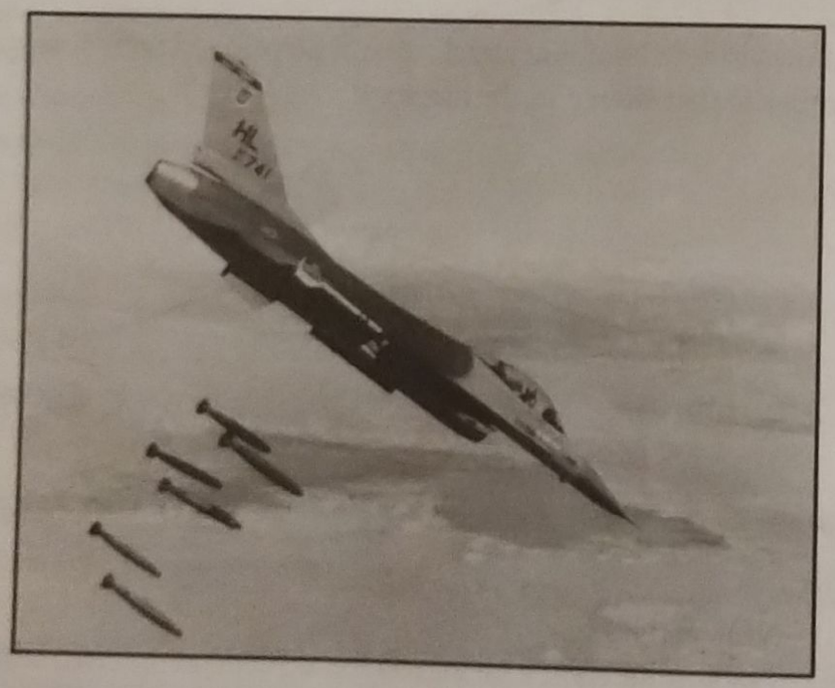
2. Attack waypoint
Action: CAS, BOMB, SEAD, S & D, etc.
Altitude: The minimum altitude assigned to the flight should not be less than that listed in the blast effect chart given earlier in this chapter. Note that the limits in the chart can be exceeded if you use a dive approach

and pull at least 4Gs out of the dive immediately after weapon release. In this case, do not start your dive from below the altitude given in the table.

Speed: Since you don't want to be over the target any longer than necessary, set the speed between 450 and 650 kts.

3. Egress waypoint: Set the same as the ingress waypoint, except put the speed at 650 kts. There's no sense in dawdling; just beware of any IR SAMs locking onto the increased heat output of your engines. Be sure to put the egress waypoint on the far side of the target (unless there is a pressing reason not to) so that the flight will not take the time to make a 180° turn directly above the target area.

High altitude bombing



A high altitude bombing run can employ either waypoint pattern given earlier. If you use the close-in attack method, place the attack waypoint directly over the target. If you are using general purpose bombs, your flight will make a high-angle dive attack to within a few thousand feet above the target. You can use the standoff waypoint pattern to keep a few miles between your flight and the target, which should help you avoid IR SAMs and AAA. Instead of placing the attack waypoint directly above the target, put the waypoint a few miles in front. The following table shows how far

the attack waypoint in the standoff pattern can be placed from the target when using various free-fall bombs released from various altitudes.

Weapon	3,000 ft	5,000 ft	10,000 ft	20,000 ft
GBU	7 nm	7 nm	8 nm	12 nm
Iron bombs, CBUs	1 nm	2 nm	3 nm	3 nm

This table represents the farthest distance the attack waypoint can be placed from the target and still allow the flight to hit its target. The table makes allowances for target acquisition and firing time.

Because the attack waypoint is in front of the target, your flight will use a shallower dive and let the momentum of the bombs carry them the rest of the way to the target. As this table illustrates, GBUs are the most effective and easiest to use weapon for this type of attack.

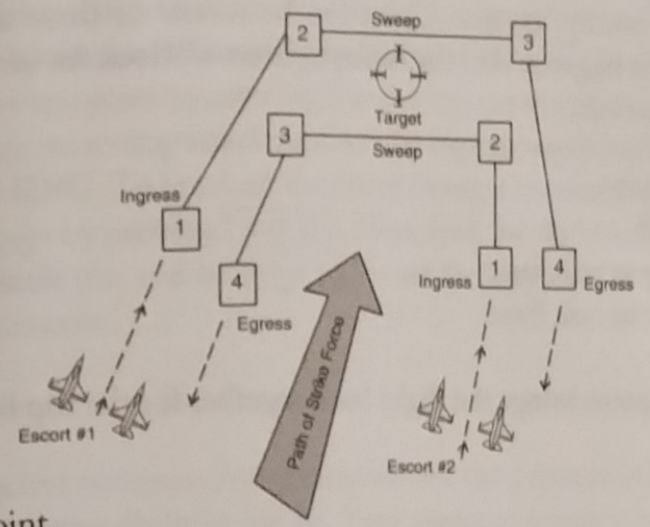
PUTTING IT ALL TOGETHER: A COMPOSITE STRIKE FORCE

The F-16 is a versatile aircraft capable of many missions, but a strike force made up of a few specialist flights often has a better chance of success than a single flight which must be equipped to carry out several different tasks during a mission. This is especially true on missions flown against well-defended targets or deep into enemy territory. Separating your aircraft into small specialized flights will allow you to better match their waypoints and armament to the tasks you need accomplished. On a strategic bombing mission, for example, there are three basic tasks which must be accomplished to successfully complete the mission. First, the strike force has to fight its way through any defending enemy aircraft. Second, the strike force must avoid or destroy any interfering enemy anti-aircraft batteries. Third, the strike force must drop enough of the correct ordnance to do the job. If there are flights specifically assigned to the first two tasks, the strike force will find the third task much easier.

The most difficult part of putting together a composite strike force is coordinating the independent flights. Because the independent flights begin the mission in the air, they have a small head start on your own flight. Therefore, you will have to fly a little faster than planned until you catch up.

Fighter escort and sweep

To help your squadron's attack aircraft make it through the cordon of enemy fighters patrolling the target, you might need to assign a flight or two to escort and sweep duties. The waypoint setup is simpler than for a standoff attack, but the escort should follow the same path as the attack aircraft for most of the flight. When the task force gets to about 20 miles from the target, the escort should accelerate ahead to sweep any enemy interceptors out of the way. If you have the resources to assign two escort flights, put them about five miles to either side of the attack flight, with one escort slightly ahead and the other slightly behind. The H Stack and Echelon formations give the best vertical and horizontal spread for an escort flight, allowing it to detect bandits in the largest possible area. If you want the flight to fly beneath enemy air search radar, however, you should assign a flat formation. The following diagram illustrates the waypoint setup for a typical fighter escort punctuated with a sweep.



1. Ingress waypoint
Action: ESCORT
Altitude: Same as task force
Speed: Same as task force

For most of the flight, the escort should remain close to the rest of the task force. When assigned an ESCORT action, the flight will only engage enemy planes which come within visual range and will not risk giving away the position of the task force by using radar. An alternative tactic is to put the escort at a higher altitude in the hope that any enemy interceptors will engage the escort and be too busy to notice the attack force passing below.

2. Sweep waypoint
Action: CAP
Altitude: 10,000 ft
Speed: 500 kts or strike force speed + 50–100 kts

Because enemy attempts at interception could disrupt the attack at the last moment, the escort fighters should accelerate ahead as the task force nears the target and clear the area of enemy planes. Keep the escort flight's waypoints a few miles away from the target so that they have time to maneuver if any SAMs are fired. While you want the fighters to distract any enemy SAM operators, you don't want your pilots to be easy targets.

3. Patrol waypoint
Action: CAP
Altitude: 10,000 ft
Speed: 500 kts

This waypoint is used to make the escort flight circle the target checking for inbound enemy fighters. There can be several of these waypoints spread around the target so that the escort fighters will look for incoming bogeys in every direction.

4. Egress waypoint
Action: CAP
Altitude: same as task force
Speed: same as task force

This waypoint brings the flight back together for the trip home.

SEAD — Suppression of enemy air defenses

A SEAD flight flown in support of another attack should be set up so that the SEAD flight hits the target area just ahead of the main strike force and just behind the fighter escort, if any. All flights in the task force should attack the target from a different direction to overwhelm and confuse the enemy defenders. Except for having a higher speed set for its attack waypoint, a SEAD flight's waypoint setup should be the same as the main attack flight. Refer to **Section VI: Electronic Warfare** for information on the specialized avionic systems, weapons and tactics of SEAD missions.

LEADING THE ATTACK

Air-to-ground

If the target is mobile, the terrain unfamiliar or you wish to make multiple attacks with different types of weapons, it might be difficult to plan out the mission with more than a straight-in, overflight approach. In some cases the flight leader must make the same decisions in flight that would normally be pre-planned. Even if your waypoints are set up for a run straight over the target, you can adjust your tactics to react to the situation by using wingman commands.

For example, consider a CAS mission planned with the waypoint pattern for the simple close-in attack. In this case, you do not need to slavishly follow the waypoint setup but can react to the situation. Suppose you spot the enemy column at right angles to your flight path, and you are carrying both Mavericks and cluster bombs. On the initial approach, you should order your wingmen to engage at Maverick range, about 5–10 nm. Instead of continuing the attack

to close range and dropping the cluster bombs, it would be more effective to break off the attack once the Mavericks had been fired and circle around to make the bomb run in line with the enemy column. To do this, press Shift V to set your waypoint counter back to the ingress waypoint. This will also tell your wingmen to stop attacking since the ingress waypoint has an action code of NOTHING. To break off the attack faster, you can also tell your wingman to disengage by pressing Shift O. Next, lead the flight into position for the cluster bomb run and hit Shift B to set the waypoint counter back to the attack waypoint.

Air-to-air

The directive commentary commands are the primary means you have for controlling your flight in the air. Your personal example is the other means since your wingmen will normally follow your aircraft. A good understanding of these wingman commands, especially the Bracket and Drag commands, can improve the success of your dogfights immeasurably. These commands are described in the *Falcon 3.0* manual. Study them; practice them.

One difficulty you might encounter in using the Bracket and Drag commands is that your wingmen must be able to pick up the enemy formation either visually or on radar to carry out the order. Your wingmen's radar usage is determined by the action assigned to the current waypoint. Your wingmen will only use radar for INTERCEPT, S&D and CAP waypoint actions. For Escort and air-to-ground action codes, your wingmen will not use radar at all until they engage an enemy plane. Escorting aircraft will rely solely on visual contact to find bogeys and will engage at about 10 nm. For CAP and S&D actions, your wingmen will use the 10–40 nm ranges of their radar but will only engage enemy aircraft within 20 nm. Only on INTERCEPT will your wingmen use the full range of their radar. They will engage aircraft within 40 nm. Thus, if your wingman acts oblivious when you instruct him to bracket, drag or even just to engage an incoming bogey, the problem could be that either your wingman has not picked up the enemy aircraft yet, either on radar or visually, or the enemy is still outside your wingman's zone of engagement. To get your wingman to engage, you might have to lead him closer to the bogey or even "vector" him in the proper direction with the break commands.

If the flight is headed toward a waypoint with a LAND action code, your wingman might also refuse to engage. In this case, you can use the Shift V key to reset the waypoint counter to a different waypoint. This will also shift your wingman into a more aggresive mode of action.

Most of the wingman commands are only followed by your immediate wingman. The commands which affect the whole flight are: engage — Shift E, disengage — Shift O, bypass current waypoint — Shift B and return to previous waypoint — Shift V. There are also three new commands which affect the whole flight: close up formation — Shift F, spread formation — Shift G and return to base — Shift H. The formation commands were described above in the section on formations. The return to base command instructs your pilots to abandon their flight plan, return to base by the most direct route and land. This command is helpful if your wingmen seem reluctant to land. However, the command cannot be recalled and you will no longer be able to control your wingmen.

Wingman and radio messages

There are a number of new radio messages in *Operation: Fighting Tiger*. Like the original radio messages, they will appear in the message bar across the top of the screen. If you have at least 1MB of EMS, you will be able to hear the messages as well if you set the Radio Messages option on the System Setup Screen to use either your sound card or the PC speaker. A few of the messages you receive will use the generic "radio static" sound, which is simply an audio cue that a message has been sent. Most messages are now fully digitized, however, so that you will not have to look away from the HUD in the middle of combat. The following list explains the new messages added for *Operation: Fighting Tiger*.

- *From AWACS: Be advised – enemy ECM activity detected.* — The AWACS has detected emissions from enemy ECM transmitters. These emissions could interfere with the operation of your radar, preventing you from achieving radar lock. Enemy ECM can also seriously degrade the performance of AMRAAM missiles.
- *Copy – Returning to base.* — This message confirms that your wingmen are carrying out your command to return to base.
- *Copy – Closing it up.* — Your wingmen are moving closer in response to your command to close up the formation.
- *Copy – Kicking out.* — Your wingmen are moving farther away in response to your command to spread the formation.
- *Closing on ship target.* — One of your wingman has acquired a ship target and is moving in for the kill.
- *Dumping air-to-ground stores!* — One of your wingmen is lightening his aircraft, probably because he has been jumped by an enemy fighter. Note that he will not dump his air-to-air weapons.

- *Harpoon away!* — One of your wingmen has fired a Harpoon missile.
- *Music on!* — One of your wingmen has turned on his ECM pod.

When you are operating in the Kurile Islands theater under the rules of engagement, you may receive the following radio messages.

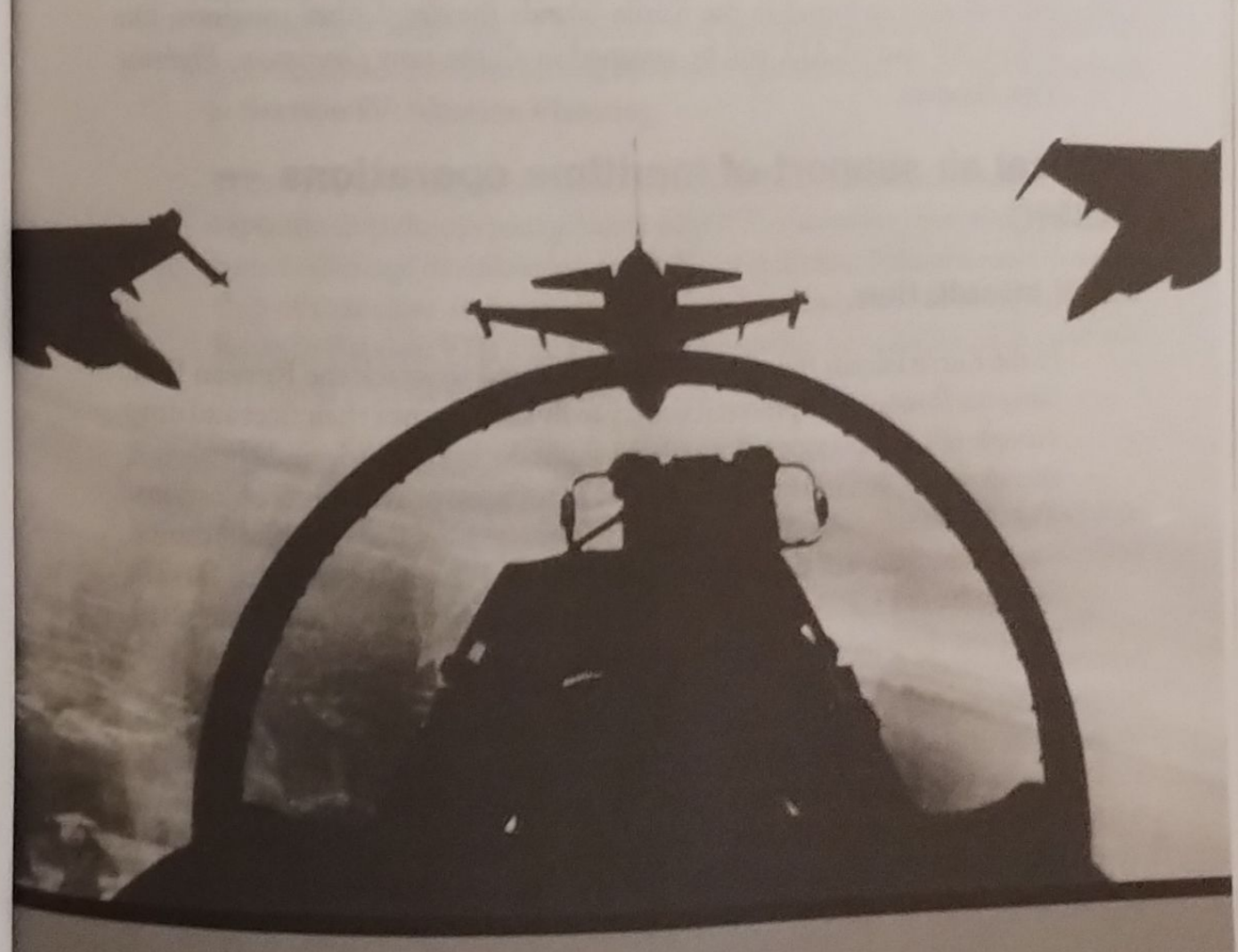
- *From AWACS: You are violating rules of engagement!* — You are violating orders given in the rules of engagement against attacking enemy forces. The rules of engagement only occur in the Kurile Islands theater.
- *From AWACS: Be advised – increased enemy air activity.* — The AWACS has detected large numbers of enemy aircraft being launched from bases around the theater or the Admiral Kuznetsov aircraft carrier. If you have violated the rules of engagement, it is likely that many of these aircraft will be headed your way.
- *From AWACS: Permission given to destroy enemy aircraft.* — This message authorizes you to destroy enemy aircraft. Note that this does not authorize you to attack other types of enemy forces.
- *You're on your own...* — One of your wingmen has lost his nerve and is heading home without you.
- *One Russian bugging out!* — One of your wingmen has persuaded a Russian pilot to leave the engagement at high speed.

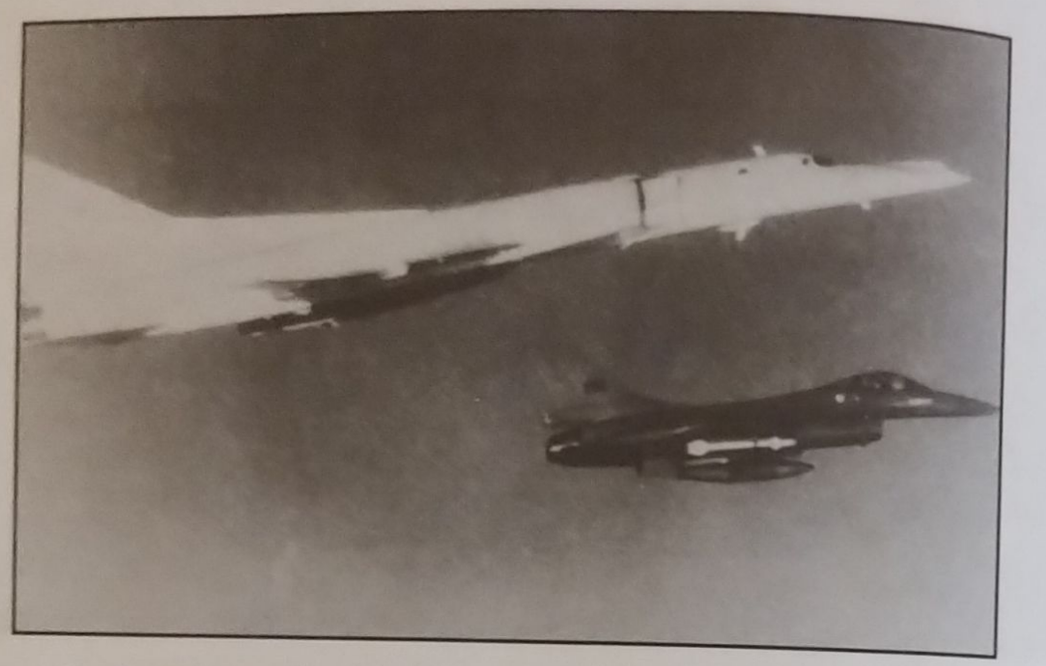
There are a couple of other radio messages that you may receive, but we will let you figure those out for yourself.

SECTION

V

TACTICAL OPERATIONS





This section describes the tactical conditions you will encounter while flying missions in *Operation: Fighting Tiger*. Several new missions have been added to those in *Falcon 3.0*. Many of these missions involve maritime operations and will only be found in the Kurile Islands theater. Other missions, like HAVCAP and SEAD, will be assigned in all the new *Operation: Fighting Tiger* theaters.

Tactical air support of maritime operations — TASMO

Naval interdiction

In the Kurile Islands theater, you will be ordered to attack the Russian fleet. Since the Russians will protest if you try to fly straight over their fleet and drop a couple of 1,000 pounders through the flight deck of the Admiral Kuznetsov aircraft carrier, you will have to find a way to suppress or avoid their objections. The chief Russian objection will be Su-27 fighters launched from the Admiral Kuznetsov. Standard fleet operating procedure calls for a cordon of Su-27s flying a barrier CAP about 80 to 100 nm from the main fleet to intercept incoming threats (such as your squadron).

If you get past this welcoming committee, your next encounter might be with the fleet's anti-submarine screen. While tasked primarily to search out and destroy submarines, the destroyers that form this screen also have effective anti-aircraft weapons. If they happen to be patrolling the area you chose as an ingress route, your mission could suddenly become a bit more interesting.

Just past the destroyers, you should meet the fleet's second major objection to your presence: the SA-N-6 Grumble. If you are armed with Harpoon missiles, which have roughly the same range as the Grumble, you should have about 15 to 20 seconds after popping up to launch altitude to acquire a target, fire your weapons and turn around before the Grumble operators are able to lock onto your aircraft and fire their missiles. If you do not have Harpoons, you will have to dodge the Grumbles to get even closer. This won't be easy, so bring an ECM pod and lots of chaff. As you approach closer to the fleet, you will find an increasing variety of ordnance hurtling in your direction. In addition to more Su-27s and Grumbles, you will encounter SA-N-9, SA-N-4 and SA-N-8 SAMs, as well as 130mm, 100mm and 30mm AAA.

The standoff technique described in **Section IV: Mission Planning** is the safest method with which to attack the Russian fleet. However, to reach the attack range of even the Harpoon missile, you will probably need to set up a composite task force with at least one fighter escort flight in addition to the attack flight. If you run out of Harpoon missiles and have to make a close-in attack on the fleet, a SEAD flight carrying HARM and Shrike missiles will also help. (If Harpoons are not available, GBU-15s and Maverick missiles are the next best weapons.) There is a new waypoint action code, ANTISHIP, specifically designed for setting up attacks against ships. This code is described in **Section IV: Mission Planning**.

When you are assigned a mission to attack the Russian fleet, you will be given a specific ship class as your primary target. To succeed in your mission, you will have to damage or sink at least one ship of this class. If there is more than one ship of that class in the enemy task force, you only need to hit one of them. Refer to **Section VII: The FSX** for information on using the SEA radar and Harpoon missile to target and destroy ships.

Amphibious assaults

In the Kurile Islands conflict, both sides will be conducting amphibious operations. You will be called upon both to attack enemy and to aid allied landings.

An amphibious assault is usually preceded by intense naval bombardment and numerous air strikes. Once the defenses of the area are softened, high-speed landing craft rush marines and equipment ashore. The Russians have an advantage in this area, using the Lebed air-cushion landing craft with about double the speed of Japanese LST landing craft. Frequently, airborne units are also dropped behind the beach defenses by helicopter or parachute. Once the

advance troops have won a secure beachhead, the larger amphibious warfare ships like the Ivan Rogov come in to drop off their large cargo of men and equipment. At this point in the battle, the landing force's main problem is keeping the flow of supplies and reinforcements sufficient to drive the offensive.

There are several missions you could be assigned involving enemy or allied amphibious operations. In general, these will be the standard combat missions, such as CAP and CAS, and your requirements for success will be the same as for a non-amphibious operation.

On CAS missions against enemy landings, you will have to attack incoming landing craft as well as any troops which have already landed on the beach before you arrived. Your first priority should be to attack enemy units already on shore. Once these troops are mopped up, go after any landing craft which have not unloaded their troops or landed yet. Mavericks are the best weapons to use against moving Lebeds. It might be difficult to hit them with anything else because of their high speed. Simply destroying a lot of enemy units might not be enough to succeed in this mission: you must also protect allied units. If you give the enemy enough time to wipe out the defending allied troops before you destroy the attacking forces, the next assault wave will simply surf ashore. In winning the skirmish, you will have lost the battle.

Because the enemy might be able to deploy anti-aircraft weapons on the beach before you reach the combat zone and will certainly have air support, assigning SEAD and fighter escort flights might be prudent.

Airborne operations

Airborne assaults are used to drop light infantry units into the enemy's rear area for many purposes, including interdicting supplies and reinforcements, disrupting command and control, enveloping front-line troops, and attacking specific high-value targets. In *Operation: Fighting Tiger*, you can be assigned a mission to interdict enemy airborne operations.

Airborne assault interdiction

Enemy airborne assault forces will be carried by large cargo planes, like the An-12. The best way to defeat the attack is to shoot down the transports before they reach their drop zone. Failing that, you will have to shoot the enemy troops out of the air as they parachute to earth or destroy them on the ground. If too many of the airborne troops reach their positions and deploy their weapons, your mission will be unsuccessful. They will also start shooting

back. Because the mission can encompass both air-to-air and air-to-ground combat, the best action code to use is S & D. However, you must keep your flight in check with the wingman commands so that they do not waste ordnance on non-mission related targets.

Tactical air reconnaissance

Tactical reconnaissance is one of the most valuable commodities on the battlefield. If you are assigned a mission to gather intelligence, you must fly within 10 nm of the assigned target waypoint to observe the enemy. Don't push your luck, however: the enemy probably won't like you poking your nose into his business and might start shooting.

High asset value combat air patrol — HAVCAP

A HAVCAP is flown to protect vital military assets from enemy attack. The HAVCAP you could be assigned is to escort the E-3 AWACS aircraft. Since the AWACS controls almost all allied air operations, its loss would devastate the allied war effort. E-3s are also quite expensive to replace. The requirement for successfully completing this mission is simple: the AWACS must survive. You can use either the ESCORT or the CAP action codes for this mission. If you use the ESCORT action code, your wingmen will engage bandits at 10 nm. If you use the CAP action code, your wingmen will be more aggressive and engage bandits at 20 nm.

Suppression of enemy air defenses — SEAD

SEAD is both a new waypoint action code and a new type of mission you can be assigned. To succeed in a SEAD mission against land-based SAMs, you will have to destroy more than a third of the anti-aircraft weapons deployed at the target site. Because of the difficulty involved in attacking ships, success on a SEAD mission targeted at maritime SAMs will require the destruction of only one anti-aircraft weapon. However, since you are in the area, you might as well wreak more destruction on the fleet, if possible, so that any subsequent missions are less hazardous. Refer to **Section VI: Electronic Warfare** for information on the threats, specialized equipment and tactics involved in a SEAD mission.

Night missions

In *Operation: Fighting Tiger*, night missions are later and darker, making it more difficult for pilots on both sides to see their adversaries. This will require you to rely more heavily on your radar and instruments to fly and fight. Because

of the limited visibility, both Padlock and Track Views will have restricted ranges at night. The start time for night missions has been changed from 9 P.M. to 11 P.M. Forget about using unguided weapons at night because it will be next to impossible to acquire targets visually.

Operating under rules of engagement

Rules of engagement are "directives issued by competent military authority under which United States forces will initiate and/or continue combat engagement with other forces encountered" [JCS Pub. 1]. Military commands routinely issue peacetime ROEs which are essentially defensive in nature. Operation plans for potential military conflicts contain more aggressive rules of engagement which are automatically invoked if the plan is put into effect.

At the beginning of the Kurile Islands campaign, the United States is not officially involved in the hostilities between Russia and Japan. Until the U.S. officially enters the conflict, you will have to operate under rules of engagement which limit your attacks on enemy forces. In the Kurile Islands theater, there are three individual rules which make up the rules of engagement. One covers your interaction with enemy naval forces, one with enemy ground forces and one with enemy air forces. For each type of force, one of the following rules will apply:

1. *Avoid any engagement with enemy forces.* If attacked, you must evade only. You may not return fire unless authorized to do so by competent authority. The AWACS will relay any authorization to fire upon enemy forces during a mission.

2. *You may defend allied assets against hostile forces.* This rule allows you to fire upon enemy forces if, and only if, the enemy commits a hostile act against you or another allied unit (or unless directed to fire by competent authority).

3. *You are free to engage and destroy enemy forces.*

These rules will be applied by the theater command according to the current level of hostilities between Russia and the United States. As U.S. forces become more deeply embroiled in the conflict, the rules will change to allow more aggressive engagement with the enemy. The rules of engagement in effect for a mission will be outlined at the end of the briefing for every mission assigned until Rule 3 applies to all types of enemy forces. Once you are allowed to freely engage and destroy all enemy forces, this part of the briefing becomes

uneccessary and is no longer given. Typical ROE instructions given in your briefing might be:

- You may freely attack all enemy air and ground forces that you encounter. In addition, you may defend friendly assets against all hostile forces.

Under this ROE, you would be allowed to engage and destroy any enemy air or ground forces, but would only be allowed to fire upon ships if they first attacked allied forces.

Violations and consequences

Any hostile act against a unit proscribed by the rules of engagement will constitute a violation. The launching of weapons targeted at an enemy unit, even if the weapons do not hit their mark, will be considered a hostile act. For example, if the engagement rule for air forces allows attacks only in self-defense, firing on an enemy aircraft will be a violation if you shoot first. Only after the enemy aircraft shoots at you or another allied unit will it become fair game. This does not give you license to attack any enemy unit, however. If an enemy aircraft shoots at you, for example, you are still not allowed to attack enemy ground or naval forces.

The immediate consequence of committing a hostile act is that enemy forces will attack more readily. They might even come looking for the pilot who initiated the hostilities. If this is the case, the AWACS crew should pick up an increase in enemy air activity. Expect to see some interceptors soon. On the other hand, if you get a message from the AWACS authorizing you to engage and destroy enemy aircraft, you can assume things have gotten out of hand and that both sides will freely engage and destroy each other's forces.

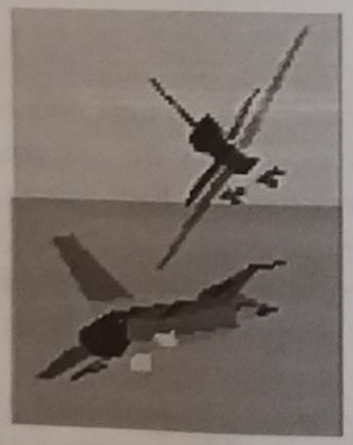
When (or if) you get back from your mission, the pilot responsible for breaking the rules of engagement will face a court-martial. If the infraction was minor, he may be let off, but otherwise he will be drummed out of the service and you will lose a good pilot.

Another possible consequence of violating the rules of engagement is that the level of hostilities between U.S. and Russian forces could escalate. Some violations are more serious than others. If you shoot down a Russian fighter who was out looking for trouble over allied territory, hostilities are less likely to escalate than if you hunt down and kill an aircraft which was minding its own business. Nor does it always take a violation of the rules of engagement — even a justified counterattack could induce the Russians to start openly attacking U.S. forces. If a number of violations on either side occur in one mission, the hostility level can rise all the way to open warfare.

Accomplishing missions under rules of engagement

While the more restrictive rules of engagement are in effect, your missions will be mostly defensive. The dilemma you will face is how to protect yourself and other allied forces without shooting down encroaching enemy aircraft. If you wait until the enemy pilot fires, your intervention will probably come too late to save his unfortunate target and your mission might not be successful. If you shoot the enemy first, you could be court-martialed.

A way out of this dilemma is to engage the enemy in a battle of nerves rather than bullets. Your aim is to convince the enemy pilot that if he doesn't abandon his current mission and bug out, life as he has known it will abruptly end. To be most persuasive, close in on the bogey's six and threaten him with severe bodily harm. Maneuver to achieve a stable firing solution — the closer you are the better — and lock your radar and perhaps an AMRAAM onto the bandit so that the his cockpit warnings light up like the electronics section of a Tokyo department store. The more inexperienced or spineless the enemy pilot, the quicker he will jettison his weapons and bug out on full afterburner. If the enemy aircraft is another fighter, the battle could become a debate to see who can provide the other with the more compelling argument to leave.



The pilot of this Russian bomber was successfully "convinced" to jettison his ordnance and leave the area.

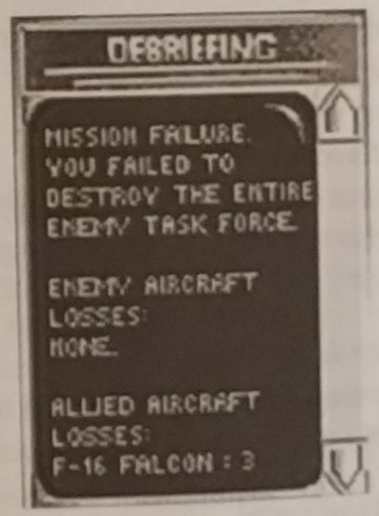
There is risk involved in this duel, however. If an enemy pilot gets behind you, he might not be able to resist the temptation to shoot. The more tense the situation, the more likely someone will fire: this is not a place for rookies. You, your wingman or the enemy pilot could lose his nerve at any time and make a fatal mistake. A pilot's skill and experience determine how likely he is to either bug out or fire against orders. For your squadron pilots, the critical skill is their dogfighting ability, so you may want to assign only pilots who are strong in this area to missions with stringent ROE. The skill of the enemy

pilots you will encounter is determined by the Enemy Logic level set on the Difficulty Levels Screen.

In addition, the Russians won't broadcast any changes in strategy on their side, so don't assume that just because the Russians did not fire upon you in earlier missions, they will not fire upon you in your current mission. Moreover, do not assume that just because you are not allowed to fire upon the Russians, they have not been authorized to fire upon you. If the AWACS authorizes you to freely fire upon enemy aircraft, you can assume that the Russians have shot down at least one U.S. aircraft during your mission and will shoot down more if you give them the chance.

MISSION DEBRIEFING

The Mission Summary portion of the debriefing contains a number of features which have been added since the original release of *Falcon 3.0*. In addition to rating your mission a success or failure, the Mission Summary will provide a brief explanation for a failure. The requirements for successfully completing each mission are given in the following section. There may be several possible reasons for failing a particular mission; the Mission Summary will only list the most obvious.



If you fail to complete your mission, you will be given one of the following or similar reasons:

- No enemy aircraft were shot down.
- You failed to destroy a single enemy unit.

If you are given one of these reasons for failing your mission, then you were assigned to attack a particular type of enemy unit but did not destroy even one.

- Too few enemy aircraft were shot down.
- You failed to completely destroy the enemy strike force.
- You inflicted only minor damage on the enemy forces.

In general, air-to-air missions require that you destroy a particular formation of aircraft or at least destroy more aircraft than you lose. Air-to-ground missions generally require that you destroy a significant percentage of the enemy troops at the assigned target.

- You failed to destroy the primary target.
- You hit the wrong target.
- You did not hit the assigned target.

You will get one of these messages if you fail to destroy the proper target at the assigned location on a bombing mission or if you do not hit a ship of the assigned class on an anti-ship mission.

- You failed to reach your assigned target.
- You failed to reach your assigned station.
- You failed to escort the aircraft all the way to the target.

Some missions require that you fly to within 10 nm of your assigned target waypoint. This is to ensure either that you attack the correct targets or that you stay with escorted aircraft all the way to the target.

- The enemy conducted successful attacks through your sector.
- None of the escorted aircraft survived.
- Allied loses were unacceptable.
- Allied ground forces were defeated.
- The resupply force was wiped out.

You will get one of these messages if you are assigned to protect allied troops or assets and the enemy succeeds in destroying them.

- The enemy operation succeeded.

You will get this message if enemy paratroopers are able to land and deploy to their assigned positions.

- Your flight was eliminated.

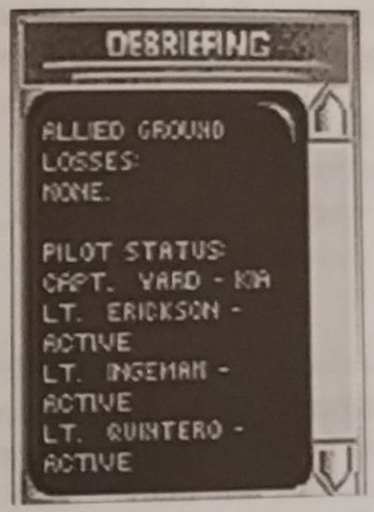
Air-to-air missions require that at least one member of your squadron survive.

- The mission ended too early.

You will get this message if you end the mission early, leaving allied forces unprotected and in danger.

The Mission Summary also tallies the enemy and allied losses which occurred during the mission and provides the post-mission status of all your squadron's

pilots who flew sorties during the mission. The status of a pilot can be active, fatigued, injured, court-martialed, MIA (missing in action) or KIA (killed in action). Fatigued pilots have been ordered to rest by the flight surgeon. They will still appear on the active roster, but will have an asterisk by their names. Fatigued pilots will not be assigned to fly a sortie unless there are too few healthy pilots for the mission. As squadron commander, you can overrule the flight surgeon by assigning a fatigued pilot to an aircraft yourself. Injured pilots will be removed from the active roster until they recuperate. Pilots with any other status will probably never fly again. You can also check the status of your pilots on the Pilot Information Screen.



MISSION REQUIREMENTS

The following section lists the requirements for successfully completing the missions you may be assigned. The mission title will be found on the Campaign Status Screen (in the Report area). The mission briefing will be similar to the typical briefing given here but will be tailored to the situation. Some of the missions require that you stay on patrol for at least 10 minutes. In reality, such missions would last much longer. As one pilot put it, however, "a combat mission is three hours of boredom and ten minutes of stark terror." We decided to give you the later ten minutes.

Scramble

Typical briefing: "AWACS has detected a force of 6 enemy aircraft approaching *point A*. Intercept and destroy these aircraft. Do not let a single enemy aircraft escape."

- Destroy enemy aircraft before they can bomb allied targets.
- Destroy all aircraft assigned in the mission briefing.
- At least one F-16 must survive the mission.

Air superiority

Typical briefing: "Lead a flight of *3* F-16s on a sweep of enemy airspace. Engage and destroy as many enemy aircraft as possible."

- Destroy more planes than you lose.
- Destroy at least one enemy aircraft.
- At least one F-16 must survive the mission.

Front line screen

Typical briefing: "Lead a flight of *3* F-16s on a defensive screen from *point A* to *point B*. Destroy any enemy aircraft encountered in this area. AWACS has reported a large strike force gathering in enemy territory. If they attempt to penetrate your sector, make sure that no enemy aircraft survive. Beware of decoys and do not stray from your assigned area. Stay on station as long as possible, but return to base when you are low on fuel or ordnance."

- Destroy any enemy aircraft encountered.
- Destroy any enemy intruders before they are able to bomb allied assets.
- Sustain your patrol for at least 10 minutes.
- At least one F-16 must survive the mission.

Combat air patrol

Typical briefing: "You are assigned to fly a combat air patrol over *point A*. Prevent enemy aircraft from destroying military assets at this site. Patrol the area until you are low on fuel or ordnance."

- Destroy any enemy aircraft attacking allied assets in your assigned sector before they can inflict casualties.
- Stay on patrol for at least 10 minutes.
- At least one F-16 must survive the mission.

HAVCAP

Typical briefing: "Rendezvous with the E-3 AWACS at waypoint 2 and relieve the escort fighters. Continue escorting the AWACS until your flight is low on fuel or ordnance."

- The AWACS must survive.
- Maintain the CAP for at least 10 minutes.
- At least one F-16 must survive the mission.

Supply CAP

Typical briefing: "A U.S. helicopter force is ferrying supplies to Japanese positions on Kunashiri Island. Provide air cover over the landing zone, then escort the helicopters back to the main island."

- At least two of the four helicopters must survive.
- Maintain the CAP for at least 10 minutes.
- At least one F-16 must survive the mission.

Amphibious landing CAP

Typical briefing: "Provide air cover for Japanese LSTs landing reinforcements on the southern tip of Kunashiri Island."

- Prevent unacceptable (50%) casualties to allied forces.
- Maintain the CAP for at least 10 minutes.

Train escort (Force Protection)

Typical briefing: "Japanese troops are moving by train to attack the Russian beachhead at Mombetsu. Intercept the train at waypoint 2 and escort it to waypoint 3."

- The train must reach its destination unscathed.

Escort transport (Milk Run)

Typical briefing: "Congratulations, you get today's milk run. A flight of C-130s are flying in supplies. Meet them at waypoint 2 and make sure they get here."

- Ensure that the transport aircraft make it safely to their destination.
- At least one F-16 must survive the mission.

Escort strike or bombers

Typical briefing: "Your mission is to escort 3 FSXs on a strike against *point A*. On approach to the target, sweep ahead to clear any enemy interceptors from the strike force's path. If warranted by enemy air defenses, assign a flight to provide SEAD."

- Provide both fighter escort and SEAD support to ensure that strike aircraft make it to their destination and back without losses.
- Your flight must fly to within 10 nm of the target waypoint.
- At least one F-16 must survive the mission.

Rendezvous escort

Typical briefing: "We have just received an SOS from a flight of *3 FSXs* headed to Memanbetsu. They are being pursued by *3 MiG-31s*. Intercept the enemy fighters and drive them from the area."

- At least one of the endangered flight of aircraft noted in the mission briefing must survive.
- At least one F-16 must survive the mission.

Airborne assault interdiction

Typical briefing: "Coded messages intercepted by Army Signal Intelligence have revealed an impending Russian airborne assault on *point A*. Intercept and eliminate the An-12s carrying the assault force. If the Russian paratroops deplane, destroy them before they can reach their positions and dig in."

- Shoot down an An-12.
- Prevent more than 75% of the enemy paratroops from deploying.

Close air support

Typical briefing: "Elements of Company C, 3rd Armored Cavalry Division have come under heavy attack from an enemy tank column. The command ALO has requested Tac air to stem the enemy thrust."

- Destroy enemy ground forces at the assigned location in a timely fashion.
- To succeed you must ensure that enemy forces take more casualties than allied.

Ground strike

Typical briefing: "Tac air reconnaissance has spotted a large concentration of enemy forces near *point A*. Lead a flight of 4 F-16s and destroy as many enemy units as possible."

- Destroy at least 20% of the enemy forces at the assigned location.

Train interdiction

Typical briefing: "The Russians are using a captured train to move troops from Mombetsu to the battle at Omui. Do not allow the train to reach its destination."

- Score at least one bomb or missile hit on the train.

SEAD

Typical briefing: "Enemy air defenses at *point A* represent a significant threat to allied air operations. Lead a flight of *3 F-16s* and destroy as many anti-aircraft weapons at this site as possible."

- Destroy at least one-third of the enemy anti-aircraft weapons at the assigned location.

Bomb

Typical briefing: "Today's ATO calls for a strike against strategic targets located at *point A*."

- Destroy the target listed by the Target button on the Set Waypoints Screen at the location assigned by the mission briefing. You may select a different target with the Target button, but you may not move the waypoint to attack targets at a different location.

Anti-ship

Typical briefing: "Attack the Russian fleet. Your primary target is *the Admiral Kuznetsov aircraft carrier*. Good luck."

- Hit at least one ship of the class assigned by the mission briefing.

SEAD anti-ship

Typical briefing: "Russian naval anti-aircraft weapons are hampering allied air operations. Lead a flight of *3 FSXs* on a strike against the Russian fleet and destroy at least one SAM or AAA system."

- Destroy at least one naval anti-aircraft weapon.

Tactical air reconnaissance

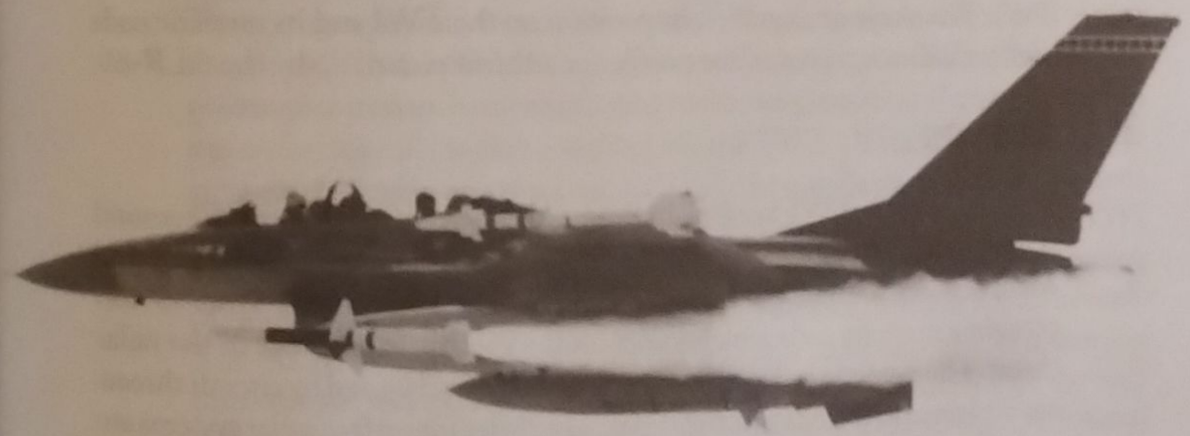
Typical briefing: "Fly to waypoint 2, then return to base."

- Fly to within 10 nm of the assigned destination.
- At least one F-16 must survive the mission.

SECTION

VI

ELECTRONIC WARFARE



INTRODUCTION

Flying an F-16, you are mostly on the receiving end of electronic warfare (EW), but there are a few things you can do to fight back. As far as the F-16 is concerned, the important part of EW involves detection and targeting signals coming from aircraft and surface radars. *Falcon 3.01* includes many new radar threats. The following section describes the measures you can take to avoid becoming the victim of one of these new threats.

THE ALR-69 THREAT WARNING SYSTEM

The F-16 and FSX are equipped with the ALR-69 Threat Warning System, a very sophisticated passive radar receiver which sifts through the electromagnetic spectrum to identify signals representing potential threats. The system can detect signals through any of seven antennas, providing full 360° threat warning. The ALR-69 can determine the radar type, the signal strength and the bearing of a threat. It can also determine if a missile has been launched and whether a threat radar is transmitting missile guidance commands. The Threat Warning Indicator (TWI) is the primary display for the ALR-69. (The LCK and LNH lights are the other system displays.) The ALR-69 information is provided to the pilot in the form of symbols displayed on the TWI. The shape of a symbol, its position on the TWI and its numeric code each provide a key part of the profile for a threat picked up by the ALR-69.

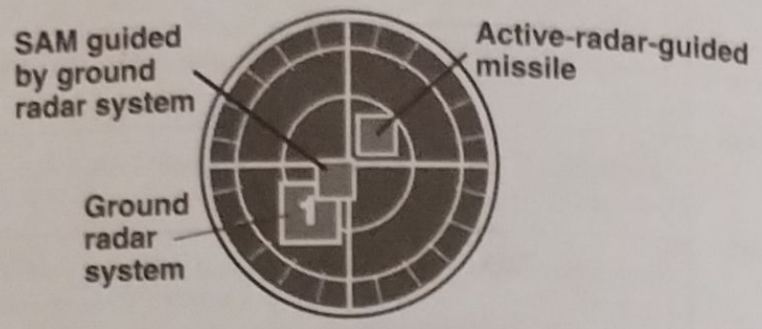
Threat type

The shape of a TWI symbol indicates the type of the threat. Diamond symbols designate aircraft radar, large squares designate surface radar and small squares designate missiles. Diamond and large square symbols also contain a numeric code which indicates the specific capabilities of the radar threat. The numeric codes indicating the types of radar used by aircraft threats are explained in the *Falcon 3.0* manual. The codes for surface radar systems are new, however, and are given in the following table.

Code	Function	Signal Type	Radars: Associated Weapons
1	Short-range SAM control	CW	Straight Flush: SA-6
2	Short-range SAM control	Pulse	Land Roll: SA-8 – Pop Group: SA-N-4 – Cross Swords: SA-N-9
3	AAA fire control	CW	ZSU-23-4 – Bass Tilt: 30mm, 100mm, 130mm
4	Long-range SAM control	Pulse	Pat Hand: SA-4 – Top Pair/Top Steer/Top Dome: SA-N-6
5	Long-range search, GCI	Pulse	Long Track, Strut Pair
6	Long-range, multi-purpose	Chirp	Sky Watch

The new codes categorize the various surface radar systems in the game by function and by the type of radar signal (continuous wave, pulse or chirp) to allow a pilot to quickly analyze the threat posed by an enemy radar system and the best means to counter or destroy it.

The missile symbols do not have codes. Missile symbols will appear in two situations. When a SAM is launched at your aircraft, the ALR-69 detects the guidance commands transmitted by the radar system and displays the missile square on top of the radar symbol on the TWI. (If there are several radar symbols stacked in one place on the TWI, the missile square will be displayed on top of the stack.) The missile symbol can also appear on the TWI independently of a radar symbol if an active-radar-guided missile is homing on your aircraft. This type of missile uses its own radar to track your aircraft. The ALR-69 is able to pick up the missile's radar and give you warning information independent of the launching system. In *Falcon 3.01* the only enemy active-radar-guided missiles you might encounter are the AA-9 Amos and the SA-N-6 Grumble. (Note: the AMRAAM is also an active-radar-guided missile. If one is somehow launched your way — perhaps in a head-to-head dogfight! — it will also appear on your TWI.)



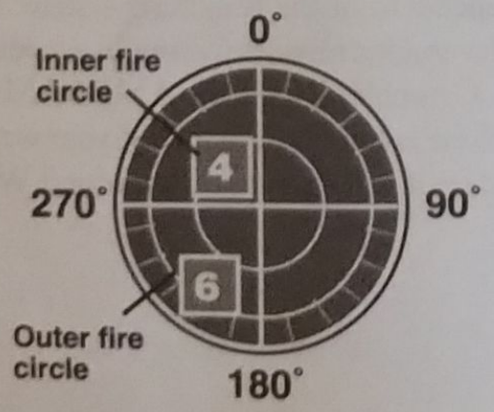
This TWI display indicates two missiles being launched at your aircraft. One missile is being guided by a ground radar system, in this case the Straight Flush (code 1); the other missile is guiding itself with an active radar seeker.

Signal strength

The TWI has two display circles (called fire rings). The symbols for very strong radar signals received by the ALR-69 are displayed on the inner fire ring. Invariably, this means that the radar emitting this signal has locked onto your aircraft (you should also see the LCK warning light up). This is because a radar with a lock will scan your aircraft constantly, while a radar which is still searching will only scan your aircraft a few times per second. This also means that you cannot use the TWI to determine how far a radar is away from your aircraft. The signal strength of a distant, locked radar will still be greater than that of a nearby, searching radar. The symbols for searching radar will be displayed in the outer fire ring.

Signal bearing

The ALR-69 displays the bearing of detected threats on the 360° circle of the TWI. The 0° point of the TWI is the top of the display. A radar threat directly in front of your aircraft will be displayed at 0° on the TWI, a threat directly behind you will be displayed at 180°.



Most of the new radar and SAM systems are associated with the Russian fleet defenses and will only be found in the Kurile Islands theater. However, the

long-range, land-based SA-4 SAM and the Long Track air-search radar have been added to the enemy inventory in all three *Operation: Fighting Tiger* theaters. The Long Track system can hand off targets to SAM systems and can be used to guide interceptor aircraft. This ties together enemy air defenses, making each component much more effective than it would be alone. **Section VIII: Reference** contains a detailed description of each radar and SAM system.

ELECTRONIC COUNTERMEASURES

The ALQ-131 ECM pod is an effective countermeasure against radar-guided SAMs, but crack SAM crews might launch their missiles anyway and attempt to "burn through" your ECM jamming. Chaff cartridges can also fool a SAM long enough for you to maneuver away. With better-trained SAM crews, you will have to combine several countermeasures to avoid SAMs.

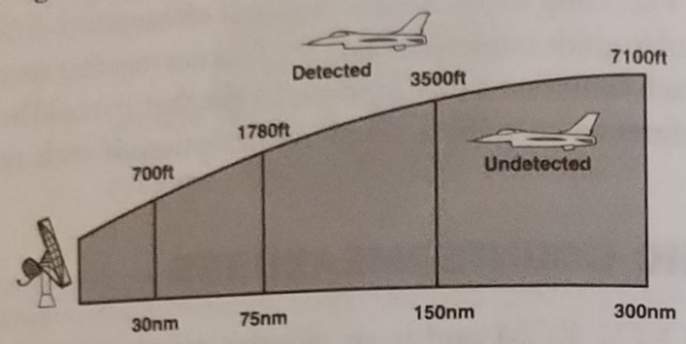
The ECM pod also works against aircraft radar and radar-guided air-to-air missiles, but it is not foolproof. Conversely, enemy planes equipped with ECM pods will use them (according to their skill level) to prevent your F-16's radar and AMRAAM missiles from functioning normally. In addition, the Sidewinder seekerhead is slaved to the F-16 radar, which may prevent the Sidewinder from locking onto an aircraft which is jamming your radar. To overcome this problem, turn off the F-16 radar by pressing R. This will free the Sidewinder to lock onto any heat source.

On the other hand, if you leave your ECM pod broadcasting, the enemy early warning aircraft will detect your presence and vector interceptors to look for you. The AWACS will notify you if enemy ECM transmissions are detected.

OTHER COUNTERMEASURES

The best countermeasure against radar detection is to fly low. Even the F-117 Stealth fighter gets most of its radar "invisibility" from its ability to fly close to the ground. The distance at which a radar can spot you drops off drastically the lower you fly. For example, the same radar that can track your plane from 150 nm at 15,000 ft can only track it from 85 nm at 2,000 ft. Underneath 500 ft, you can usually avoid detection until you get to within about 20 nm of a radar site.

The following diagram illustrates the range of the Russian Sky Watch radar against targets flying at various altitudes:



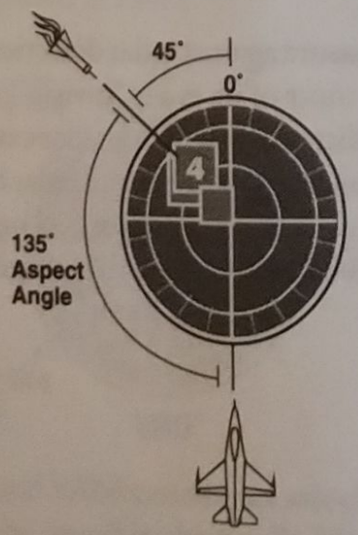
Sky Watch radar range. All surface radar systems show a similar drop-off in range with decreasing target altitude.

Pulse radar can be fooled by turning perpendicular to the source. Thus, if you must climb out of the radar shadow of the earth to acquire a target guarded by a pulse radar system, it is better to make the climb at right angles to the radar so that you remain undetected for as long as possible.

The ultimate countermeasure is destroying the radar emitter. Shrike and HARM missiles are designed for this task. Shrikes are an older weapon and can only lock onto continuous wave emissions. The HARM missile was developed to make up for the Shrike's shortcomings and is effective against all radar types.

Dodging SAMs

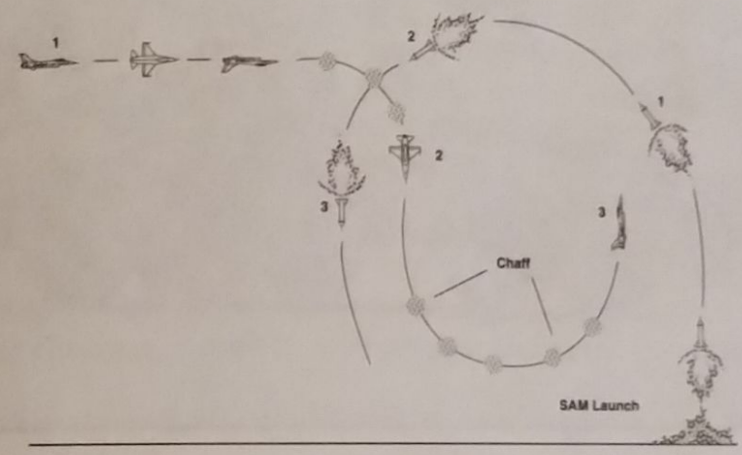
The countermeasure of last resort against incoming SAMs is to dodge while dumping flares and chaff. Although you can't outrun a SAM, you can out turn one. One way to dodge a SAM is to fly toward it with about a 135° aspect angle and break hard across the path of the missile when it gets close.



Approaching a SAM at a 135° aspect angle. In this situation, you would break left.

To achieve a 135° aspect ratio with an incoming missile you must fly almost straight at it. Maneuver your aircraft so that the missile square is positioned either 45° left or right of the TWI 0° point. When the missile gets close — but not too close — break. Timing is everything.

Another dodge is to lead the SAM toward the ground, pulling up at the last second. Since the SAM is less manueverable than you are, it should smash into the ground.



1. Fly toward the SAM until you see the SAM level out or, if it's a long-range SAM, it gets within visual range.

2. Roll to invert your aircraft and break straight toward the ground, dumping chaff as you go.

3. Just before you hit the ground, pull up as hard as you can. Keep dumping chaff.

There is some hazard involved with both of the above techniques.

ENEMY SAM OPERATORS

While you might only see the flames and hurtling steel, behind the controls of every surface-to-air missile is a human being. SAM operators are not perfect. Like the rest of us they get overworked, they can focus too sharply on one target and they worry about dying too soon. The less experienced the SAM crew, the more likely they will suffer from target overload and target fixation. They will take longer to get their systems operational when an air raid warning is sounded and will take longer to fire their missiles once they have acquired a target. The more work you create for the SAM crews, the less effective they will be. Having multiple flights crossing the target from different directions can confuse the operators, making it difficult for them to get off even a single good shot. Inexperienced crews will tend to concentrate all their fire on the flight leader, which is you.

SUPPRESSION OF ENEMY AIR DEFENSES — SEAD

When flying on SEAD missions, you might notice that SAM crews can also be cagey. They might not turn on their radar until you pass by so they get a good shot at your six. This makes SEAD missions a very dangerous game. You are using your plane as bait, trying to induce the SAMs to turn on their radar so you can drive an anti-radiatation missile down their throat.

SEAD missions can be flown in support of another task force or independently as part of a campaign to destroy enemy anti-aircraft weapons (Wild Weasel). Setting up a supporting SEAD mission is described in **Section IV: Mission Planning**. On an independent SEAD mission you will have more resources and your tactics will not be limited by those of another strike force.

The main weapons you have for hunting SAMs and AAA are your ALR-69 Threat Warning System and your HARM and Shrike anti-radiation missiles. To attack your aircraft, the enemy must use their radar. When they do, the ALR-69 will tell you where they are. For true Wild Weasel pilots, the appearance of an enemy radar blip on the TWI is not a threat but an opportunity. Few things are more satisfying than seeing the TWI blips die out as your missiles hit home.

SEAD action code

SEAD is also the name of a new waypoint action code designed specifically for air defense suppression missions. The action is similar to CAS, but instead of attacking any type of ground unit, the flight will focus its attacks on anti-aircraft weapons and radar systems. The flight will use any air-to-ground weapons that you give them, not just the HARM or Shrike anti-radiation missiles. These anti-radiation missiles are the best weapons for the job, but if your mission is to suppress the air defenses of target in support of air strike, you should carry additional weapons. This is because an F-16 can only load up to four HARMs or Shrikes, but can carry a slew of Mavericks and bombs. These extra weapons will be needed to keep the enemy SAM operators ducking for cover while the rest of the strike force hammers the primary target. When you are hunting SAMs independently of a strike force, you have the luxury of using the anti-radiation missiles to pick off the radar systems from long range. When you are flying in support, you have to accompany the strike force to close range anyway, so it makes sense to carry short-range weapons.

There are three distinctly different types of targets against which SEAD missions can be flown: permanent SAM sites, mobile land-based SAMs and AAA, and maritime SAMs and AAA. The SEAD action code can be used to set up an attack against any of these targets.

Permanent SAM sites and radar installations

Missions against permanent installations can be flown using the standoff attack method and HARM missiles. The key to success is to stay low and to avoid any concentrations of smaller anti-aircraft defenses on the way to the target. When you get within range of the target, you might have to pop up to induce the enemy radar operator to paint your aircraft so that your anti-radiation missiles will have a signal to home in on. Then it will be a race to see who can get the missiles off first. You have a slight advantage in this game because your missiles are fire-and-forget and because the radar site can't maneuver. After you take out the radar of a permanent SAM site, close in and destroy the missile launchers for good measure — but beware of short-range SAMs and AAA which might be at the site as well.

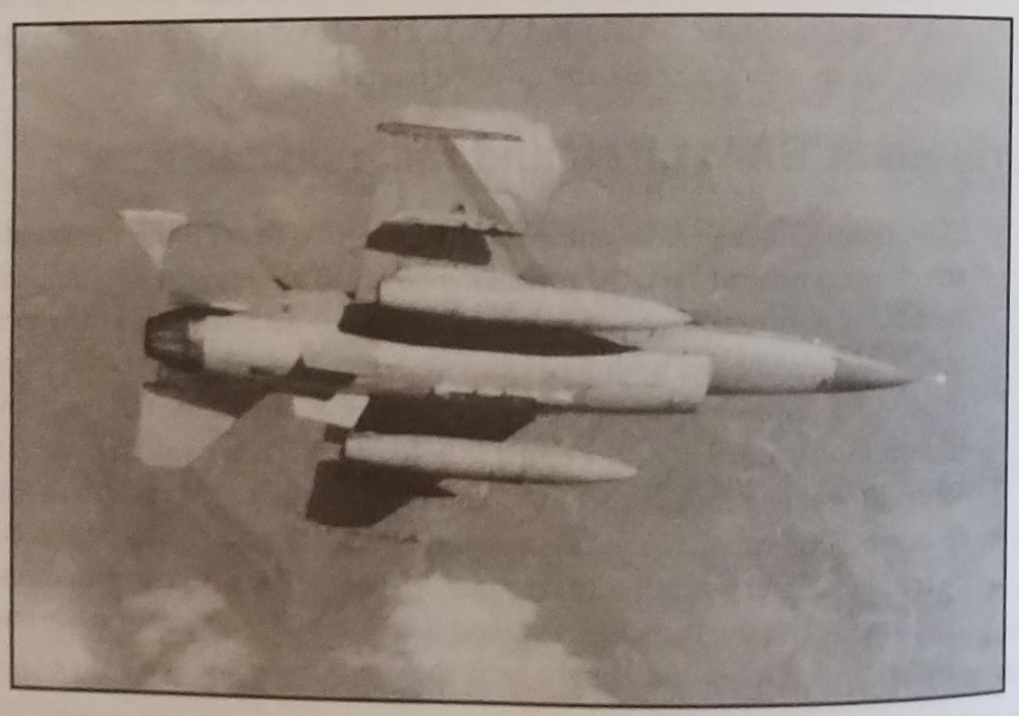
Mobile land-based SAMs and AAA

Mobile land-based SAMs and AAA are used both for the point defense of valuable military and civilian assets and to overwatch moving troop and supply convoys. As a point defense, the anti-aircraft weapons will be scattered around

a site. To destroy them, you will have to wait for them to turn on their radar and use HARM or Shrike missiles. If they refuse to turn on their radar or have none, you will have to hunt them down one by one with Mavericks, bombs, LAUs or guns. When anti-aircraft weapons are protecting a moving convoy, they will be deployed within the convoy itself. In these circumstances, the air defenses are usually easy to find.

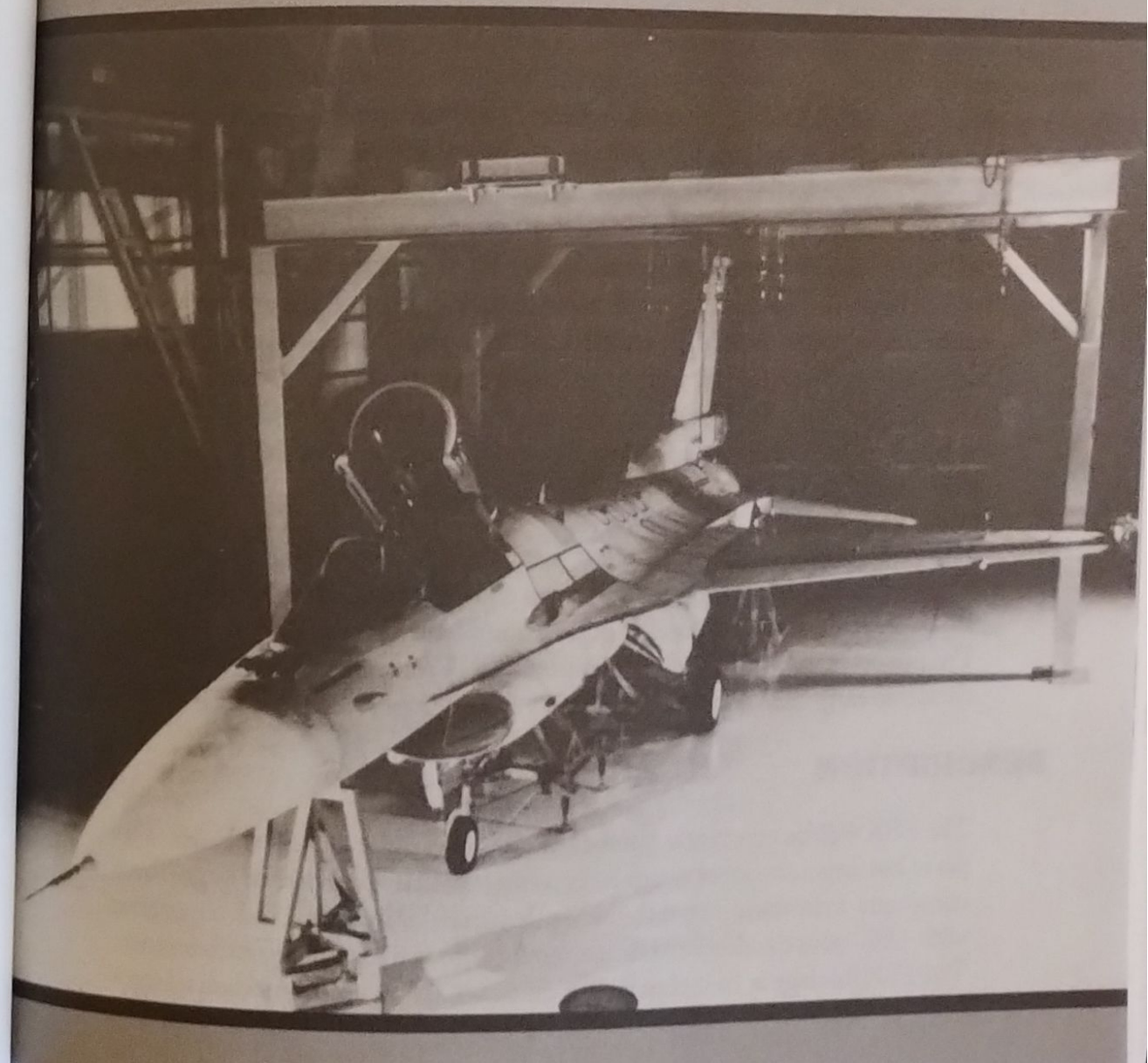
Maritime SAMs and AAA

Maritime anti-aircraft weapons pose a significant threat. The main difference between the deployment of maritime and land-based anti-aircraft weapons is the sheer volume of naval anti-aircraft weapons located in a small area. The Russian fleet has just about every kind of SAM you can think of and lots of heavy AAA to boot. Your best tactic for attacking the anti-aircraft capabilities of the Russian fleet is to look for ships that are isolated from the main force or are on the periphery.



SECTION VII

THE FSX



"This will be the Zero fighter of the modern era..." — General Kiyoshi Matsumiya

INTRODUCTION

The FSX is a highly modified variant of the F-16 Fighting Falcon under joint development by General Dynamics and Mitsubishi Heavy Industries (who made the original Zero). Any squadron you assign to the Kurile Islands theater will fly the FSX aircraft. The Kurile campaign takes place after 1999, the year the production of the FSX is scheduled to begin, and should be a good test of the new aircraft's combat capabilities. The following chapter discusses the background of the FSX, its capabilities, and how to use the FSX's SEA radar and Harpoon anti-ship missiles.

BACKGROUND

In reality, the FSX still faces challenges to its completion. It has already hit a couple of bumps which have made the project two years late and well over budget, and faces a few more down the road. The main obstacles the project had to overcome were objections by the U.S. Congress to the transfer of defense and aerospace technologies to Japan. The Department of Defense still has not approved the transfer of the source code for the flight and weapons control computers, although a deal has been worked out to mollify the Congress on other issues by giving the American company, General Dynamics, a greater share of the production contracts. Because of the overdrawn budget and the conditions added to the deal by the U.S. Congress, the FSX project faces political challenges in Japan as well.

The FSX project began as an effort to replace Japan's aging F-1 fighter. Mainly for financial reasons, the F-16 was chosen as a base model over the F-15, the F/A-18 and a proposal to build a completely new all-Japanese plane. The project began in 1987. The current plan calls for the first test flight of the aircraft in early 1996 and the completion of the first production FSX in 1999.

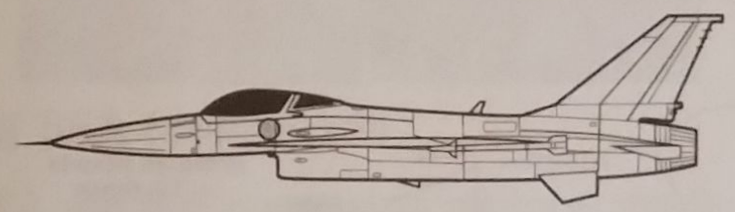
DESCRIPTION

The FSX will incorporate a number of changes to the standard F-16. The plane has larger, stronger wings built using Japanese expertise in large-form composite manufacturing technology. Increased wing area and an engine with 4,000 pounds additional thrust should enhance the FSX's performance. The FSX fuselage is 16 inches longer than an F-16 to accommodate a larger

fuel tank and has a slightly larger vertical stabilizer. The aircraft should have a top speed of about Mach 2 and a 9-ton payload.

The FSX will incorporate a positive-pressure breathing system which helps prevent G-induced loss of consciousness. Without such a system, a pilot must make a tiring "strain maneuver" when he begins to pull high Gs to prevent blood from pooling in his legs and to maintain proper breathing. The positive-pressure system doubles the amount of time a pilot can withstand high Gs, while reducing fatigue. The system works by increasing the air pressure to the lungs as the G force rises and also by providing a counter-pressure to the chest to help the heart pump blood to the brain and eyes. Increased G tolerance should prove helpful to pilots of the nimble FSX.

The U.S. model of the FSX flown in *Operation: Fighting Tiger* uses the same avionics as a stock F-16, which means that the HUD symbology, the CCIP bombsight, the gun funnel, the radar and other systems work exactly the same. The only difference is the addition of a SEA radar mode coupled to the Harpoon missile. These similarities should allow Falcon drivers to quickly convert to the FSX. Because the aircraft's primary task is maritime and ground attack, the FSX engine is tuned for low altitude performance. One additional feature should appeal to those pilots who enjoy flying low on CAS and mud-moving missions: the stronger construction will allow the FSX airframe to endure a few more Shilka hits before disintegrating.



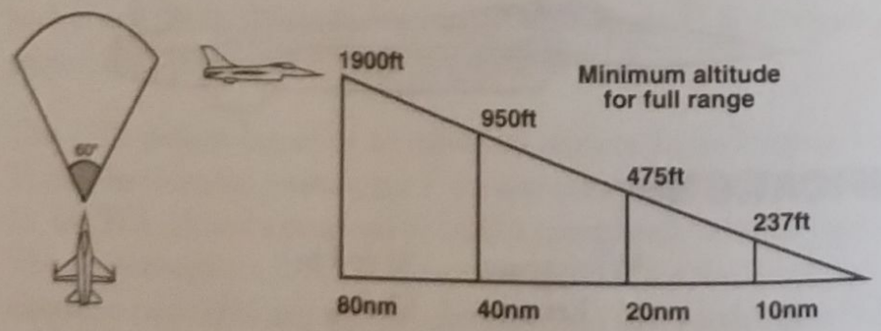
SPECIFICATIONS

Wing span:	31 ft 3 in
Length:	51 ft
Empty weight:	16,137 lbs
Internal fuel:	8,000 lbs
Max weight:	40,113 lbs
Static thrust:	29,000 lbs
Max speed:	1,100 kts
Corner velocity:	422 kts
G limit:	9.4+

USING THE HARPOON MISSILE AND SEA RADAR

The Harpoon is a long-range anti-ship missile which can help you keep a comfortable distance between your plane and an enemy ship bristling with air defense weapons. The missile contains both an inertial navigation system (INS) and an active radar seeker. The launching platform, which can be an aircraft, a ship or a submarine, programs a set of coordinates near the intended target into the missile INS prior to launch. Once fired, the INS guides the missile to these coordinates and switches on the radar seeker. The Harpoon then searches for a target and, finding one, guides itself to impact. The missile can be programmed to ignore some potential targets so that high-value assets can be attacked amidst groups of less important or damaged vessels. The Harpoon is a very reliable missile, but is subject to countermeasures, including ECM, chaff rockets and the CADS-1 high-speed, automatic missile defense system found on many Russian vessels.

The FSX radar features a sea-scanning mode which automatically feeds targeting information to the Harpoon INS. This radar mode is automatically switched on when the Harpoon is selected as a weapon, but can also be manually turned on with F4. The designation "SEA" will appear on the REO display whenever the SEA radar mode is selected. The radar has selectable search ranges of 10, 20, 40 and 80 nm, and scans an angle of 60° in front of the aircraft. The curvature of the Earth will decrease the maximum range of the radar at low altitudes.

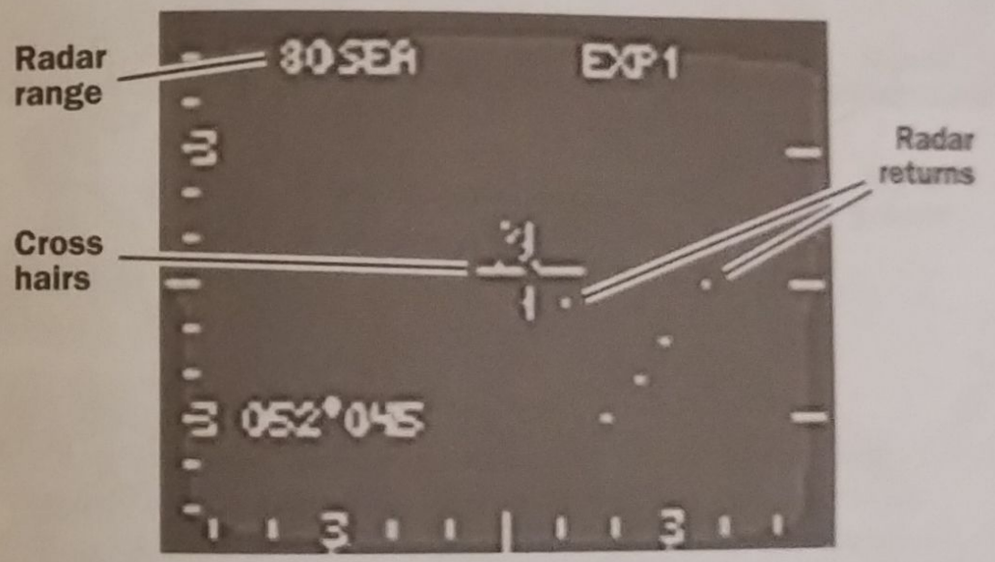


SEA radar scan angle and range. When flying at low altitude, an aircraft's radar can be blocked by the curvature of the Earth. To detect ships at a range of 80 nm, the aircraft must be at least 1900 ft above sea level.

The area of the ocean surface scanned by the SEA radar is determined by the range setting. Radar returns from surface vessels within this area are displayed as blips on the REO: the bigger the ship, the brighter the blip. The radar is capable of displaying three intensity levels: the dimmest intensity is used for ships below 20,000 tons in displacement, the middle intensity for ships

between 20,000 and 40,000 tons, and the brightest intensity for ships larger than 40,000 tons. The shape of the blip matches the ship's radar signature, but the variation in the radar signatures of the different ships is fairly small, except at close range.

The SEA radar has a set of cross hairs to designate the target programmed into the Harpoons INS. If a target has not been designated, the cross hairs will be centered on the REO. The weapons control computer will automatically program the Harpoon so that the missile radar comes on at the optimum distance from the target and skips over any intervening ships. The radar also has an expanded mode which will magnify the display 4x, centered on the selected target.



Selecting SEA radar mode

The SEA radar mode is automatically switched on when the AGM-84A Harpoon is selected as a weapon. The radar mode can also be switched on and off manually with F4 (unless an optically guided weapon like the Maverick is superseding the use of the REO).

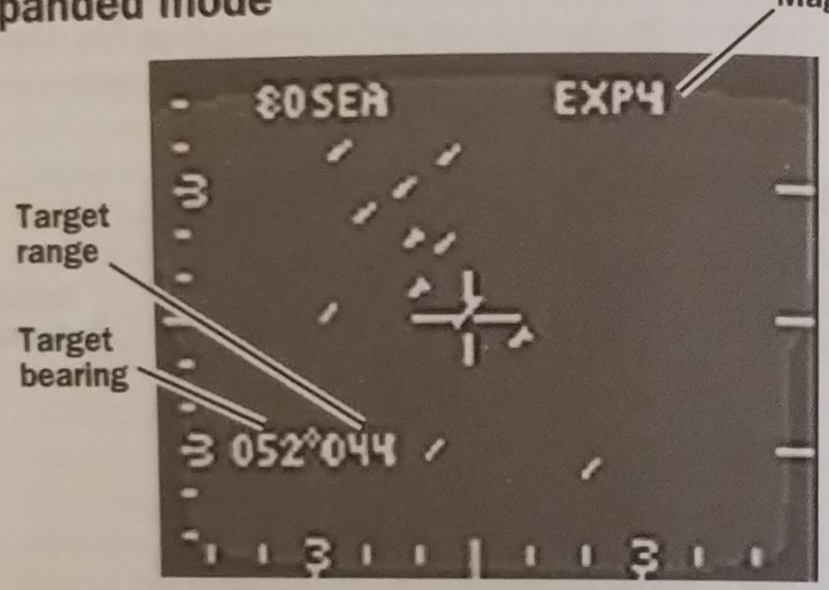
Changing ranges

The SEA radar range can be cycled through 80, 40, 20 and 10 nm settings with the F8 key. Pressing Shift F8 will cycle through the ranges in the opposite direction.

Designating targets

The cross hair designator can be cycled through all the targets appearing on the REO with the [T] key. The bearing and distance of the selected target will be displayed on the lower left portion of the REO, and the target will be programmed into the Harpoon's attack computer. Switching targets will automatically reprogram the missile.

Expanded mode



The expanded mode of the SEA radar can be engaged once a target has been designated in the 1x mode. Pressing [F1] will magnify the REO display 4x, centered on the designated target. In expanded mode, the target designator switch ([T]) will only cycle through targets located in the magnified area, and the display will remain centered on the original target. If the plane banks or otherwise maneuvers so that the magnified area can no longer be scanned by the radar, the radar will switch back to 1x magnification. The radar can be manually switched back to 1x magnification by pressing [F2].

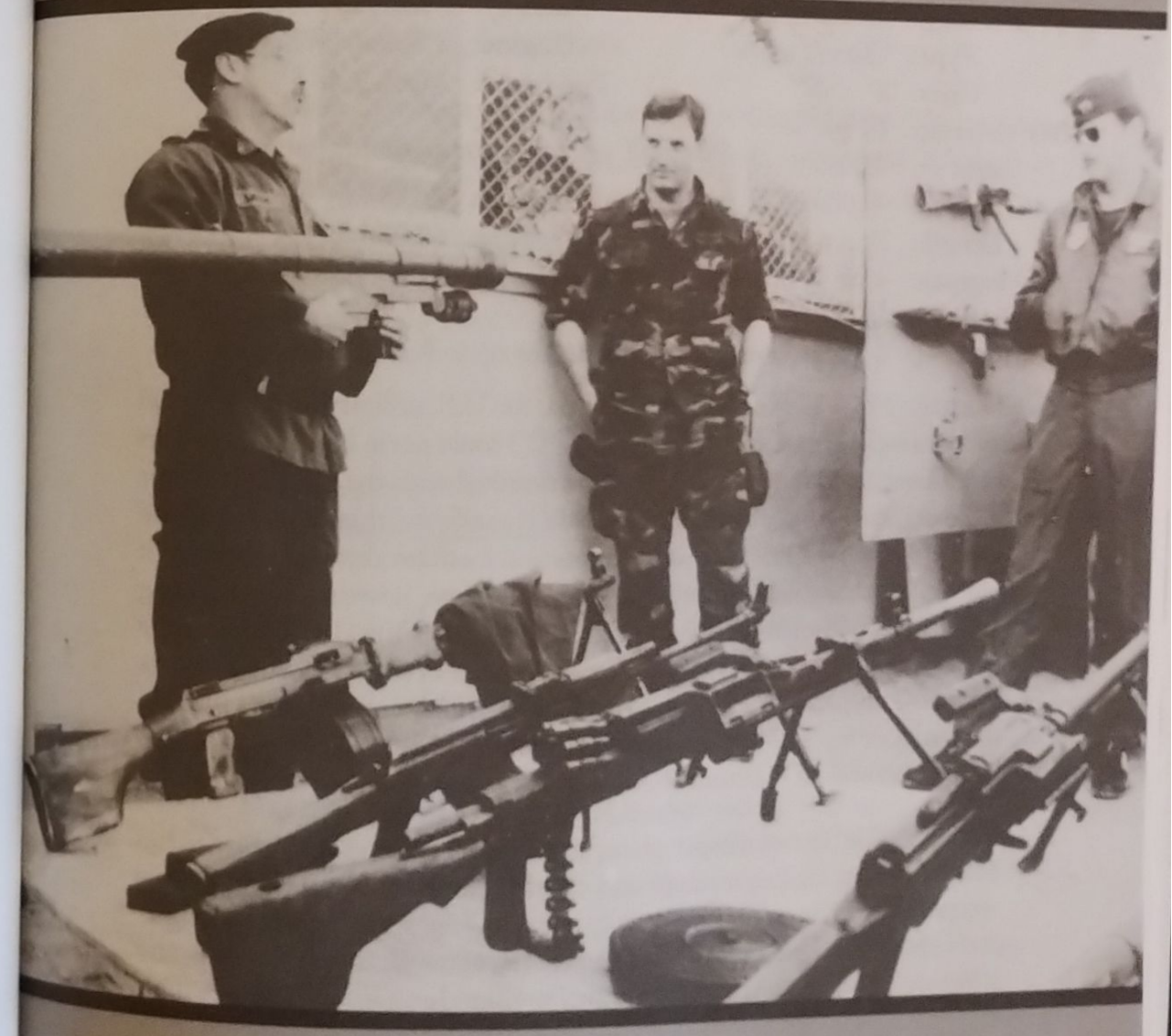
ANTISHIP action code

There is a special waypoint action code, ANTISHIP, for ordering a flight to attack ships. This action code is described in **Section IV: Mission Planning**. For more information on naval strike missions, refer to **Section V: Tactical Operations**.

SECTION

VIII

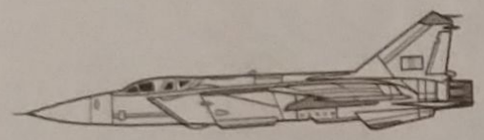
REFERENCE



TECHNICAL SPECIFICATIONS

Aircraft

MiG-31 Foxhound



Owner: Russia
Type: Class A fighter
Crew: 2
Best turn rate: 18° per sec
Min wt: 48,000 lb
Max wt: 90,000 lb
Length: 70 ft 6 in
Wingspan: 45 ft 11 in
Maiden flight: 1977
Armament: 1 30mm cannon and mounts for up to 8 A-A missiles

Made by: Mikoyan
Engine: 2 Solovyov Type D
Top speed: Mach 2.4
G limit: 9
Max range: 2,485 nm
Combat radius: 1,135 nm
Ceiling: 60,000 ft
Radar type: 1

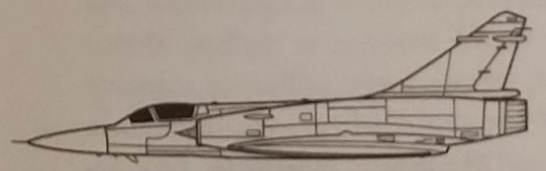
When the MiG-25 Foxbat first appeared, the U.S. military feared it would outclass anything the United States or NATO could put in the air. These fears were allayed in 1976 when a Soviet pilot defected with the plane, giving the U.S. a chance to closely inspect the aircraft. It was found that the MiG-25 did not perform as well as was thought at low and medium altitudes, although it was very capable at high altitudes. At the same time, however, the defecting Soviet pilot told of work on an improved version of the MiG-25 with a stronger frame and more powerful engines capable of supersonic flight at low altitude. This plane, named the MiG-31 Foxhound, entered into service in the Soviet air force in 1983 and was first photographed by an F-16 pilot who intercepted one off the coast of Norway in 1986.

The Foxhound is not a super-plane, however. High-end performance has been sacrificed for longer range and endurance. The aircraft also carries a weapons officer, incurring the penalty of extra weight and size over a single-seat fighter. The Foxhound is a dedicated interceptor. Its most notable feature is its radar, which is the most advanced of any found on a Russian fighter. This

radar has a reported search range of 165 nm, a track range of 145 nm and can engage six targets simultaneously. The radar is sophisticated enough to sort out low-flying aircraft from ground clutter, giving the aircraft full look-down, shoot-down capability. The radar is designed to work closely with the AA-9 Amos missile which has an estimated range of over 80 nm. The Foxhound can carry up to eight of these missiles, although the plane has usually been seen carrying four AA-9s and from two to four shorter range missiles. The aircraft also sports a forward-looking IR detector/tracker with a range of 10 nm which can feed targeting information to its IR missiles, effectively extending their range. The Foxhound is similar to the F-14 Tomcat armed with Phoenix missiles, although the MiG-31 has never been launched from a carrier and is not quite as capable. Like his counterpart in the Tomcat, the Foxhound pilot prefers to destroy his foe at long range. At dogfight ranges, he would have to rely on his short-range missiles and a 30mm cannon. The Foxhound can almost turn with a stock F-16 at low and medium altitudes, but will outperform it at higher altitudes.

The best way to beat a MiG-31 is to sneak inside the range of its Amos missiles and defeat it in a turning battle. The MiG-31 is deployed exclusively by the Russian Republic.

Mirage 2000



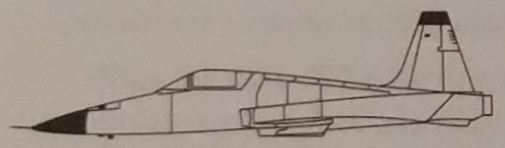
Owner: India
Type: Class A fighter
Crew: 1
Best turn rate: 21° per second
Min wt: 16,535 lb
Max wt: 36,375 lb
Length: 47 ft 1 in
Wingspan: 29 ft 6 in
Maiden flight: 1978
Armament: 2 30mm cannon, 5 fuselage attachment points and 4 wing hardpoints

Made by: Dassault-Breguet
Engine: 1 SNECMA M53-5
Top speed: Mach 2.35
G limit: 9
Max range: 1,800 nm
Combat radius: 435 nm
Ceiling: 59,000 ft
Radar type: 1

The Mirage 2000 is perhaps the only modern aircraft designed for sheer flight performance. The trade-offs made to achieve this performance were a relatively short combat radius and poor ride comfort at low altitude. Poor ride comfort is more significant than it might sound because it increases pilot fatigue and hampers ground attack capabilities. The aircraft is of the same generation of technology as the F-16, but while it may outhandle an F-16, it has inferior radar and avionics. Like the F-16, the Mirage 2000 uses fly-by-wire controls and a flight computer which adjusts the aircraft's control surfaces for optimum performance. The Mirage 2000 was designed as an interceptor and air superiority fighter, but ground attack capabilities were added later.

The most obvious feature of the Mirage 2000 is its delta-wing, tail-less configuration. It carries a pulse-doppler radar with a range of 54 nm and up to eight AA missiles, the most capable being the Matra Super 530 with semi-active radar homing (SARH). The aircraft also carries two 30mm cannons with 125 rounds per gun. The Mirage 2000 has been exported to many countries including India, which calls the plane "Divine Thunder" and uses it as a fighter.

F-5E Tiger II



Owner: South Korea
Type: Class B fighter
Crew: 1
Best turn rate: 20° per second
Min wt: 8,085 lb
Max wt: 25,225 lb
Length: 48 ft 2 in
Wingspan: 26 ft 8 in
Maiden flight: 1972
Made by: Northrop
Engine: 2 GE J85
Top speed: Mach 1.6
G limit: 8
Max range: 1,300 nm
Combat radius: 125 nm
Ceiling: 51,000 ft
Radar type: 3
Armament: 2 20mm cannon, 5 hardpoints and 2 wingtip missile rails

The F-5 Tiger is a light tactical fighter which serves in the air forces of over 21 countries. Although the F-5 is technologically obsolete, Northrop has kept

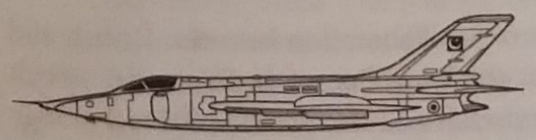
the aircraft alive by tailoring it to the needs of its customers. The plane's miniscule price compared to more advanced aircraft hasn't hurt sales either. The aircraft can carry a credible weapons load, but it suffers from short combat legs and a basic, limited-range radar set. The bulk of South Korea's air force consists of F-5s.

J-7 Airguard

Owner: Pakistan, North Korea
Type: Class B fighter
Crew: 1
Best turn rate: 16° per second
Min wt: 11,629 lb
Max wt: 16,600 lb
Length: 48 ft 10 in
Wingspan: 23 ft 6 in
Maiden flight: 1965
Made by: Chengdu
Engine: 1 Chengdu WP-7b
Top speed: Mach 2.05
G limit: 8
Max range: 1,203 nm
Combat radius: 324 nm
Ceiling: 60,000 ft
Radar type: 3
Armament: 2 30mm cannon and 4 wing hardpoints

The J-7 is a Chinese version of the MiG-21 Fishbed. It incorporates improvements over the Soviet original reflecting the more favorable military and technological contacts between China and the West. These improvements include a modern HUD and weapons-aiming computer, better avionics, as well as better combat performance and handling in general. The aircraft can carry both Chinese and Western AA missiles including Sidewinders and Matras. The aircraft has been sold to many countries including Pakistan and North Korea. Although it is capable of carrying air-to-ground weapons, it is used almost exclusively as a fighter.

Q-5 Fantan

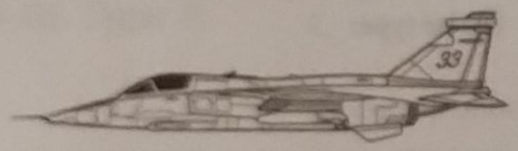


Owner: Pakistan, North Korea
Type: Attack
Crew: 1
Best turn rate: 15° per second
Made by: Nanchang
Engine: 2 Shenyang WP-6
Top speed: Mach 1.5
G limit: 7.5

Min wt: 14,317 lb
Max wt: 26,455 lb
Length: 53 ft 4 in
Wingspan: 31 ft 10 in
Maiden flight: 1972
Max range: 1,080 nm
Combat radius: 324 nm
Ceiling: 52,000 ft
Radar type: 5
Armament: 1 23mm cannon and 10 hardpoints for external stores

Like the J-7, the Q-5 Fantan is a Chinese copy of a Soviet-designed aircraft, in this case the MiG-19. The Q-5 shows a similar improvement in capability over the Soviet original because of China's access to Western technology. While its grandfather plane is obsolete, the Fantan has been constantly upgraded over the years in response to a strong export demand, mainly from Pakistan and North Korea. The plane is most commonly configured as an attack aircraft and can carry up 4,400 lbs of ordnance.

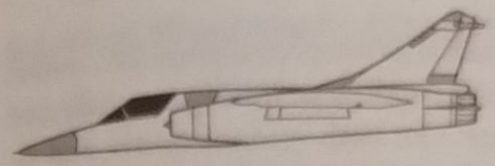
Jaguar



Owner: India
Type: Attack
Crew: 1
Best turn rate: 15° per second
Min wt: 15,432 lb
Max wt: 34,612 lb
Length: 55 ft 2 in
Wingspan: 28 ft 6 in
Maiden flight: 1976
Made by: SEPECAT
Engine: 2 Rolls Royce Mk 804
Top speed: Mach 1.6
G limit: 8.6
Max range: 1,902 nm
Combat radius: 460 nm
Ceiling: 50,000 ft
Radar type: 5
Armament: 2 30mm cannon and 5 hardpoints for external stores

The Jaguar strike fighter, the result of a collaboration between British and French aerospace companies, forms the backbone of India's aerial assault force. India purchased several of the aircraft and also licensed the technology to build their own. It can be employed in a variety of air-to-ground missions, including CAS, Wild Weasel, anti-shipping and precision strikes, and can carry almost 20,000 lbs of ordnance and fuel. The Jaguar is deployed by Britain, France and India.

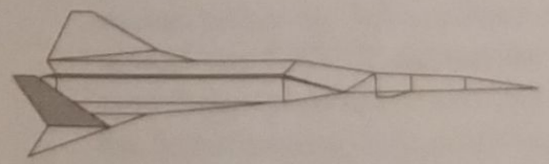
Mitsubishi F-1



Owner: Japan
Type: Attack
Crew: 1
Best turn rate: 15° per second
Min wt: 13,893 lb
Max wt: 30,200 lb
Length: 58 ft 7 in
Wingspan: 25 ft 10 in
Maiden flight: 1968
Made by: Mitsubishi
Engine: 2 Ishikawajima TF40-801A
Top speed: Mach 1.6
G limit: 8.6
Max range: 750 nm
Combat radius: 250 nm
Ceiling: 50,000 ft
Radar type: 5
Armament: 1 23mm cannon, 5 hardpoints and 2 wingtip missile rails

The F-1 design was closely based on the SEPECAT Jaguar, and many components, including the engines, were built by Mitsubishi under license. The aircraft was designed to perform air defense, land attack and anti-ship missions. It can carry substantially less ordnance and fuel than the Jaguar, but is otherwise an adequate, if unimpressive, attack craft.

Unmanned Fighter (UMF)



Owner: Russia
Performance: unknown

The limiting factor of aircraft performance is the human pilot, who can withstand only 9 Gs of force. The structural G limit is much higher for many modern fighters. The Mirage 2000, for example, can take over 13 Gs. Since the G limit translates directly to dogfighting ability, it is obvious that if you remove the human factor you will have the potential for a much better fighter. In addition, if the cockpit, instrumentation and life support systems can be eliminated, the aircraft will not only be significantly lighter, but significantly cheaper to produce. The problem, of course, is replicating the pilot's knowl-

edge, instincts and analytical ability with machine intelligence. While this is not possible today, it probably will become possible in the near future.

The UMF is an experimental aircraft, and Russia is expected to use the conflict with Japan to test the design under combat conditions. Intelligence collected on the vehicle suggests that it can be used for both reconnaissance and air superiority. It can probably be launched from large aircraft such as Tu-22s, from the carrier Kuznetsov and from ground installations. It is estimated that the Russians possess fewer than a dozen of these vehicles. The strengths of the UMF are expected to be a tight turn radius, quick acceleration and a blazing top speed. It is also thought to incorporate elements of stealth technology which would give it a small radar signature. The UMF's weaknesses are estimated to be a somewhat crude combat control computer and its inability to work with other aircraft as a team, which should make it susceptible to decoy tactics like the drag-and-bag or bracketing. It was learned by the CIA that an earlier prototype ran amok with radar missiles and shot down a few of its escorts. If this bug has not been fixed, it is possible that the UMF will be armed only with short-range IR missiles and guns. It is not known whether the UMF receives any remote guidance or command signals.

Ships

Admiral Kuznetsov Aircraft Carrier

Owner: Russia
Crew: 1,600
Displacement: 65,000 tons
Beam: 124 ft
Radar: Sky Watch, Cross Swords and Bass Tilt
Air defenses: SA-N-9, SA-N-8, 30mm AAA
Other weapons: 60 MiG-29s, Su-27s and Su-25s; 12 SS-N-19 anti-ship missile launchers; and CADS-1, an automatic anti-missile defense system
Maiden voyage: 1989
Propulsion: 4 steam turbines
Length: 1,000 ft
Speed: 30+ knots

The construction of the Admiral Kuznetsov aircraft carrier marked a significant departure in Soviet naval strategy. Prior to the Kuznetsov's commission, the Soviet navy was designed primarily for coastal defense and the submarine interdiction of other nation's fleets. While the Soviets had a few aircraft carriers before the Admiral Kuznetsov, these were only capable of launching second-rate V/STOL fighters like the Yak-38. In contrast, the Kuznetsov can carry three squadrons of navalized MiG-29s and Su-27s, the finest aircraft in the Soviet arsenal. Such a fighting force would have allowed the Soviets to project their military might around the world at a moment's notice and to compete for control of the high seas with other world powers. While this may not have fit all that well with the common Soviet strategy of fomenting insider insurgency campaigns, the carrier is quite suitable to a more capitalist mode of power projection which the remade Russian Republic will undoubtedly explore in the next few years.

The Kuznetsov's main weapons are the aircraft it carries. The MiG-29s will likely be used for any ground attack missions, the Su-27s most likely for air superiority and BARCAP. The ship also carries several medium- and short-range SAMs capable of shooting down both aircraft and incoming anti-ship missiles. Like many modern Russian and U.S. naval vessels, the Kuznetsov

employs a high-speed cannon which automatically tracks and fires at incoming anti-ship missiles. The first thing you will notice about the Kuznetsov is its radar. The carrier employs the Sky Watch radar system, which has a large phased-array antenna capable of detecting aircraft up to 300 nm away. The Kuznetsov first left port in 1989. Three more carriers of similar design are under construction.

Kirov-Class Battle Cruiser

Owner: Russia
Crew: 800
Displacement: 28,000 tons
Beam: 92 ft
Maiden voyage: 1978
Propulsion: Nuclear and steam
Length: 813 ft
Speed: 30+ knots
Radar: Top Pair/Top Steer/Top Dome, Cross Swords and Bass Tilt
Air defenses: SA-N-6, SA-N-9, SA-N-8, 130mm and 30mm AAA
Other weapons: 3 helicopters, 20 SS-N-19 anti-ship missile launchers, CADS-1, 3 ASW rocket launchers and 10 torpedo tubes

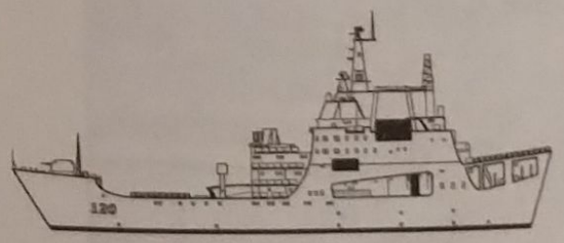
Apart from aircraft carriers, the Kirov-class battle cruisers are the largest warships built by any nation since World War II. The cruisers possess a full complement of anti-ship, anti-submarine and anti-aircraft weaponry. Its most formidable anti-aircraft weapon is the SA-N-6 Grumble, with a range of 54 nm, but it also bristles with medium- and short-range SAMs and AAA. Its 130mm gun alone can hurl more than a half-ton of shrapnel over a distance of almost 10 miles every minute. The Kirovs possess long-range air-search radar with a range of up to 150 nm. Four Kirovs have been built to date.

Udaloy-Class Destroyer

Owner: Russia
Crew: 250
Displacement: 8,100 tons
Beam: 63 ft
Maiden voyage: 1980
Propulsion: Diesel
Length: 531 ft
Speed: 35+ knots
Radar: Cross Swords and Bass Tilt
Air defenses: SA-N-9, SA-N-8, 100mm and 30mm AAA
Other weapons: 2 helicopters, 20 SS-N-14 anti-submarine missile launchers, 3 ASW rocket launchers and 8 torpedo tubes

The Udaloys are large anti-submarine destroyers employed to defend and screen larger ships like the Kuznetsov and Kirovs. Its air defenses include short-range SAMs and AAA. There are currently 13 Udaloys in the Soviet navy, and additional ships are being built at a rate of one per year.

Ivan Rogov-Class Amphibious Warfare Ship

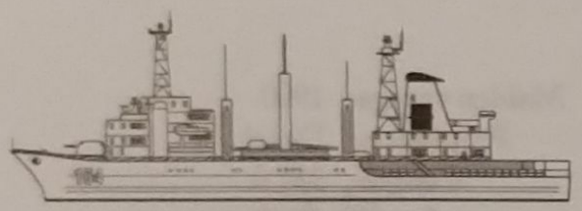


Owner: Russia
Crew: 200
Displacement: 13,000 tons
Beam: 79 ft
Maiden voyage: 1976
Propulsion: Diesel
Length: 518 ft
Speed: 23 knots
Radar: Pop Group and Bass Tilt
Air defenses: SA-N-4 and 30mm AAA

The Rogov and its class companions are the largest amphibious warfare ships in the Soviet navy. Each can carry a naval infantry battalion with full

equipment. The ships can either launch assault landing craft and amphibious tanks offshore or unload its cargo directly onto the beach. The Rogov carries two Lebed landing craft. Short-range SAMs and AAA provide the Rogov with a modest degree of air defense, but it normally relies on the protection of ships like the Kirov. Three Rogov-class ships have been built to date.

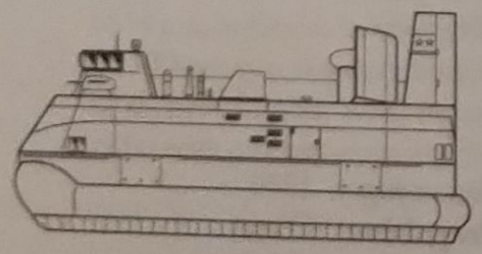
Berezina-Class Replenishment Oiler



Owner: Russia
Crew: 600
Displacement: 36,000 tons
Beam: 85 ft
Radar: Pop Group
Maiden voyage: 1978
Propulsion: Diesel
Length: 695 ft
Speed: 22 knots
Air defenses: SA-N-4
Other weapons: 2 ASW helicopters and 2 RBU-1000 ASW rocket launchers

The Berezina-class replenishment oilers are the largest and most capable supply ships in the Russian navy. These ships carry an estimated 16,000 tons of fuels, 500 tons of fresh water and 3,000 tons of munitions and provisions. They are capable of transferring fuel and supplies to Russian naval vessels while underway.

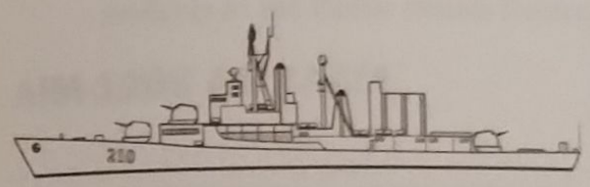
Lebed-Class Air-cushion Landing Craft



Owner: Russia
Crew: 8
Displacement: 85 tons
Beam: 36 ft
Air defenses: 30mm AAA
Maiden voyage: 1976
Propulsion: Diesel
Length: 81 ft
Speed: 70 knots

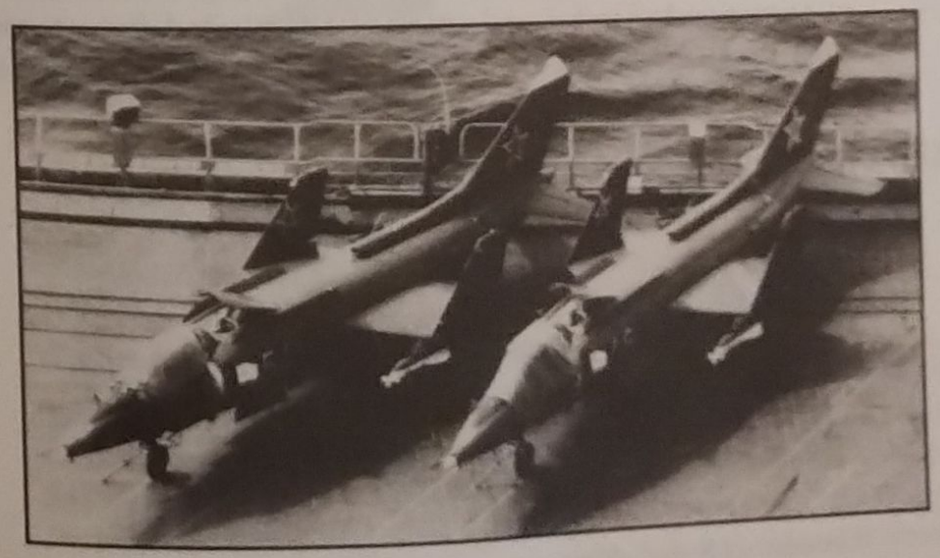
Two Lebed assault landing craft are carried by the Rogov-class amphibious warfare ship. Each Lebed can carry 2 tanks, 120 troops or 45 tons of cargo. Its only armament is a 30mm Gatling gun.

Allen M. Sumner-Class Destroyer



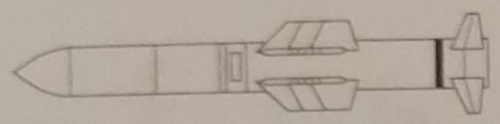
Owner: Japan
Crew: 270
Displacement: 3,320 tons
Beam: 41 ft
Maiden voyage: 1944
Propulsion: Diesel
Length: 390 ft
Speed: 30 knots
Weapons: 1 ASW helicopter, Stinger SAMs, 127mm and 30mm AAA, and torpedo tubes

Japan's Allen M. Sumner destroyers were originally owned by the U.S. Navy and used in World War II. The Sumner employs Stinger shoulder-launched SAMs and its 127mm and 30mm guns for air defense.



Weapons

AGM-84A Harpoon

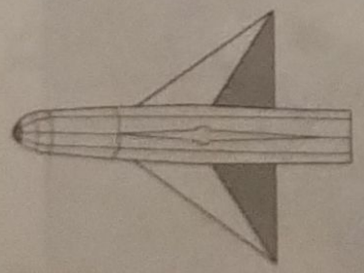


Owner: U.S. and allies
Guidance: INS and active radar
Speed: Mach .75
Weight: 1,145 lb
Type: Anti-ship
Range: 57 nm
Warhead: 500 lb penetration
Drag factor: 23

The Harpoon is a versatile anti-ship missile which can be launched by aircraft, ships and submarines. The launching platform passes activation coordinates to the weapon before firing. The missile then skims above the wave tops until it reaches the activation point, whereupon it turns on its radar and attacks the programmed target. Once launched, the missile needs no further guidance from the launch platform.

It should take two to four Harpoon hits to sink a ship the size of the Kuznetsov or Kirov, one hit to sink a Udaloy. These ships are not defenseless, however, and can employ chaff rockets and automatic anti-missile defenses to spoof or destroy the Harpoon. The Harpoon is only available in the Kurile Islands theater.

AIM-9R Sidewinder

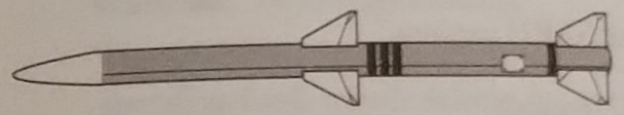


Owner: U.S. and allies
Seeker: All-aspect imaging infrared
Speed: Mach 2+
Type: Air-to-air
Range: 10 nm
Warhead: 25 lb blast fragmentation

Weight: 208 lb
Drag factor: 15

The AIM-9R is the latest version of the Sidewinder, featuring improved control and guidance circuits for better target acquisition and resistance to ECM. While its range is the same as an AIM-9M, it should have a better kill ratio. It is too new to have been combat tested, however. The AIM-9R is only available in the Kurile Islands theater.

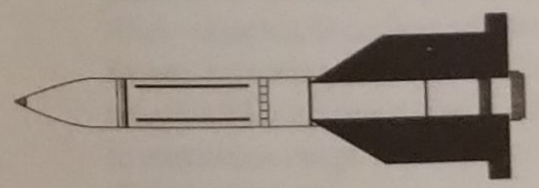
AIM-120E AMRAAM



Owner: U.S. and allies
Seeker: Active radar
Speed: Mach 4+
Weight: 335 lb
Type: Air-to-air
Range: 25 nm
Warhead: 40 lb high explosive
Drag factor: 18

The AIM-120E is an upgraded version of the AMRAAM, featuring an improved radar seeker with a longer range and better electronic counter-countermeasures (ECCM). The AIM-120E is only available in the Kurile Islands theater.

AA-9 Amos

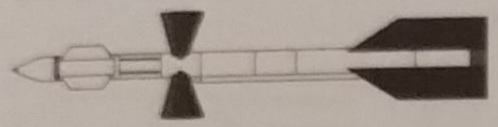


Owner: Russia
Seeker: Mid-course SARH, terminal phase active
Speed: Mach 4.5
Range: 45–85 nm
Warhead: 176 lb blast fragmentation

The AA-9 Amos is a standoff interception missile dedicated to the MiG-31. It is similar in capability and mission to early versions of the American AIM-54 Phoenix. The missile tracks its target by homing in on a signal bounced off the target by the MiG-31 radar. When it gets to within about 15 nm from the target, the missile switches on its own active radar to guide itself through impact. The weapon is one of the largest A-A missiles ever built and is

normally equipped with a large conventional warhead (but can be armed with a nuclear device).

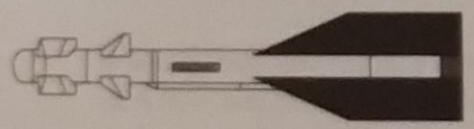
AA-10 Alamo



Owner: Russia
Seeker: All-aspect IR and SARH versions
Speed: Mach 3
Range: 5 nm (IR) 18 nm (SARH)
Warhead: 86 lb blast fragmentation

The Alamo is the newest Russian medium-range missile and can be carried by all the top-line Russian fighters. The missile is produced in both radar and IR versions, although the IR version probably has a much smaller effective range. A weakness of the radar version is that the launching aircraft must keep the target illuminated with its radar until the missile hits.

AA-11 Archer



Owner: Russia
Seeker: All-aspect IR
Speed: Mach 3
Range: 8 nm
Warhead: 45 lb blast fragmentation

The Archer is the newest Russian dogfighting missile, comparable to the AIM-9M in capability. It can be carried by the Su-27, the MiG-29 and the MiG-31.

Surface-to-air missiles

SA-N-9

Owner: Russia
Guidance: Cross Swords radar
Range: 8 nm
Ceiling: 39,000 ft
Speed: Unknown
Warhead: 44 lb fragmentation

The SA-N-9 is an advanced short-range, radar-guided SAM capable of anti-aircraft and anti-missile defense. It is normally fired from vertical launch tubes fitted below the deck.

SA-N-6 Grumble

Owner: Russia
Guidance: Top Pair/Top Steer/Top Dome radar
Range: 54 nm
Ceiling: 100,000 ft
Speed: Mach 6
Warhead: 200 lb fragmentation, nuclear capable

The SA-N-6 Grumble, the most potent SAM you will encounter, is the primary air defense weapon of the Soviet fleet. The system is carried by the Kirov-class battle cruisers and provides the ship with a defensive umbrella of 54 nm, matching the range of allied standoff weapons like the Harpoon. The missile's Mach 6 speed gives it a flight time of about one minute from launch to maximum range. Like the SA-N-9, the Grumble is launched from below-deck vertical tubes.

SA-N-8 Gremlin

Owner: Russia
Guidance: IR
Range: 3 nm
Ceiling: 18,000 ft
Speed: Unknown
Warhead: 5.5 lb high explosive

The SA-N-8 is an improved and navalized version of the SA-7 Grail.

SA-N-4 Gecko

Owner: Russia
Guidance: Pop Group radar
Range: 8 nm
Ceiling: 39,000 ft
Speed: Mach 3
Warhead: 40 lb fragmentation

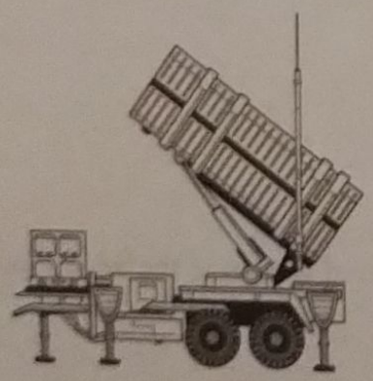
The SA-N-4 is a navalized version of the SA-8. Its associated radar, the Pop Group, is virtually identical to the land-based Land Roll radar used with the SA-8. The SA-N-4 is found on the Rogov-class amphibious warfare ships.

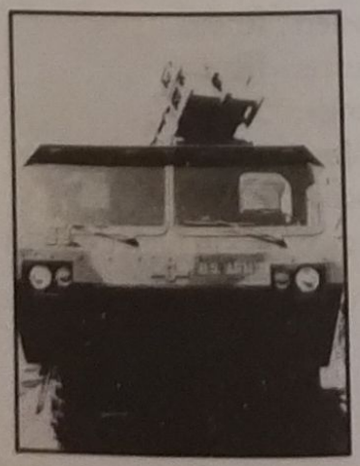
SA-4 Ganef

Owner: Russia and client states
Guidance: Mid-course command beamrider, terminal SARH
Range: 44 nm
Ceiling: 80,000 ft
Speed: Mach 2.5
Warhead: 250 lb blast fragmentation

The SA-4 Ganef is a medium- to long-range SAM deployed in permanent batteries in the *Operation: Fighting Tiger* theaters. The Long Track radar passes target information to the SA-4 battery, which uses the Pat Hand radar for missile guidance and target illumination. During the first two-thirds of its flight, the missile is controlled by operator commands. Only in the last phase of its flight does the missile need target illumination. This allows expert SAM operators to ambush unsuspecting aircraft since a TWI might only pick up the incoming missile after it is too late. Flight time from missile launch to maximum range is less than two minutes.

Patriot missile





Owner: U.S.
Guidance: Track via missile (SARH)
Range: 44 nm
Speed: Mach 3
Warhead: 198 lb fragmentation

The Patriot missile, famous for busting Scuds in the Gulf War, was designed to replace older, less capable U.S. SAM systems. Highly praised at first, the Patriot's effectiveness was questioned after the war. While the Patriot missiles managed to hit all but two Scuds which came within range, fragments of these missiles still caused considerable damage. Part of this criticism results from asking the Patriot to do more than that for which it was designed. The Patriot, intended as a point-defense, anti-aircraft system, does not have a large enough warhead to reduce a large ballistic missile to completely harmless pieces. The Iraqi Scuds were so poorly built and inaccurate anyway that it is arguable that the Patriots did little but randomly change the path of a randomly aimed missile. Thus, from the standpoint of reducing Scud damage, the Patriot was only partially successful, although no one can know how much damage the Scuds would have done if not for the Patriots. Technically, however, the Patriot did quite well, scoring hits on 45 of 47 Scud missiles that flew within range of a Patriot battery. In fact, one Scud broke into 14 pieces during flight, and each piece was hit by two Patriots. Ideally, the Patriot can protect localized, high-value targets, but it can't control where the fragments of its successes will fall.

A Patriot battery consists of a phased-array radar, an Engagement Control Station (ECS) and up to eight 4-tube missile launchers. The system uses a CW radar emitter to illuminate incoming targets. Both the Patriot missile and the ground-based phased-array antenna pick up radar reflections from the target and transmit position data to the ECS. The ECS compares the two data sources and guides the missile along the most efficient intercept course. If either the missile or the ground radar receivers are jammed, the ECS can use one source independently to guide the missile. The system fires automatically and includes an IFF interrogator to avoid shooting down friendly aircraft.

Radar systems

Sky Watch

Owner: Russia
Range: 300 nm
Type: Chirp
TWI code: 6
Purpose: Air search, GCI, tactical surveillance

The Sky Watch is a large phased-array radar mounted on the Kuznetsov aircraft carrier. The system utilizes chirp technology and a set of four large, fixed antenna to extend the range over the horizon. A chirp radar gets its name by continually varying the radar frequency, which extends the range and makes the demodulated radar signal sound like a bird's chirp. Normally the range of a pulse radar is limited by the distance the radar signal can travel in the time between the pulses. Beyond this range, the radar operator cannot tell whether a radar return is reflecting off a distant target illuminated by the first pulse or reflecting off a closer target illuminated by the second pulse. A chirp radar can more easily distinguish between two such returns because the second pulse has a different frequency than the first pulse. Chirp radars are also less susceptible to ECM.

The Sky Watch is used for many purposes aboard the Kuznetsov, including air search, intercept control, early warning and target handoff to SAM and AAA systems.

The radar signal can be targeted by the AGM-88 HARM missile. Shrike missiles, however, are ineffective. The TWI will display a boxed "6" when a Sky Watch radar paints your aircraft. While the maximum range of the radar is quite long, the range of the Sky Watch, like that of other radars, drops off sharply with decreasing target altitude.

Top Pair/Top Steer/Top Dome

Owner: Russia
Range: 150 nm
Type: Pulse
TWI code: 4
Purpose: Air search, acquisition and fire control for SA-N-6

The Top Pair/Top Steer/Top Dome radar group is the primary air search radar of the Kirov-class battle cruiser. It also provides target acquisition and guidance control to the SA-N-6 long-range SAM.

Cross Swords

Owner: Russia
Range: 22 nm
Type: Pulse
TWI code: 2
Purpose: Acquisition and fire control for SA-N-9

The Cross Swords radar is used on many ships to provide target acquisition and guidance commands to the SA-N-9 SAM.

Pop Group

Owner: Russia
Range: 25–40 nm
Type: Pulse
TWI code: 2
Purpose: Acquisition and fire control for SA-N-4

The Pop Group radar is similar to the Land Roll land-based radar and provides limited search, target acquisition and guidance commands to the SA-N-4 SAM.

Bass Tilt

Owner: Russia
Range: 12 nm
Type: CW
TWI code: 3
Purpose: AAA fire control

The Bass Tilt system is a common AAA fire control radar found on most armed Russian vessels.

Long Track

Owner: Russia and client states
Range: 93 nm
Type: Pulse
TWI code: 5
Purpose: Tactical surveillance, GCI, target handoff to SA-4, SA-6 and SA-8

The Long Track radar is a common system used for many air defense purposes. Its primary function is tactical surveillance, and it can use target information to vector interceptors or pass the information to various SAM

systems. Without the Long Track's air search capabilities, many enemy SAM systems are limited to visual acquisition of their targets. The Long Track is mounted on a tracked vehicle, giving it the mobility to keep up with advancing ground forces. Like that of all radar, the range of the Long Track is degraded against low altitude aircraft, especially if they can use terrain features for additional radar masking.

Pat Hand

Owner: Russia and client states
Range: 50 nm
Type: Pulse
TWI code: 4
Purpose: Acquisition and control of SA-4

The Pat Hand radar is used with the SA-4 SAM. The system tracks both the target and the missile, providing mid-course guidance commands. When the target is within range of the missile's radar receiver, the SA-4 homes in on the radar signal reflecting off the target. Using data from a nearby Long Track radar, an experienced operator can avoid painting the target until the last minute to reduce the time the target has to react to the missile threat. While the radar equipment is mounted on a tracked vehicle, it cannot operate while moving and is normally deployed in permanent or semi-permanent sites.

Straight Flush

Owner: Russia and client states
Range: 20 nm
Type: CW
TWI code: 1
Purpose: Fire control and guidance for SA-6, limited search

The Straight Flush radar is used to fire and control SA-6 SAMs. The system normally uses data from a Long Track radar, but the Straight Flush does have a limited search capability. The Straight Flush is mounted directly on the SA-6 launcher.

Land Roll

Owner: Russia and client states
Range: 10 nm
Type: Pulse
TWI code: 2
Purpose: Fire control, guidance of SA-8

The Land Roll is used to fire and control SA-8 Geckos. It can guide two missiles simultaneously at closely spaced targets. Like the Straight Flush, the Land Roll relies on the Long Track radar for target search. Without the Long Track, the SA-8's range is curtailed sharply. The Land Roll is mounted on the turret of the SA-8 launch vehicle, forming a self-contained system which can fire on the move.

Gun Dish

Owner: Russia and client states
Range: 10–12 nm
Type: CW
TWI code: 3
Purpose: ZSU-23-4 fire control

The ZSU-23-4 AAA system sports an integral fire control radar which increases the deadliness of its aim. The radar also allows ZSU-23-4 gunners to anticipate the ingress of aircraft coming from outside visual range.

GLOSSARY

38th parallel - A convenient cartographic line used to divide Korea in half at the end of World War II. The Soviet Union accepted the surrender of all Japanese forces north of the 38th parallel; the U.S. accepted the surrender of all Japanese forces to the south.

AAA - Anti-Aircraft Artillery.

Air cushion landing craft - The military name for a Hovercraft used to land troops and equipment under fire.

Airguard - J-7 aircraft.

Alamo - AA-10 missile.

Amos - AA-9 missile.

Amphibious warfare ship - A ship designed to launch amphibious assault craft as well as unload men and equipment directly onto the beach.

AMRAAM - Advanced Medium-Range Air-to-Air Missile.

APC - Armored Personnel Carrier.

Archer -AA-11 missile.

ATF - Advanced Tactical Fighter.

AWACS - Airborne Warning And Control System.

Bandit - An aircraft identified as hostile.

Bass Tilt - A Russian naval radar system used for AAA fire control.

BDA - Battle Damage Assessment — The determination of the amount of damage done during a mission.

Beaming - The act of turning perpendicular to a pulse-doppler radar system so that it cannot detect your aircraft.

Bogey - An unidentified radar contact.

Bracket - A two-ship air combat tactic of flying to both sides of an oncoming enemy formation so that the enemy pilots must expose their six to one of the two friendlies.

CADS-1 - Combined Air Defense System — a Russian navy anti-missile defense system similar to the U.S. Phalanx system. It combines a high-speed cannon with short-range SAMs, has an integral radar and fires automatically.

CAP - Combat Air Patrol.

CAS - Close Air Support.

Center of gravity - An important strategic or political site, the loss or destruction of which robs a nation of its will to fight.

Chaff - A radar decoy consisting of fine metallic strips fired from aircraft and ships.

Chirp radar - An advanced radar system which constantly changes its frequency of operation to extend its range. The demodulated signal sounds like a bird's chirp, hence the name.

Close-in attack - An attack launched from within a few miles of a target.

Cluster bomb - A bomb consisting of a housing and many submunitions which disperse over a large area for maximum lethality.

Command beamrider - A missile that follows a command beam emitted by a ground station.

CW - Continuous Wave — a type of radar.

Control zone - The area behind an aircraft in which an attacking aircraft can match any evasive maneuver.

Corner speed - The lowest speed at which an aircraft can pull maximum Gs.

Cross Swords - A Russian naval radar system associated with the SA-N-9.

Directive commentary - Inflight commands which you can give to one or all of your wingmen.

DMZ - DeMilitarized Zone — a 4-kilometer-wide buffer zone separating the armies of North Korea and South Korea.

DPRK - The Democratic People's Republic of Korea — North Korea.

Drag - A two-ship air combat tactic which involves one aircraft accelerating ahead to turn in front of an oncoming foe in an attempt to lure him into a tail chase. The other aircraft lags behind so that it can close in on the enemy pilot's six when he turns to chase the first aircraft.

Drag Index - An index proportional to the drag coefficient that hanging a given piece of high-explosive or external store will add to your aircraft.

Durandal - The brand name for a popular anti-runway munition.

ECCM - Electronic Counter-CounterMeasures — electronic measures taken to counter someone else's measures to counter your measures to counter...

ECM - Electronic CounterMeasures — electronic methods for defeating enemy offensive or defensive weapons and radar systems.

Egress - n. The way out. **v.** Leave.

Elbow - The control zone.

EW - Electronic Warfare — radar, radar jamming, etc.

Etorofu - One of the disputed Kurile Islands.

Fantan - Q-5 aircraft.

Flare - An infrared-guided missile decoy.

Fly-by-wire controls - Aircraft controls with signals transmitted by electronic rather than hydraulic or mechanical means.

Foxhound - MiG-31 aircraft.

Free-fall bomb - A bomb which drops solely by its own weight.

FROG missile - Free Rocket Over Ground — a designation for Russian unguided surface-to-surface missiles.

FSX - A heavily modified variant of the F-16 Fighting Falcon built by Mitsubishi Heavy Industries.

Ganef - SA-4 SAM.

GBU - Guided Bomb Unit.

GCI - Ground Controlled Intercept.

Gecko - SA-N-4 and SA-8 SAMs.

General Dynamics - Manufacturer of the F-16 Fighting Falcon.

General purpose bomb - A bomb for all seasons.

Glide slope - The proper descent angle for landing.

Gremlin - SA-N-8 SAM — an improved, naval version of the SA-7 Grail.

Grumble - SA-N-6 SAM.

GSD - Glide Slope Deviation — the deviation of an aircraft from the optimum glide slope.

Gun Dish radar - The fire control radar for the ZSU-23-4 Shilka.

GW - Gross Weight.

Habomai - One of the disputed Kurile Islands.

HARM - High-speed Anti-Radiation Missile.

Harpoon - AGM-84A anti-ship missile.

HAVCAP - High Asset Value Combat Air Patrol.

High-level attack - An attack initiated from above 10,000 ft.

Hokkaido - The northernmost major Japanese island.

Hovercraft - The trademark for a vehicle which rides upon a cushion of air.

HUD - Head-Up Display.

IFF - Identify Friend or Foe — a device which sends out a coded signal to other aircraft. The IFF device in the other aircraft will automatically reply if they are programmed with the same code (which is changed daily) and will be designated as friendly.

ILS - Instrument Landing System.

Ingress - n. The way in. **v.** Enter, approach.

INS - Inertial Navigation System.

Iron bomb - A simple bomb consisting of a metallic casing filled with high explosive. Any of the Mk series of bombs.

JSDF - Japanese Self-Defense Force.

Kashmir - A province shared by India and Pakistan.

Kim Il Sung - Democratic dictator of the Democratic People's Republic of Korea.

Kim Jong Il - Son of Kim Il Sung.

Kunashiri - One of the disputed Kurile Islands.

Land Roll - A Russian radar system associated with the SA-8.

LD - Localizer Deviation — the deviation of an aircraft from the optimum approach heading to a runway.

Long Track - A Russian tactical surveillance radar.

Low-level attack - An attack initiated from below 10,000 ft.

LST - Landing Ship, Tank — an amphibious landing craft designed to carry tanks and soldiers to shore.

MacArthur, Douglas - U.S. general during World War II and the Korean War.

Maverick - One of the AGM-65 series of air-to-ground missiles.

MiG - An aircraft developed by the Mikoyan and Gurevich design bureau. "Mig" is also a colloquial Russian adjective used to describe anything which is unusually swift.

NATO - North Atlantic Treaty Organization.

Ordnance - Weapons.

OTH radar - Over-The-Horizon radar — an extended range radar capable of detecting targets beyond the normal radar horizon.

Pat Hand - A Russian radar system associated with the SA-4.

Phased-array radar - A radar system utilizing several antenna receivers to improve reception.

Pop Group - A Russian radar system associated with the SA-N-4.

Pop-up maneuver - A quick climb from a low altitude to an altitude sufficient to acquire the intended target.

REO - Radar/Electro-Optical Display.

Retrograde maneuver - Any maneuver involving an advance away from enemy forces.

ROE - Rules of Engagement — "directives issued by competent military authority under which United States forces will initiate and/or continue combat engagement with other forces encountered" [JCS Pub. 1].

ROK - Republic of Korea — South Korea.

SAM - Surface-to-Air Missile.

SARH - Semi-Active Radar Homing — a missile guidance system which requires a separate radar system to provide target illumination for the missile to home in on.

Scramble - A quick reaction mission flown to intercept an incoming attack force.

Scud - NATO code name for a type of Russian-designed ballistic missile.

SEAD - Suppression of Enemy Air Defenses.

SEATO - South East Asia Treaty Organization.

Shikotan - One of the disputed Kurile Islands.

Shrike - AGM-45A anti-radiation missile.

Sidewinder - One of the AIM-9 series of air-to-air missiles.

Sky Watch - An OTH radar deployed on the Admiral Kuznetsov aircraft carrier.

Slick - A general purpose bomb without high-drag fins.

Spoof - To jam, decoy or evade.

Standoff attack - An attack initiated from outside the range of effective defensive response.

Stinger - A U.S. short-range infrared-guided SAM.

Straight Flush - A Russian radar system associated with the SA-6 SAM.

Strain maneuver - A chest-tensing technique used by pilots to partially counteract the effects of high-G forces.

TASMO - Tactical Air Support of Maritime Operations.

Terrain masking - The use of terrain features to evade radar detection.

Tiger II - F-5 aircraft.

Top Pair/Top Steer/Top Dome - A group of Russian naval radar systems associated with the SA-N-6.

Tsushima Strait - The passage between South Korea and Japan.

TWI - Threat Warning Indicator.

Umbrella, air defense - The range and ceiling of a given air defense weapon.

UMF - Un-Manned Fighter.

V/STOL - Vertical/Short Takeoff and Landing.

VASI lights - Visual Approach Slope Indicator lights.

Waypoint - The points along a flight plan at which speed, altitude and action are set.

Wild Weasel - A dedicated SEAD aircraft with specialized electronics for the detection and destruction of anti-aircraft weapons.

Winchester - Out of weapons.

Yalta conference - A conference at the end of World War II which divided Europe and Asia between the Soviet and American empires.

Yeltsin, Boris - First president of the Russian Republic.

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IX

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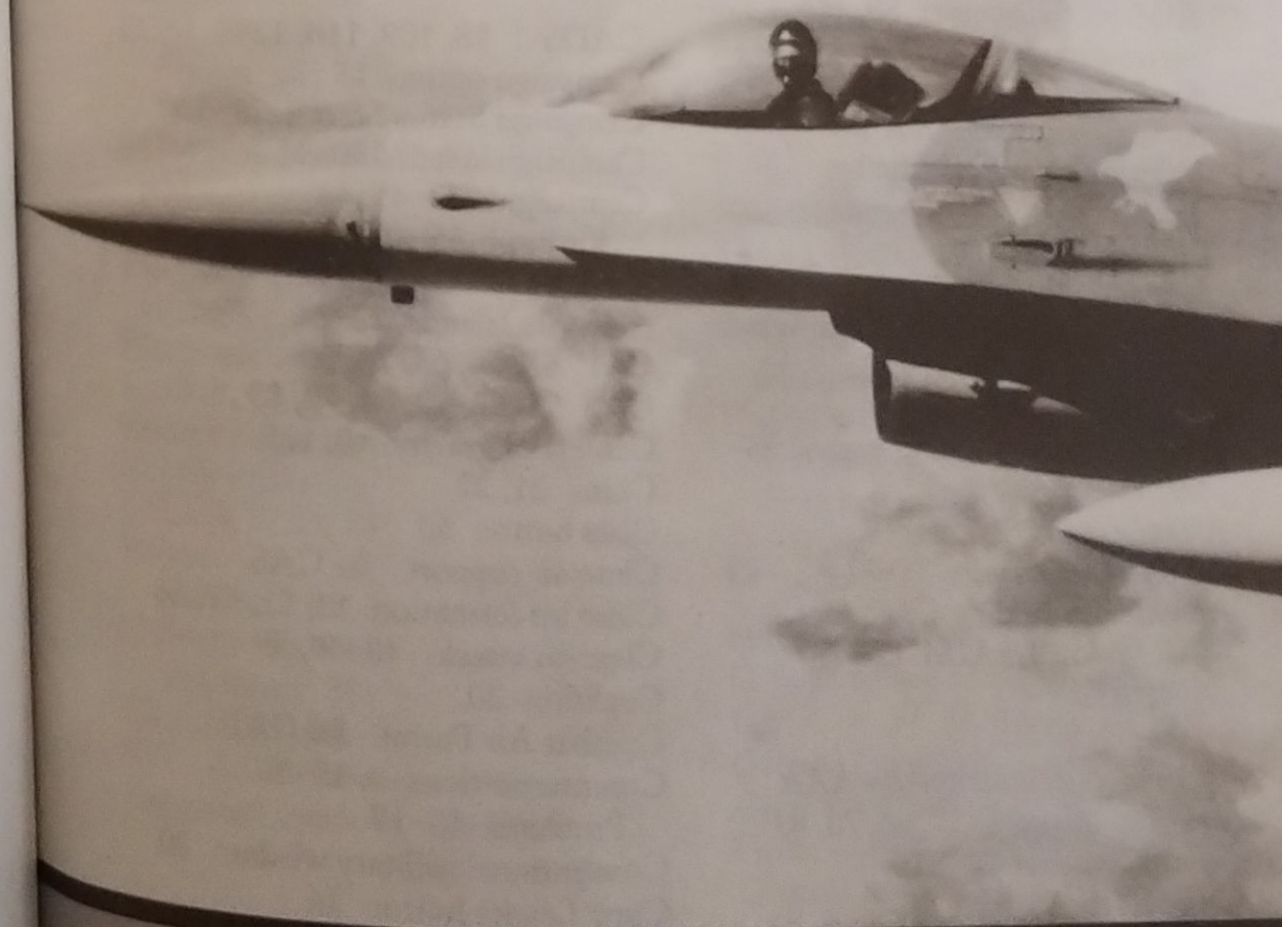

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